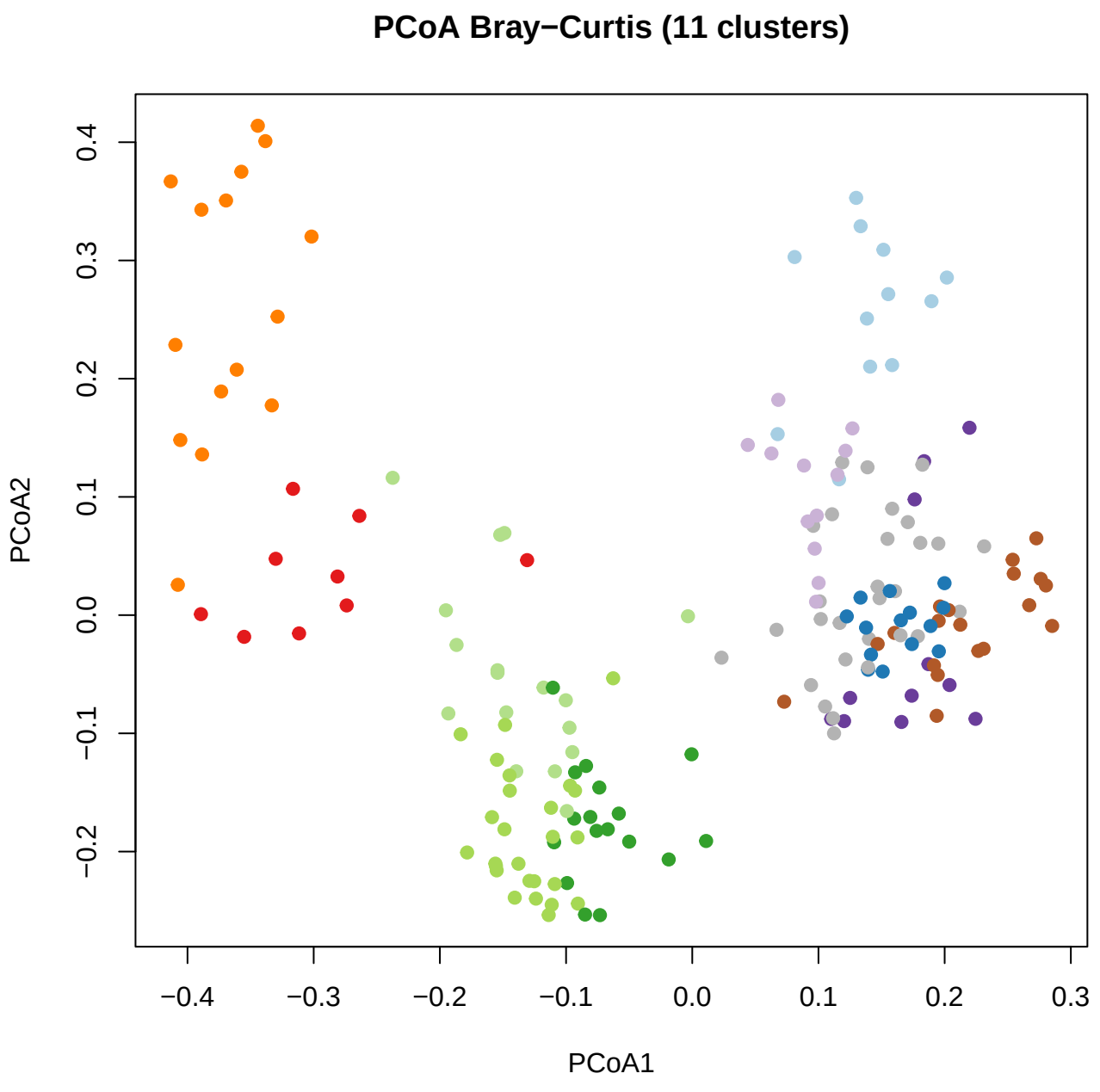
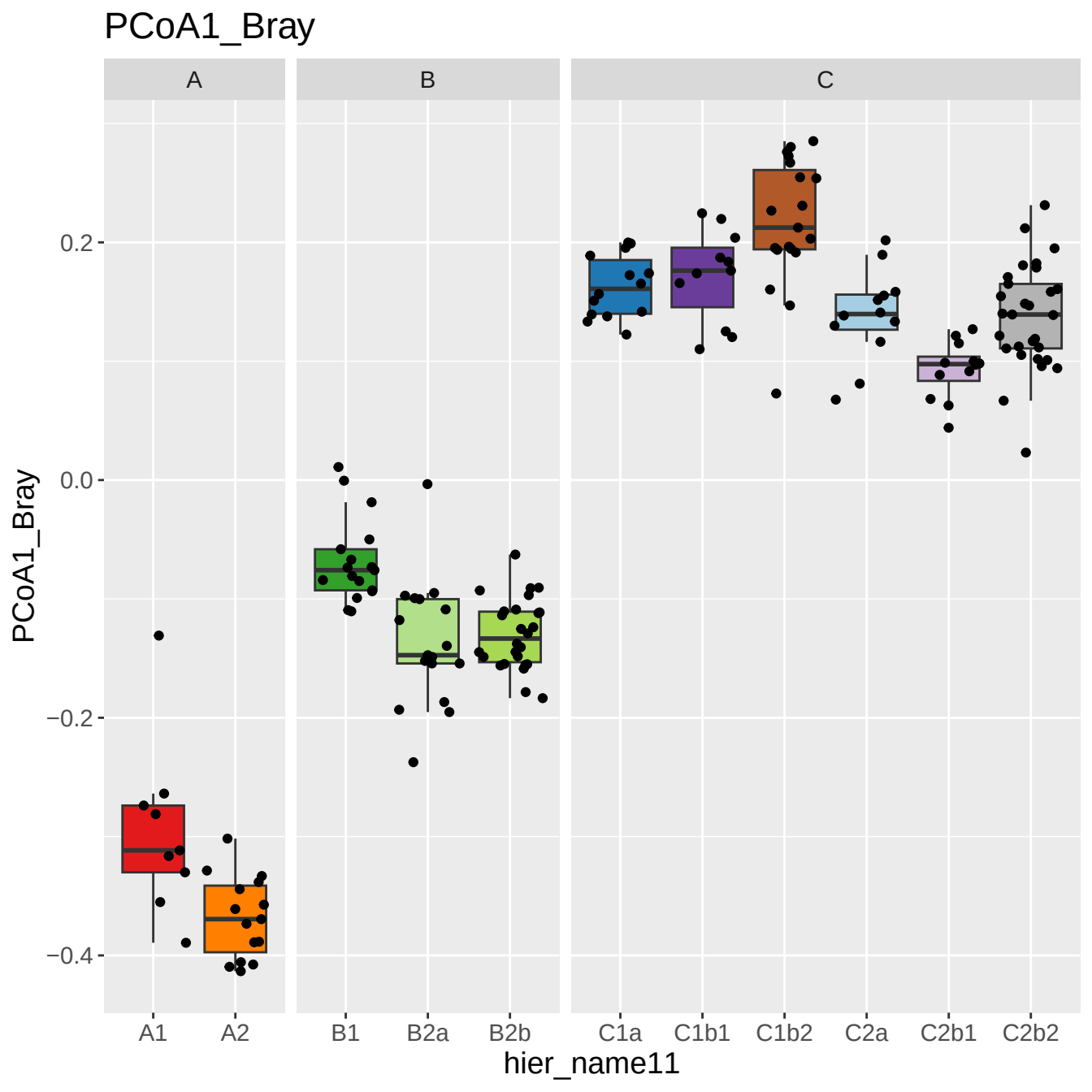
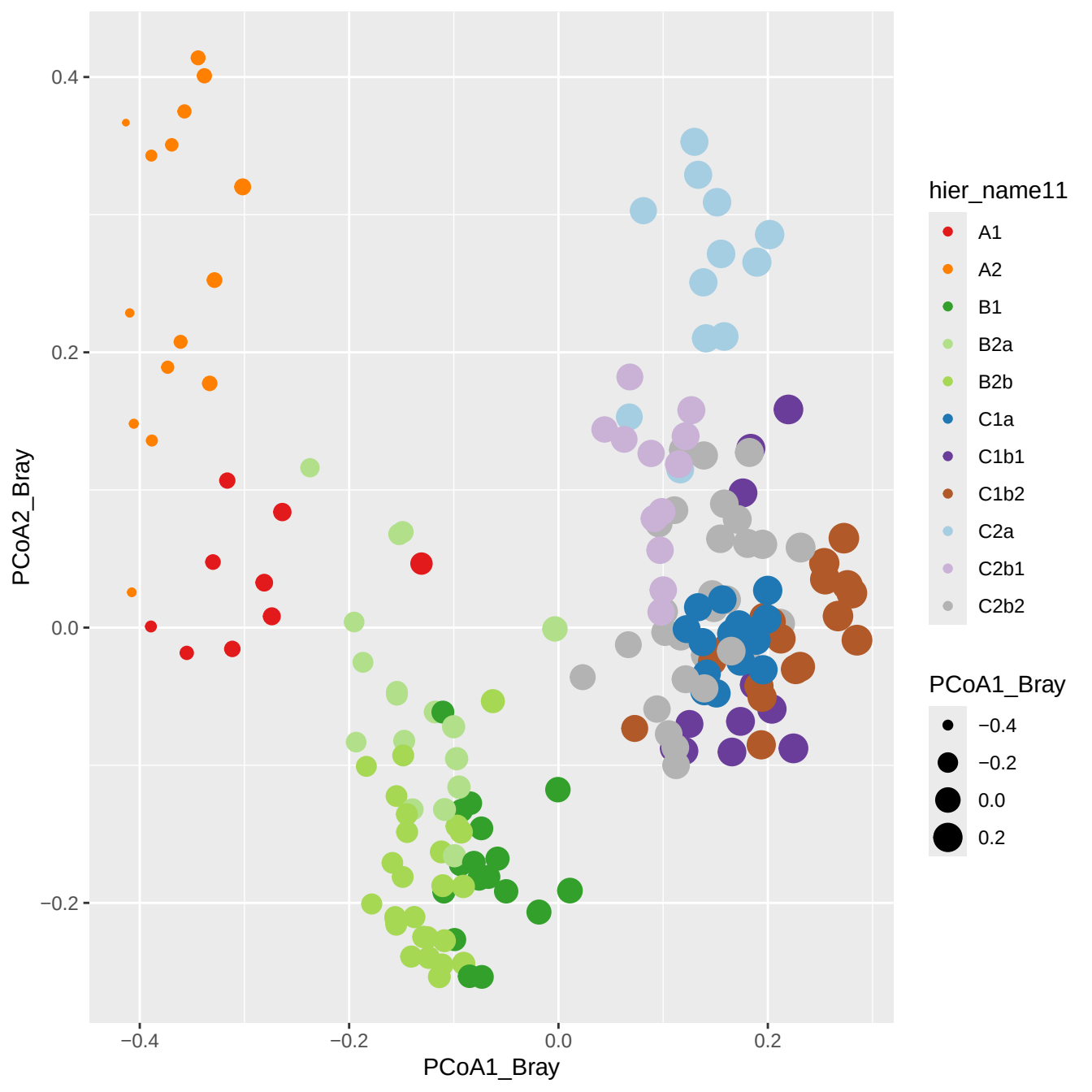
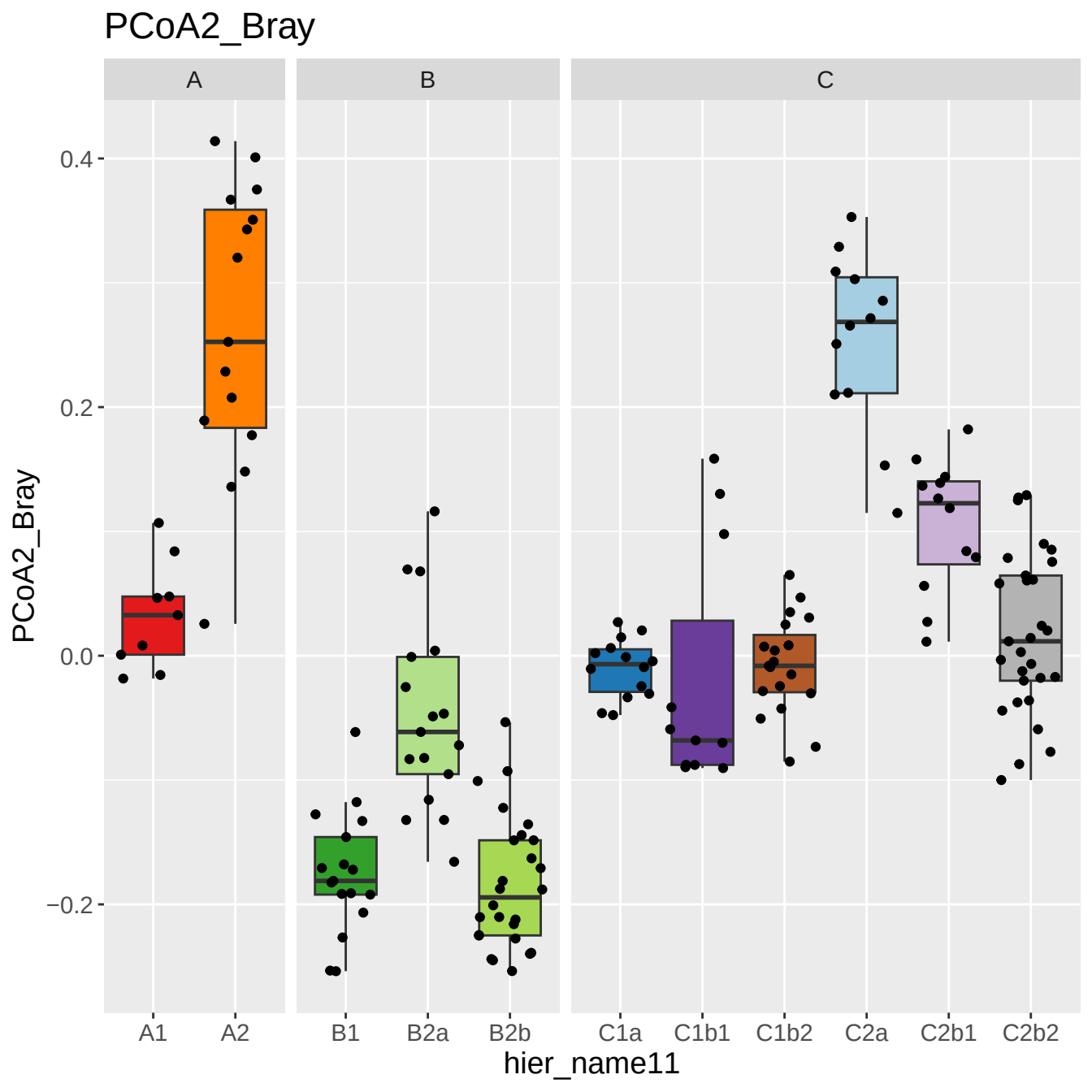
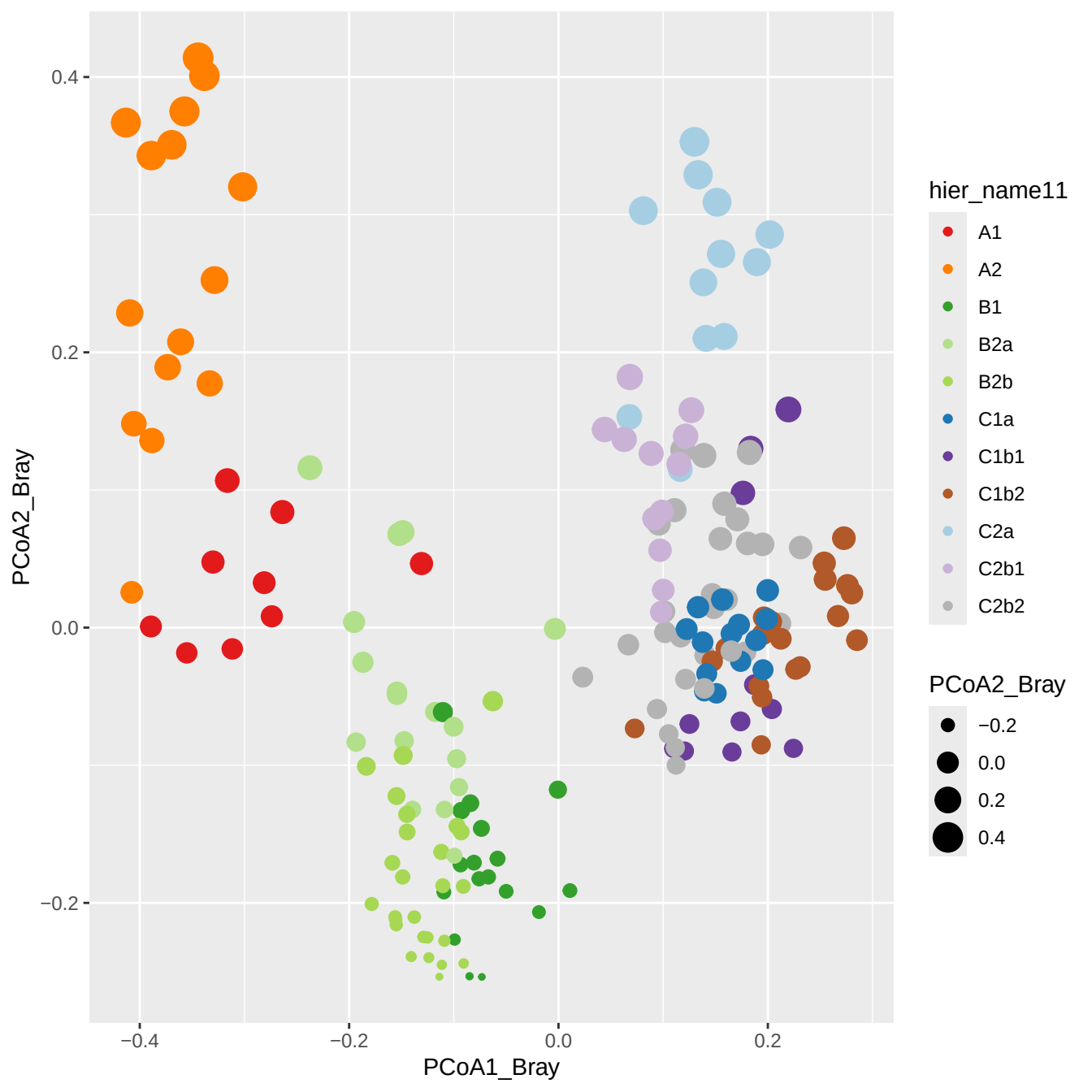
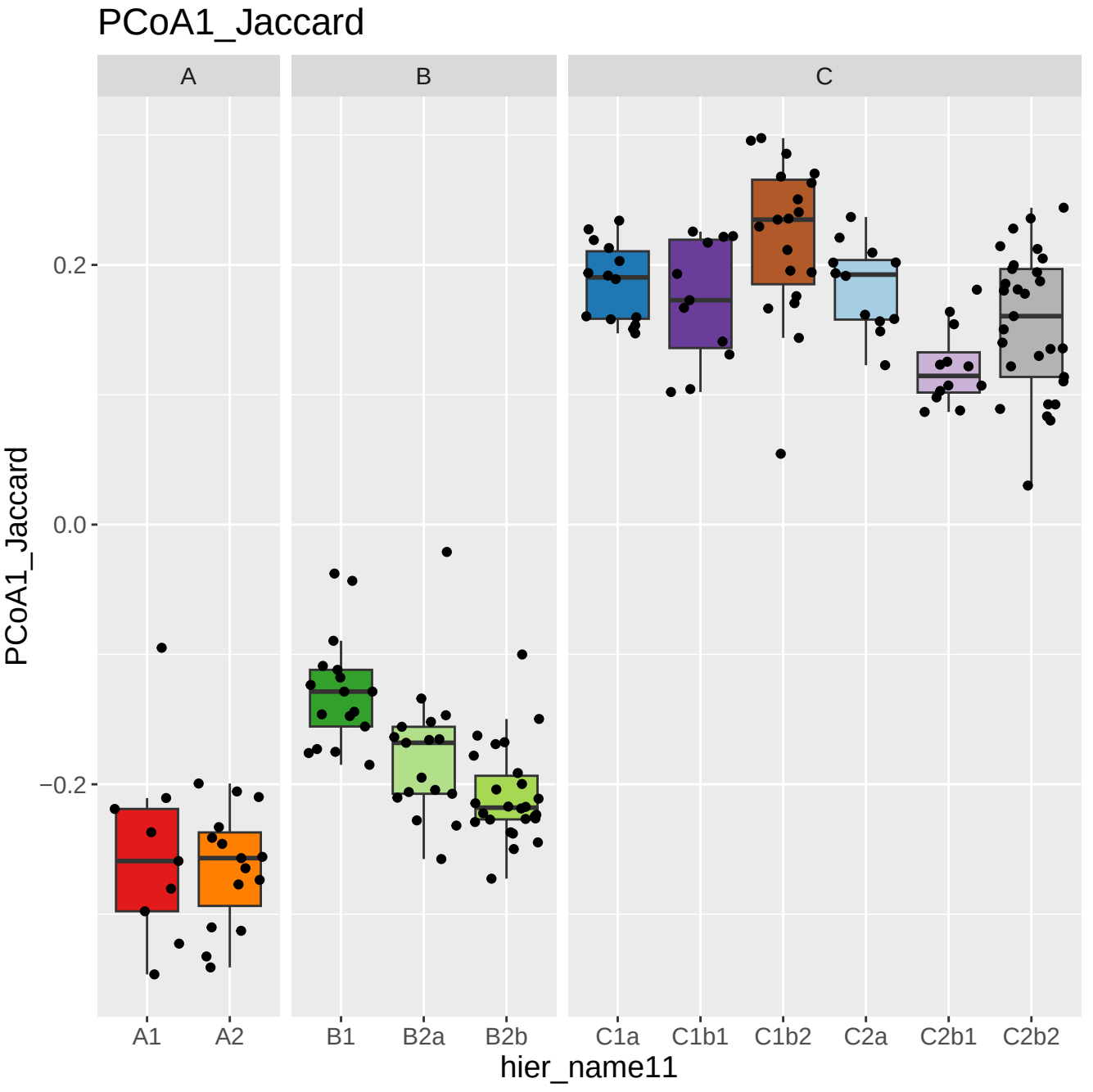

PCoA Bray−Curtis (11 clusters)
PCoA2
PCoA1
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.4
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3

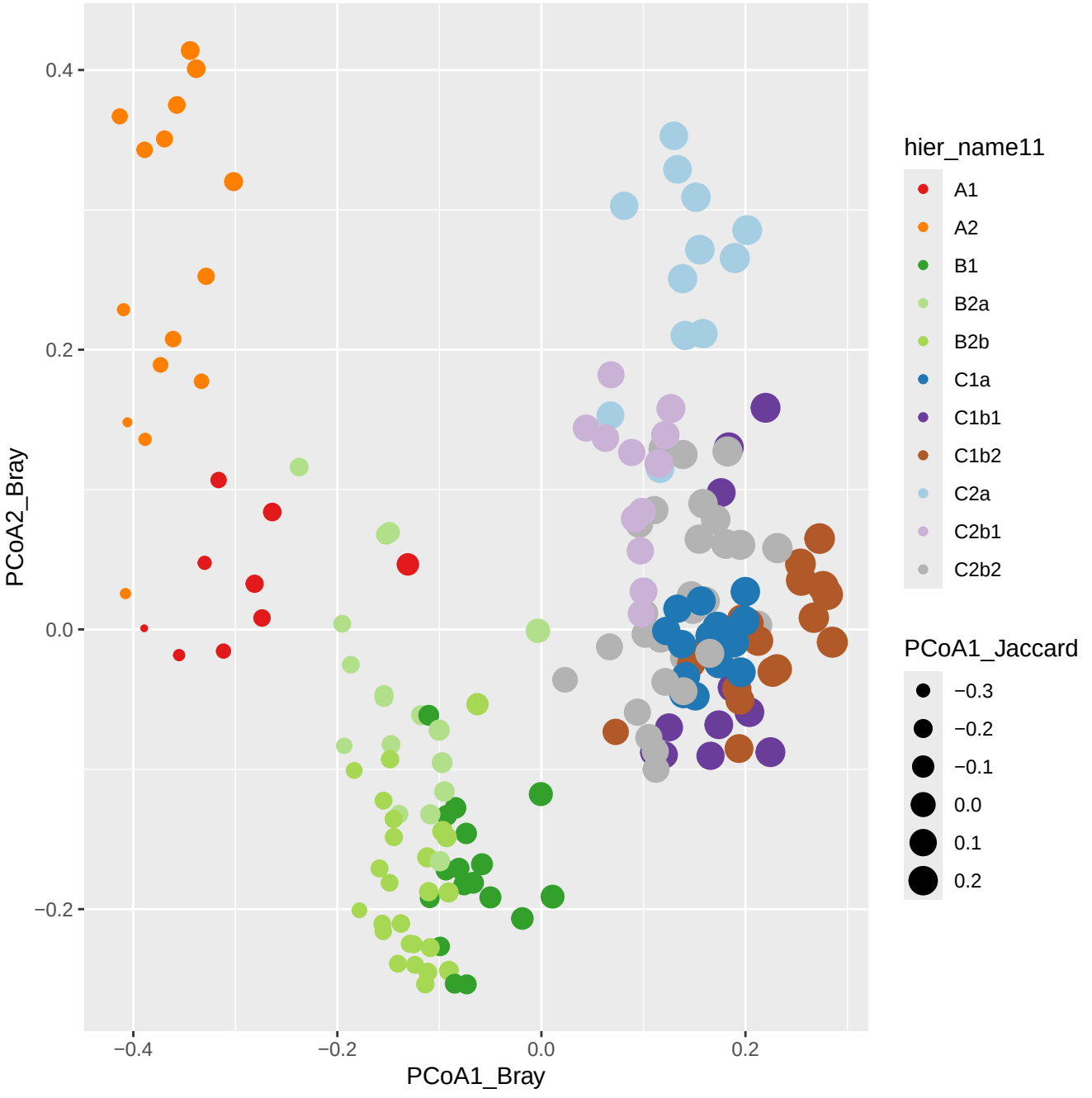

PCoA1_Bray
A
B
C
0.2
0.0
−0.2
−0.4
PCoA1_Bray
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

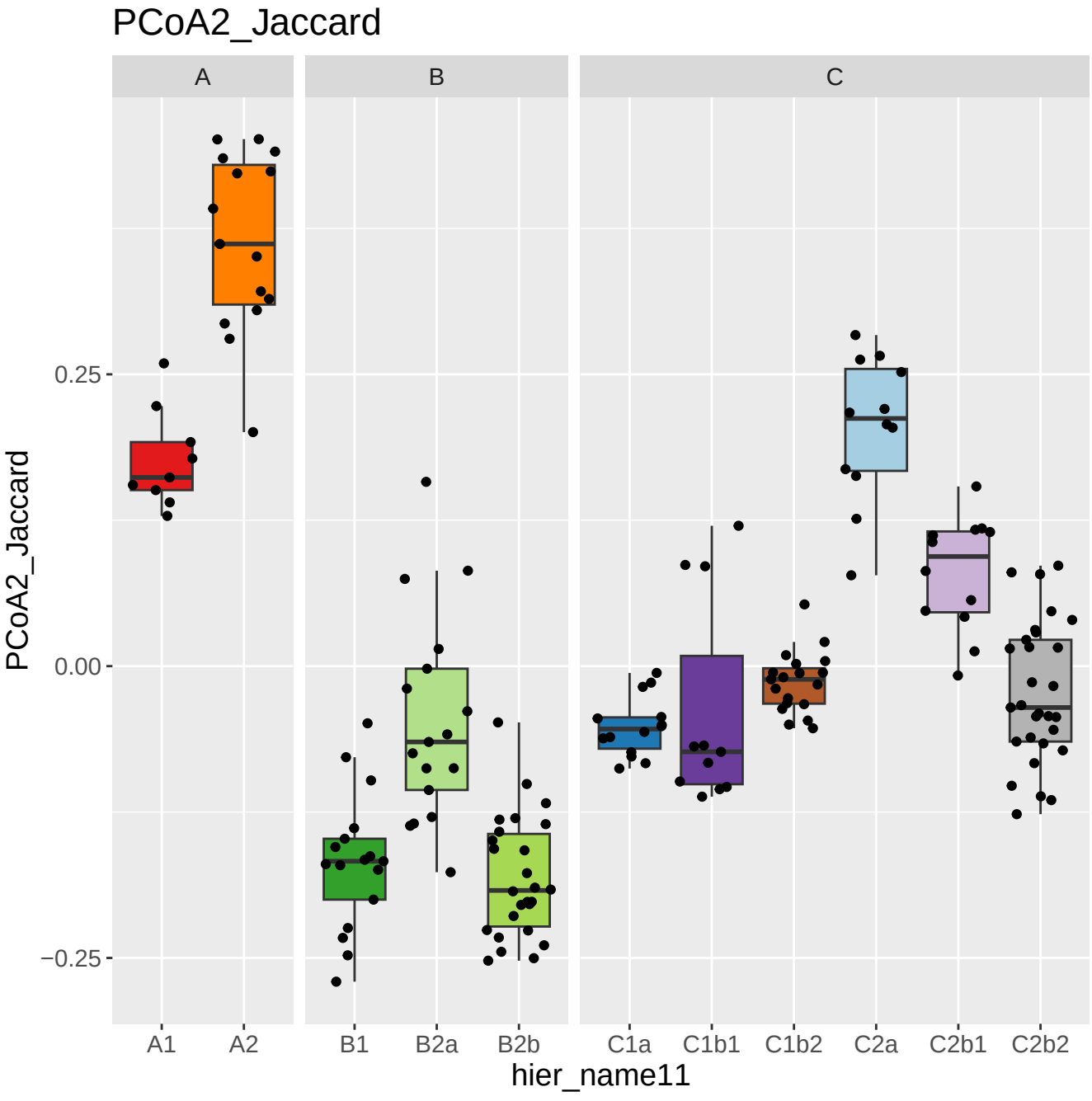

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
PCoA1_Bray
−0.4
−0.2
0.0
0.2

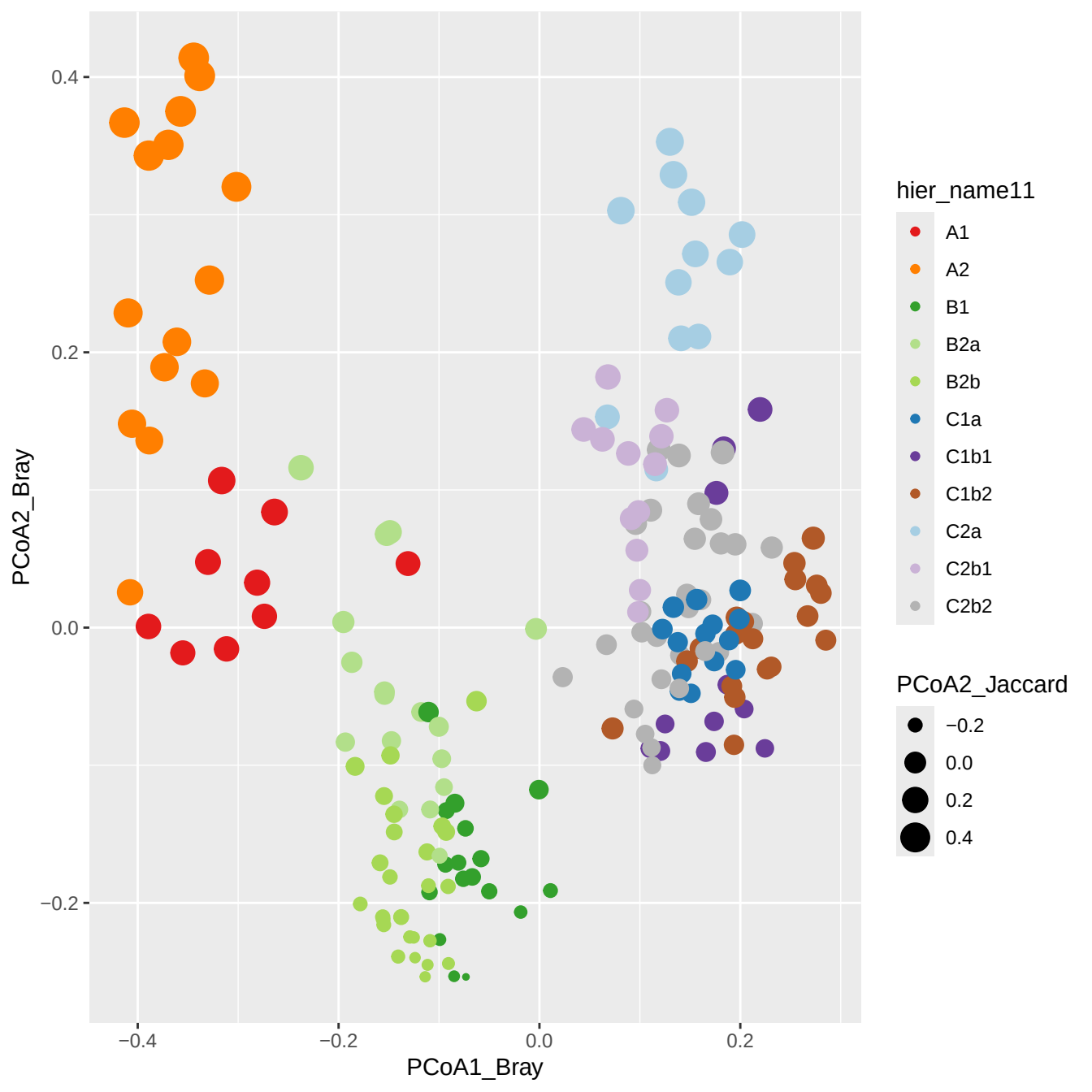

PCoA2_Bray
A
B
C
0.4
0.2
0.0
−0.2
PCoA2_Bray
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

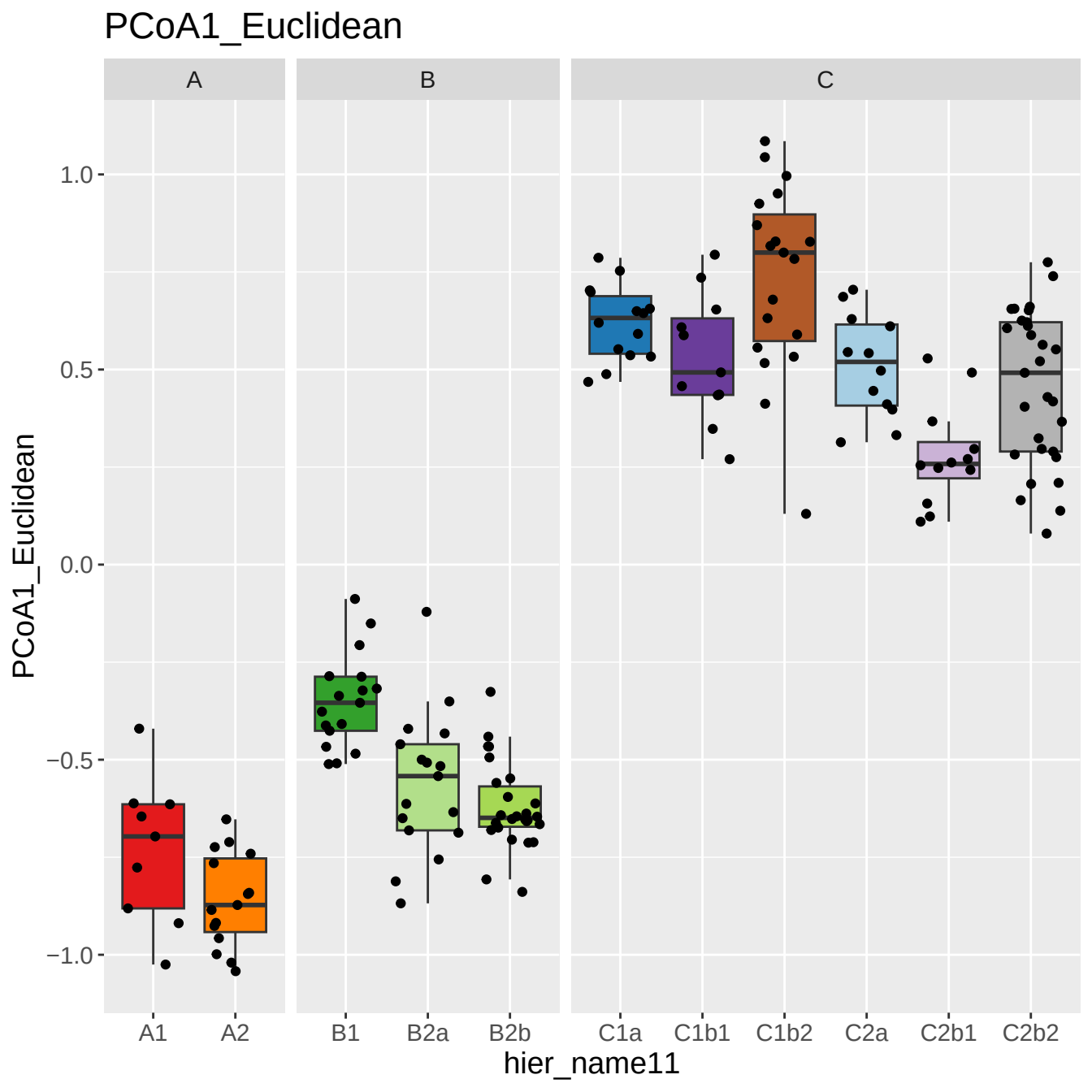

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
PCoA2_Bray
−0.2
0.0
0.2
0.4

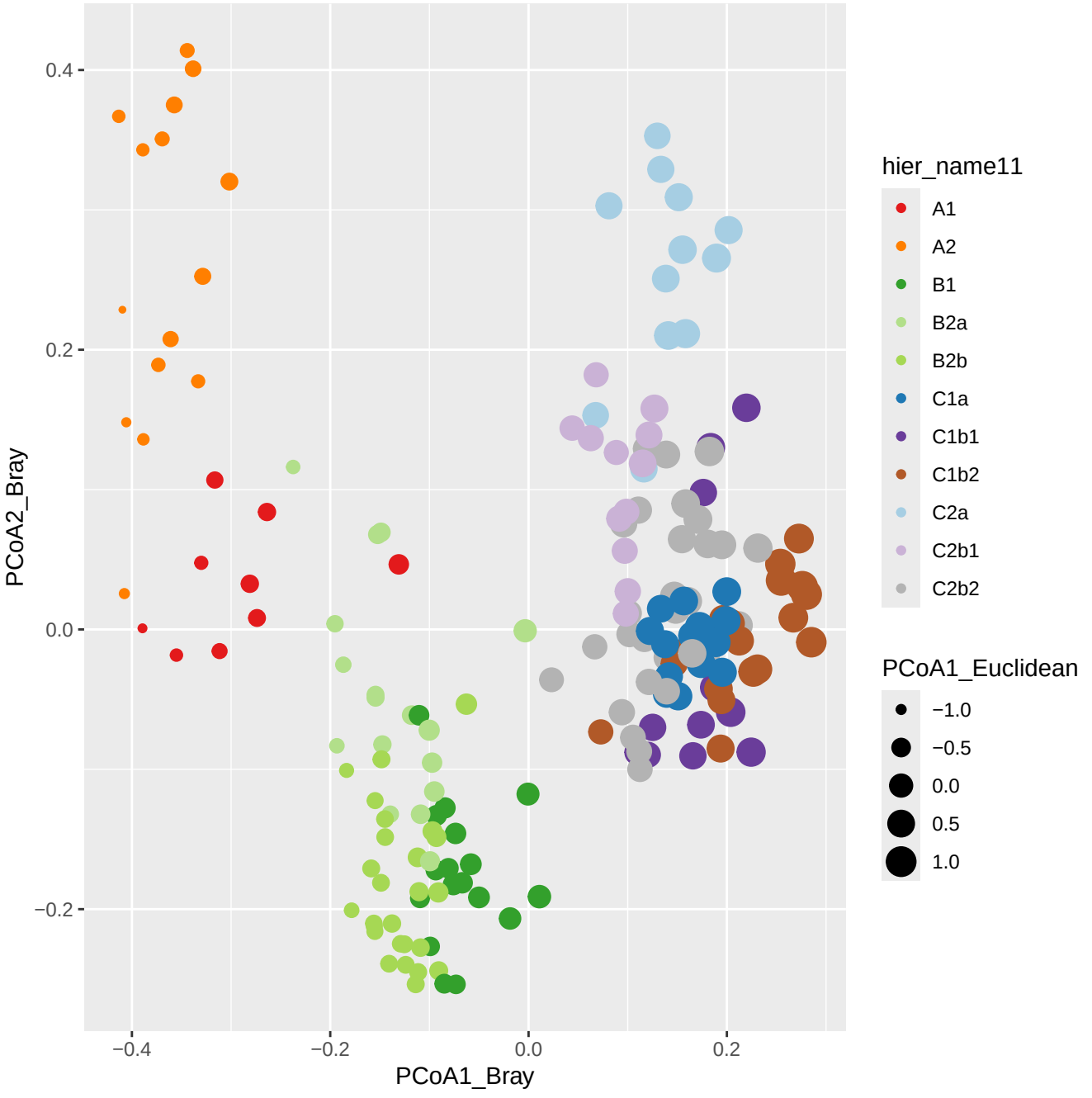

PCoA1_Jaccard
A
B
C
0.2
0.0
−0.2
PCoA1_Jaccard
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

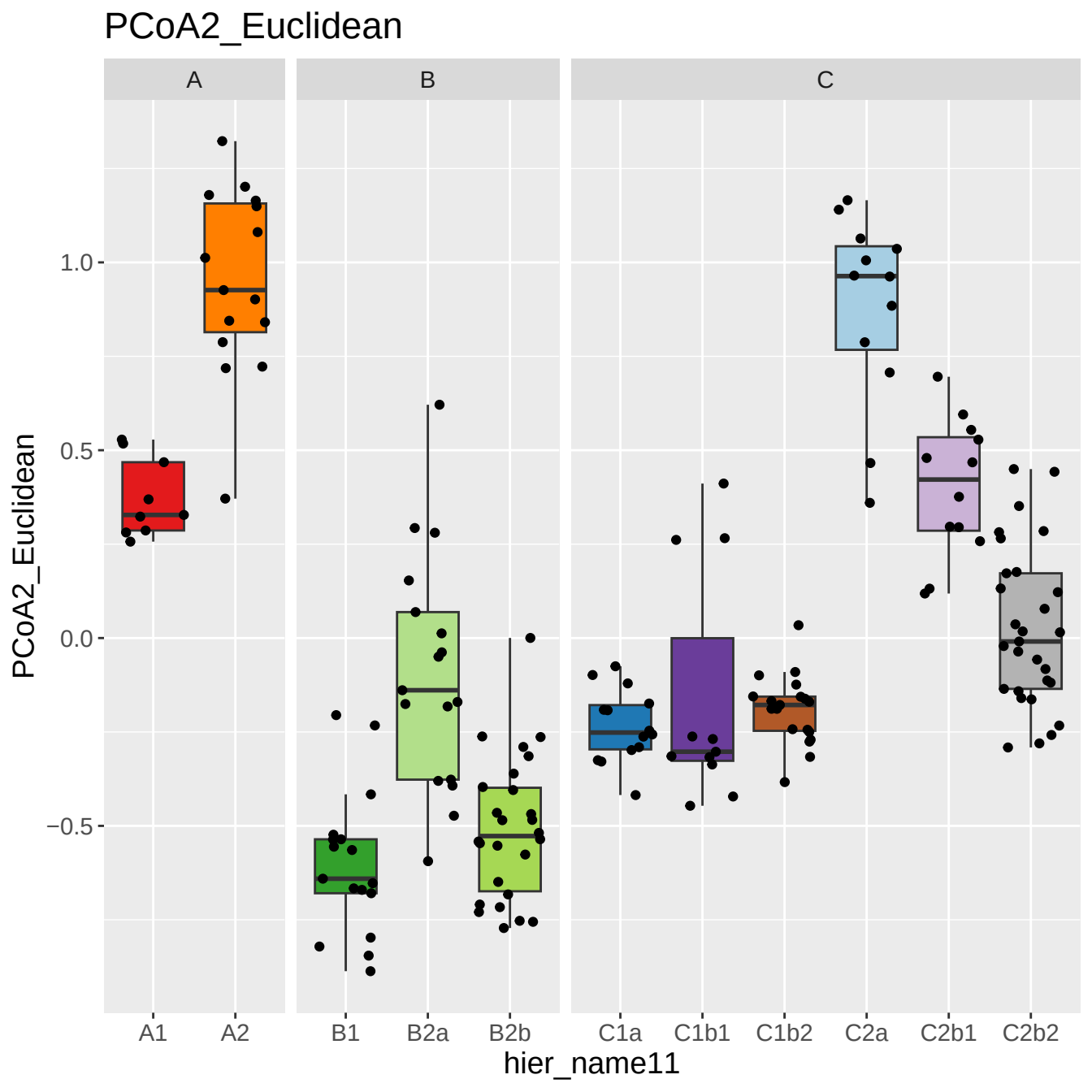

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
PCoA1_Jaccard
−0.3
−0.2
−0.1
0.0
0.1
0.2

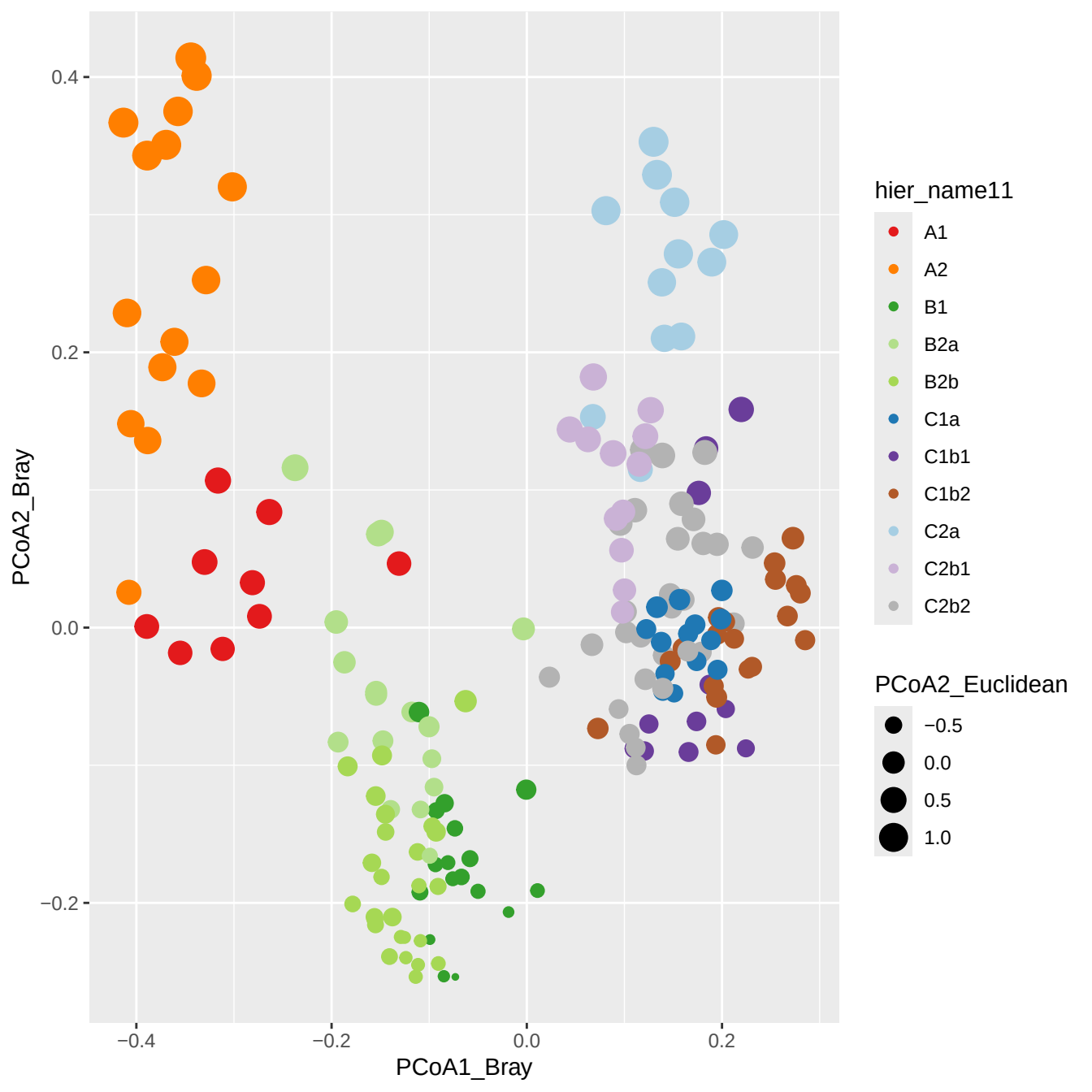

PCoA2_Jaccard
A
B
C
0.25
0.00
−0.25
PCoA2_Jaccard
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

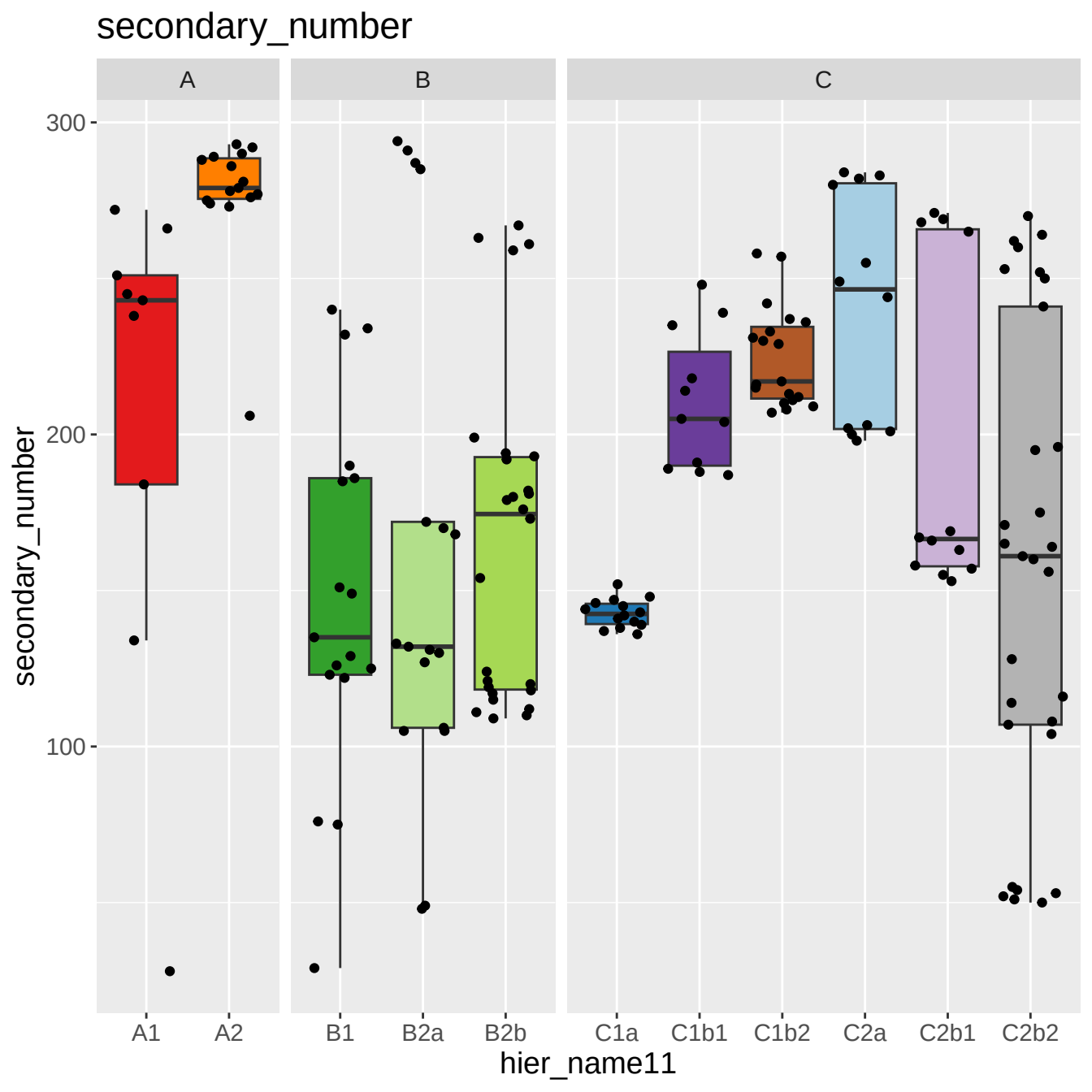

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
PCoA2_Jaccard
−0.2
0.0
0.2
0.4

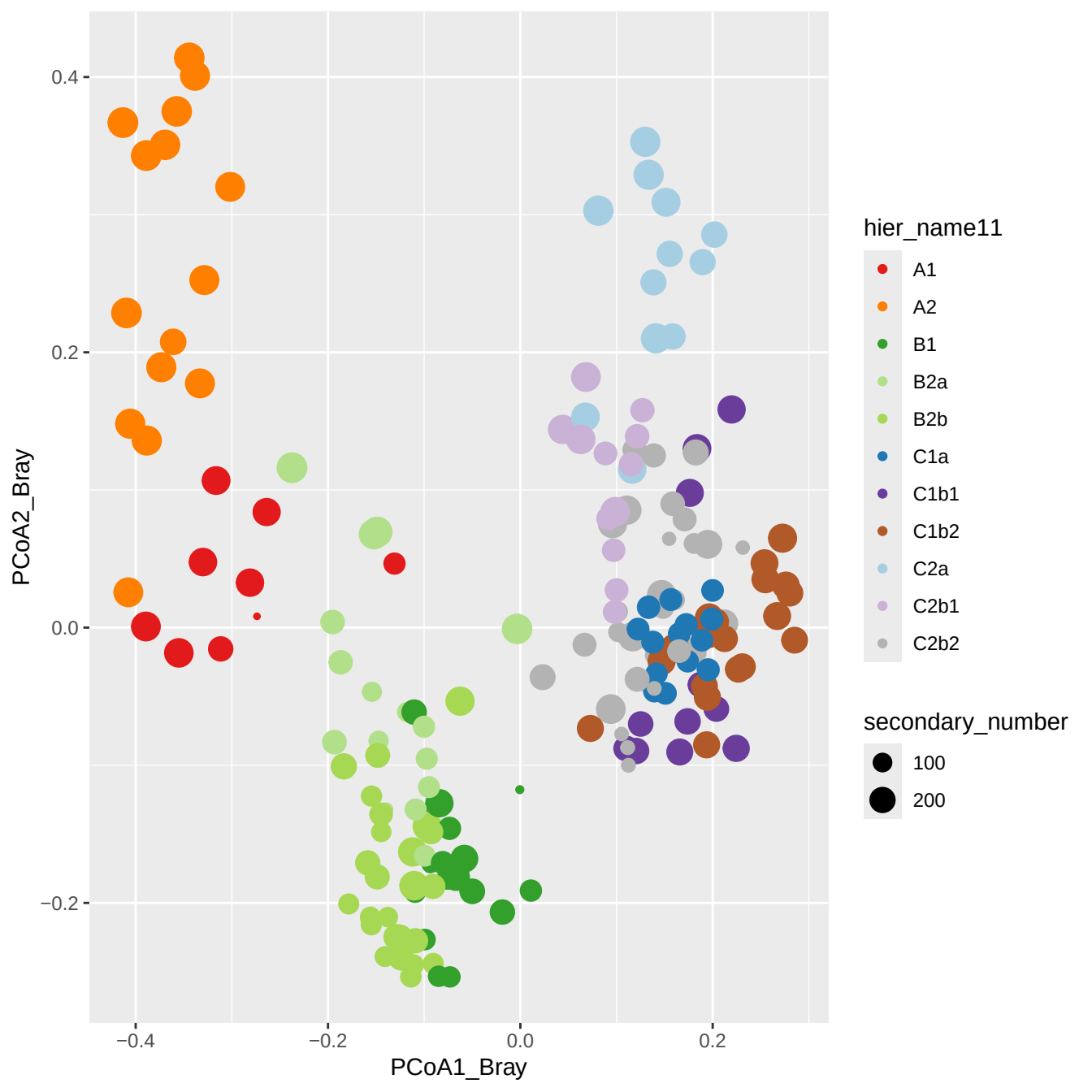

PCoA1_Euclidean
A
B
C
1.0
0.5
0.0
−0.5
−1.0
PCoA1_Euclidean
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

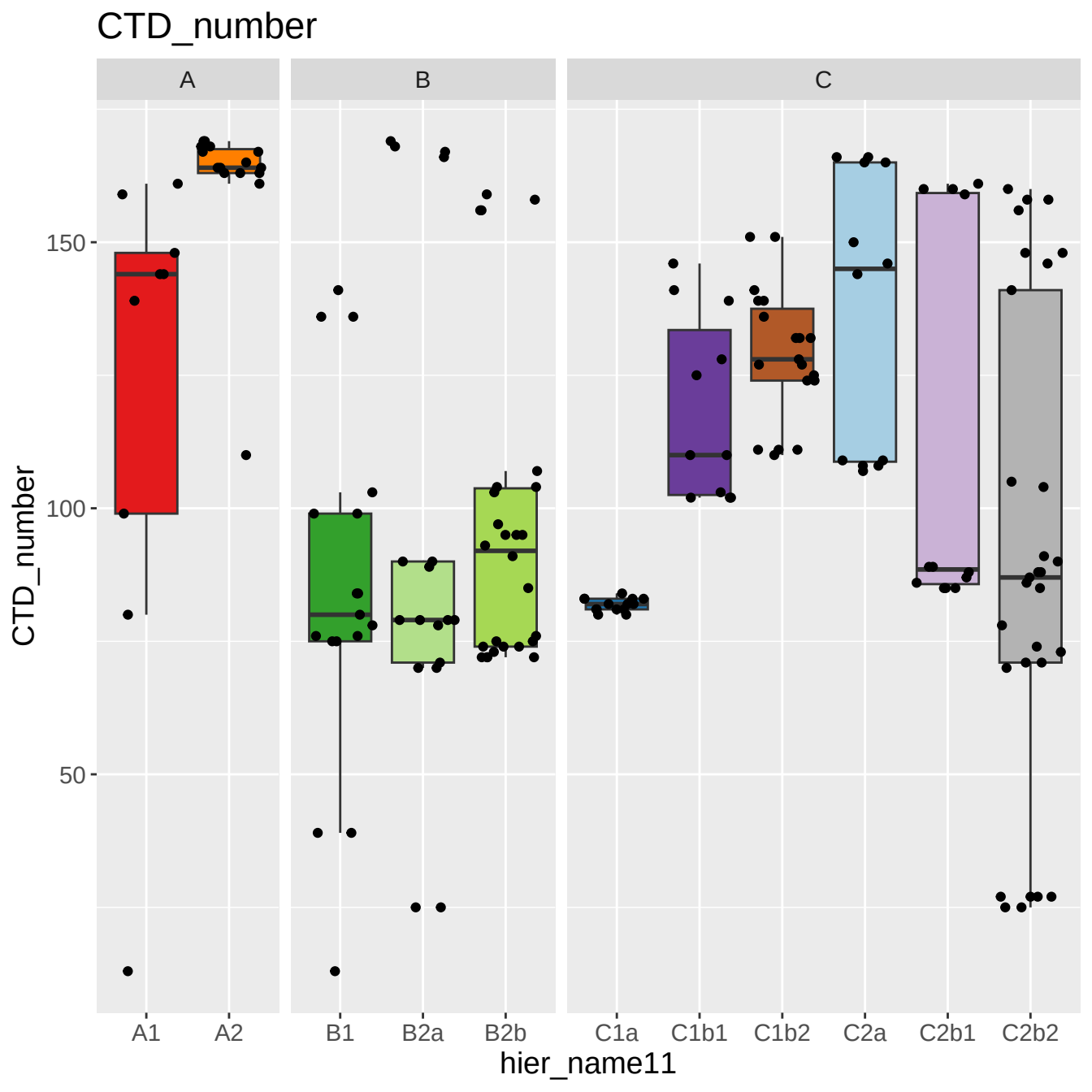

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
PCoA1_Euclidean
−1.0
−0.5
0.0
0.5
1.0

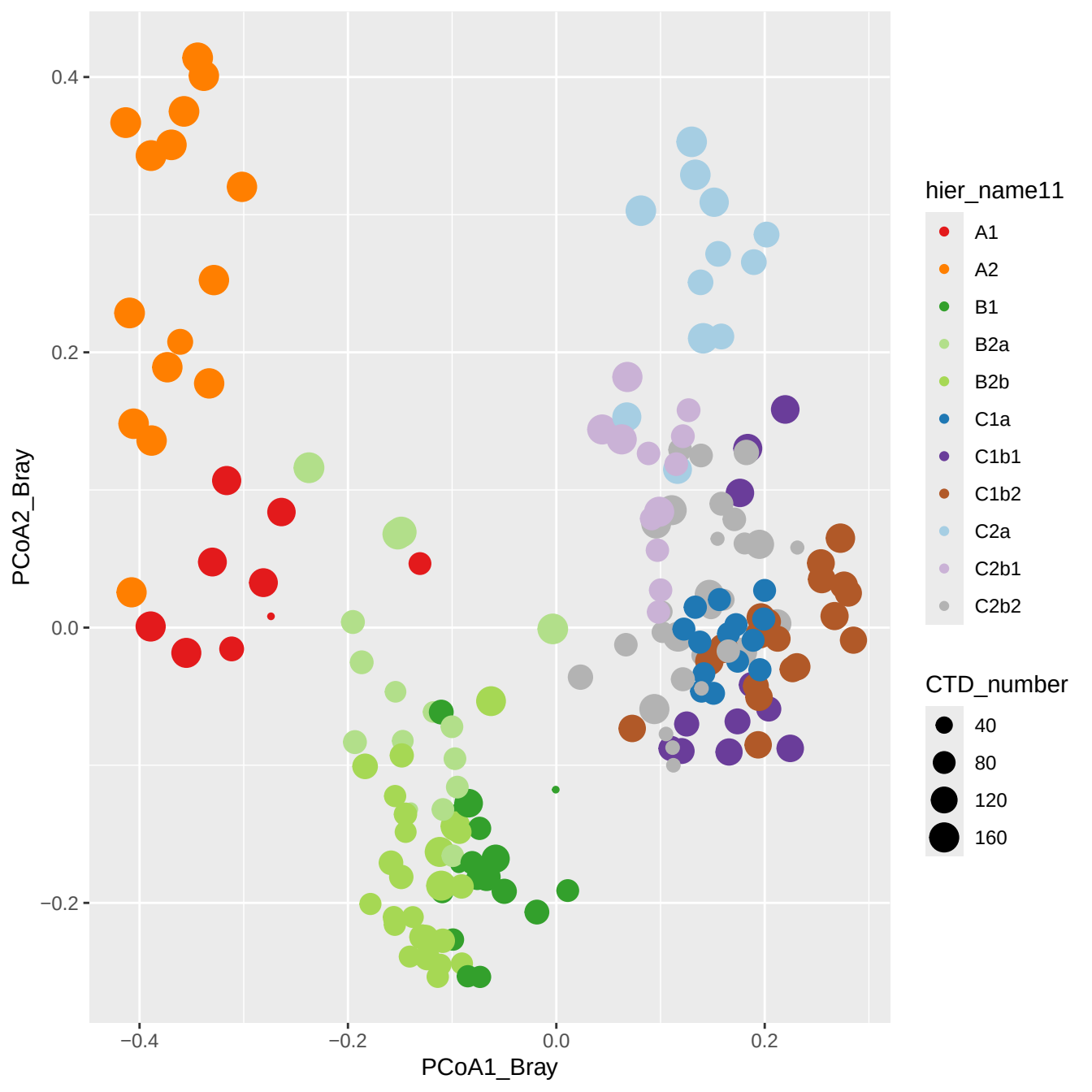

PCoA2_Euclidean
A
B
C
1.0
0.5
0.0
−0.5
PCoA2_Euclidean
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

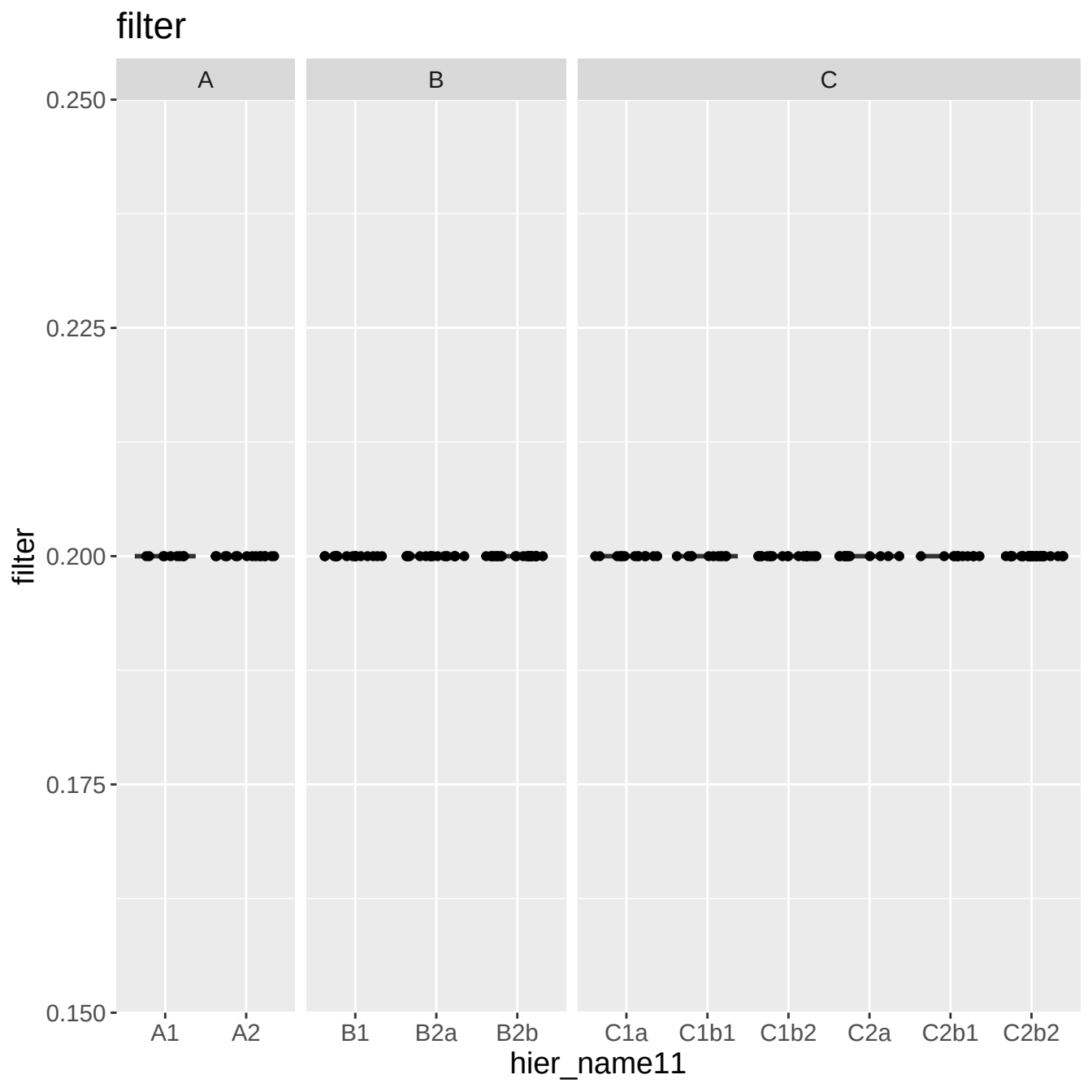

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
PCoA2_Euclidean
−0.5
0.0
0.5
1.0

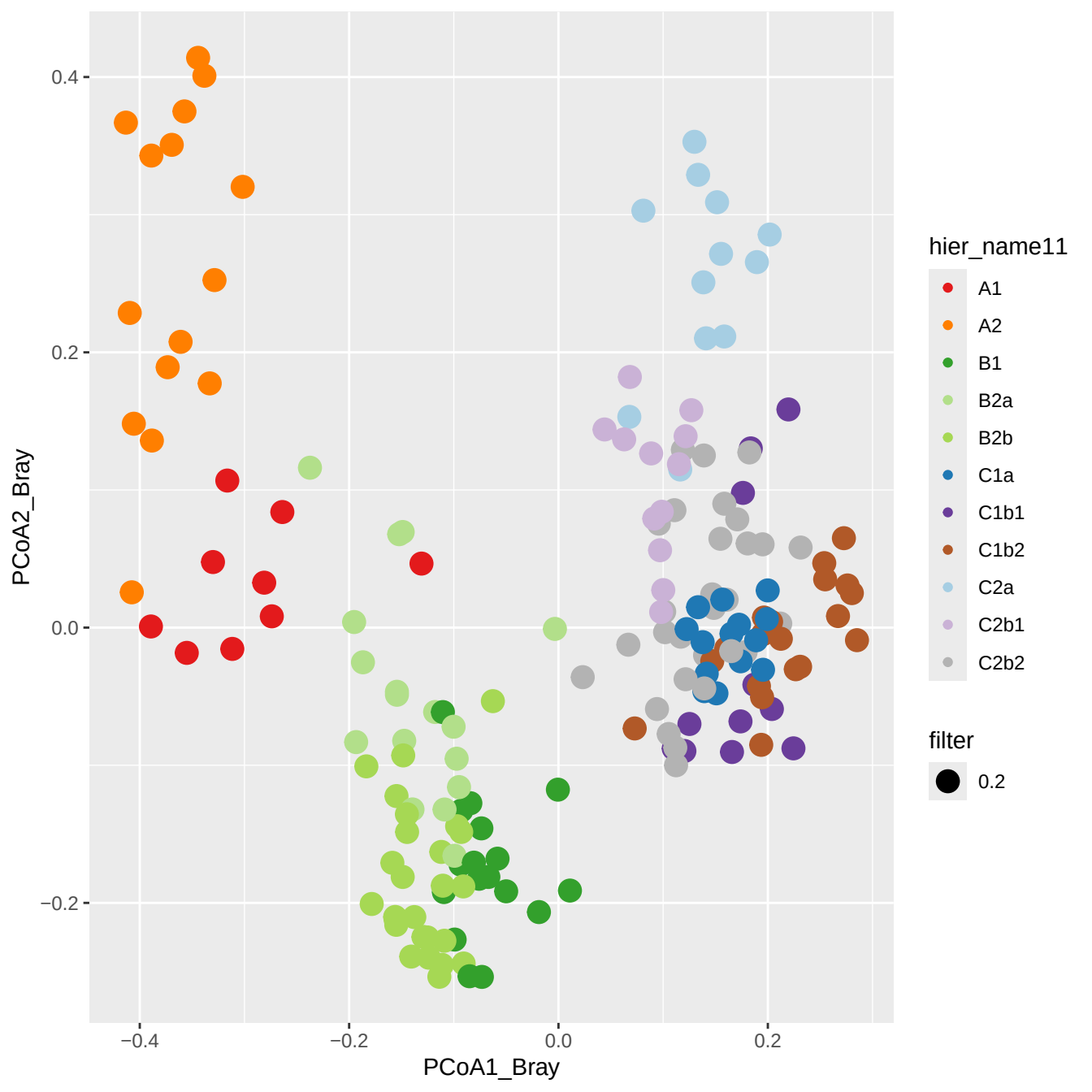

secondary_number
A
B
C
300
200
100
secondary_number
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

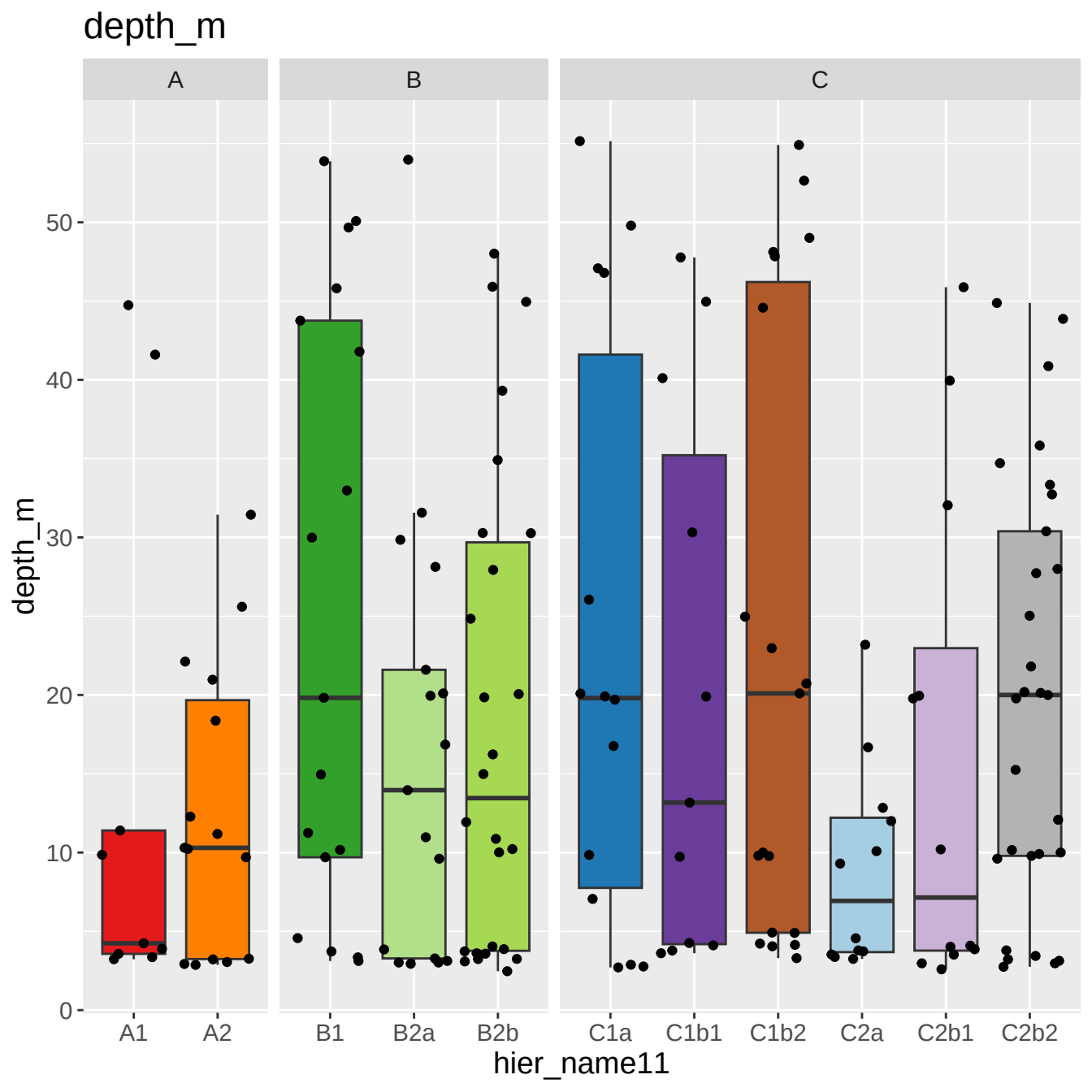

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
secondary_number
100
200

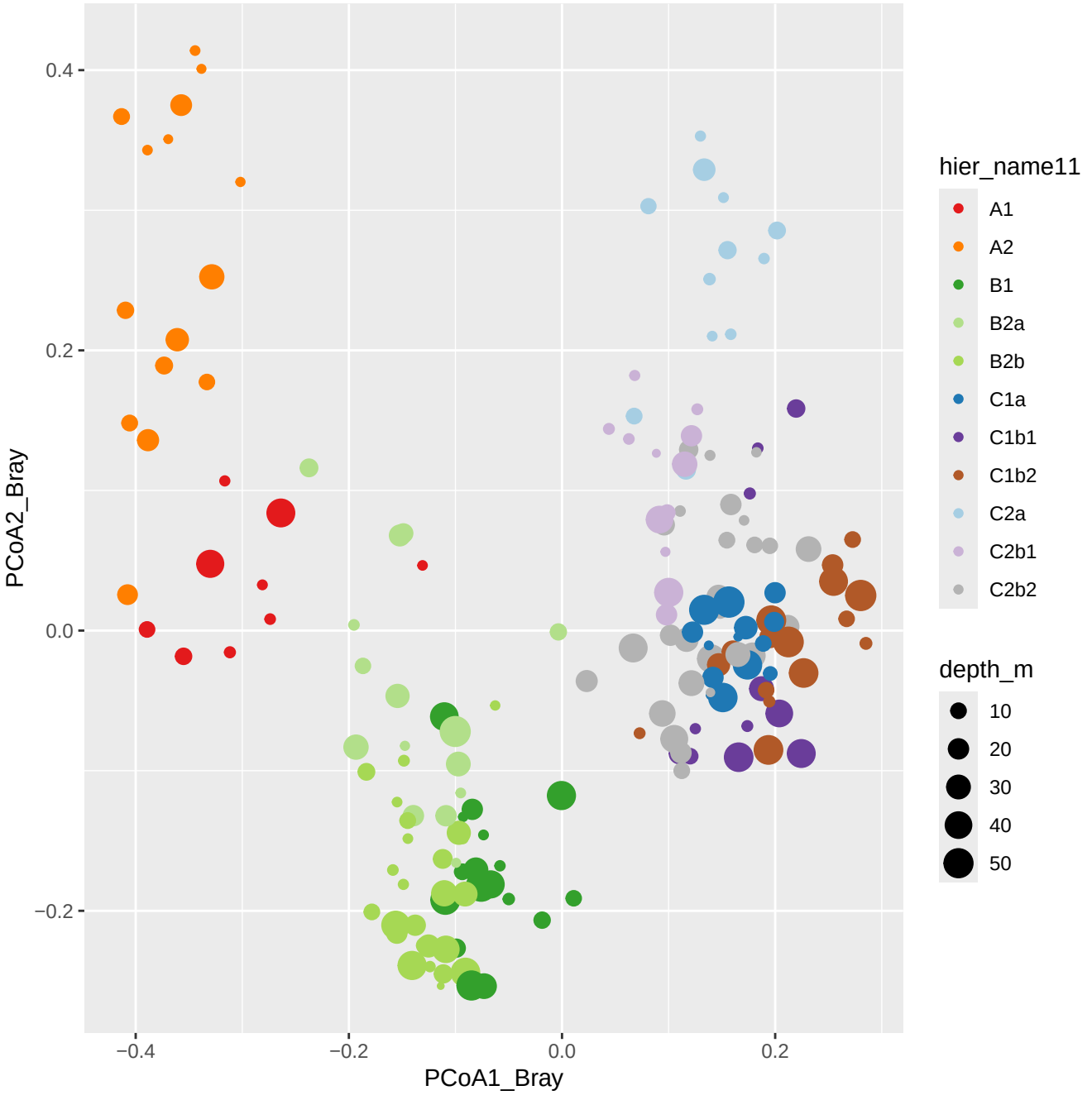

CTD_number
A
B
C
150
100
50
CTD_number
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

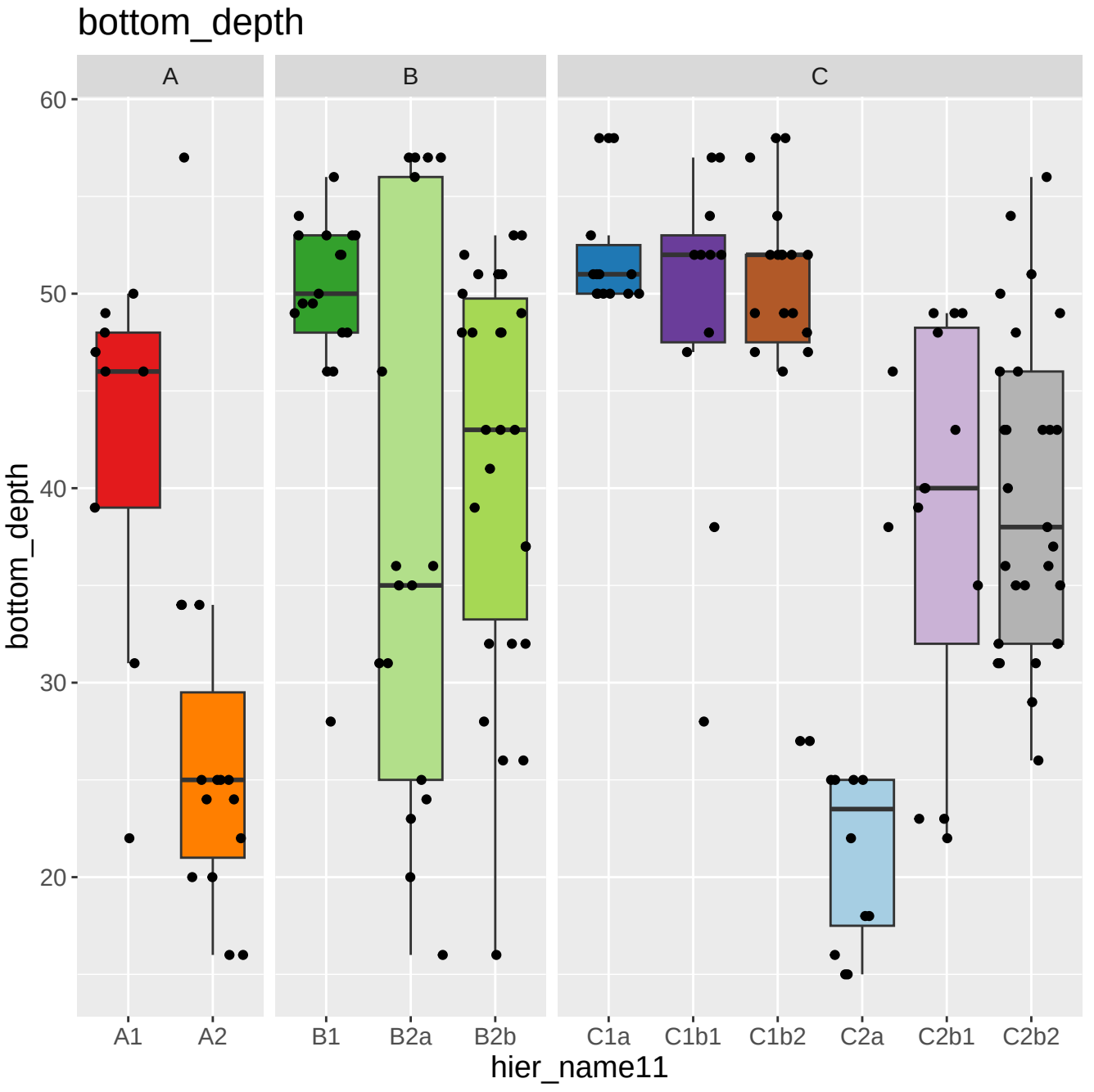

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
CTD_number
40
80
120
160

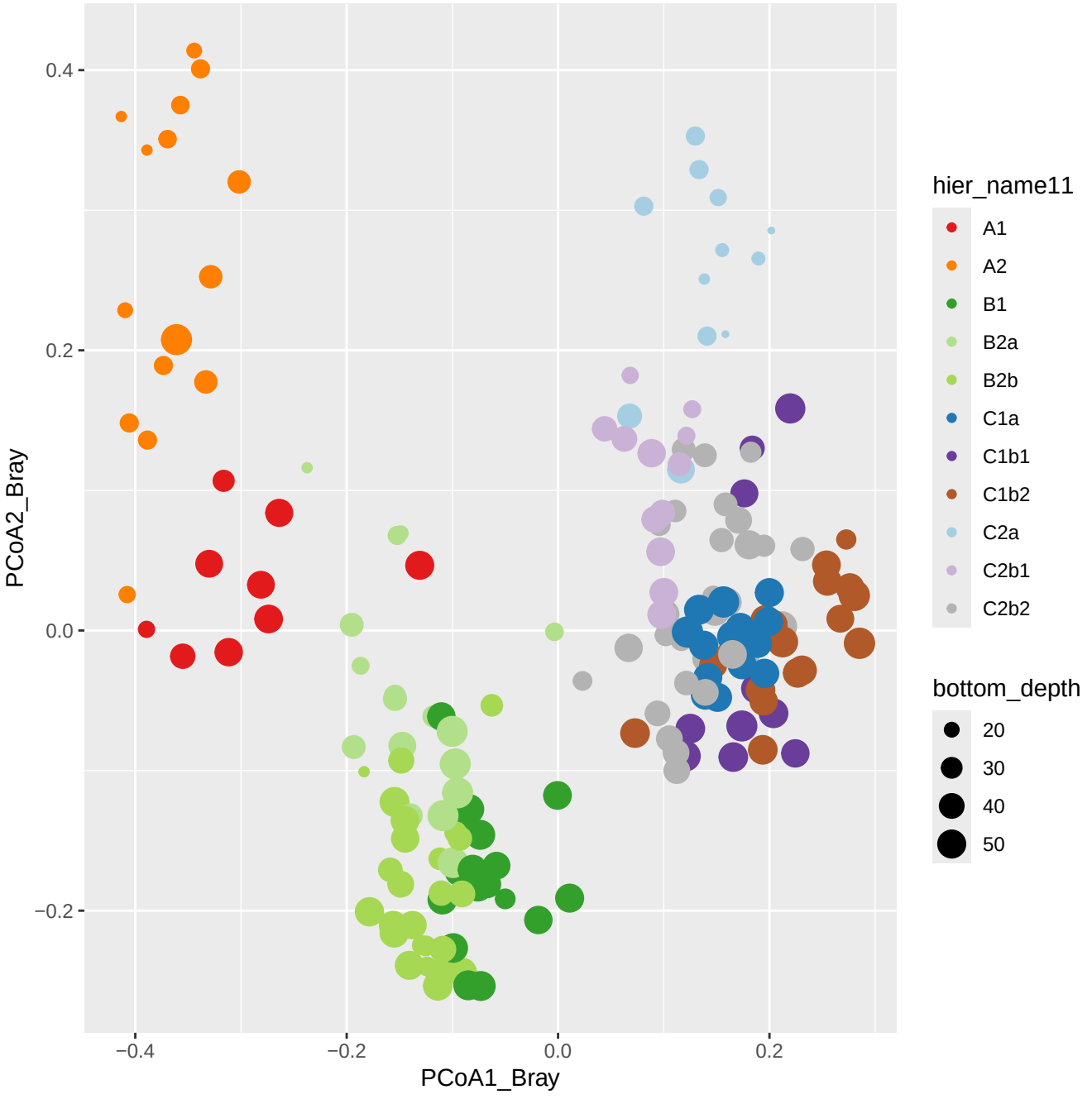

filter
A
B
C
0.250
0.225
0.200
0.175
0.150
filter
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

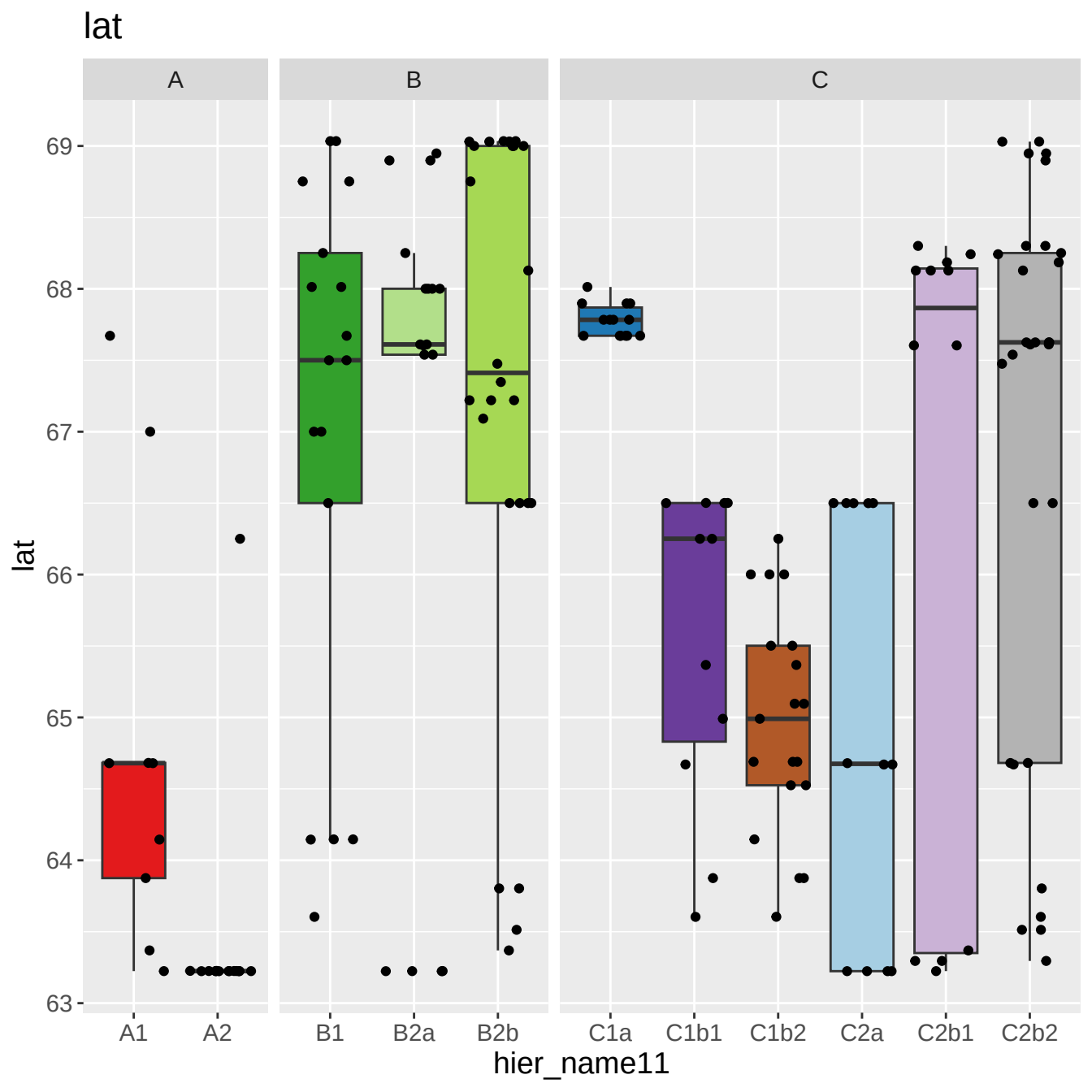

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
filter
0.2

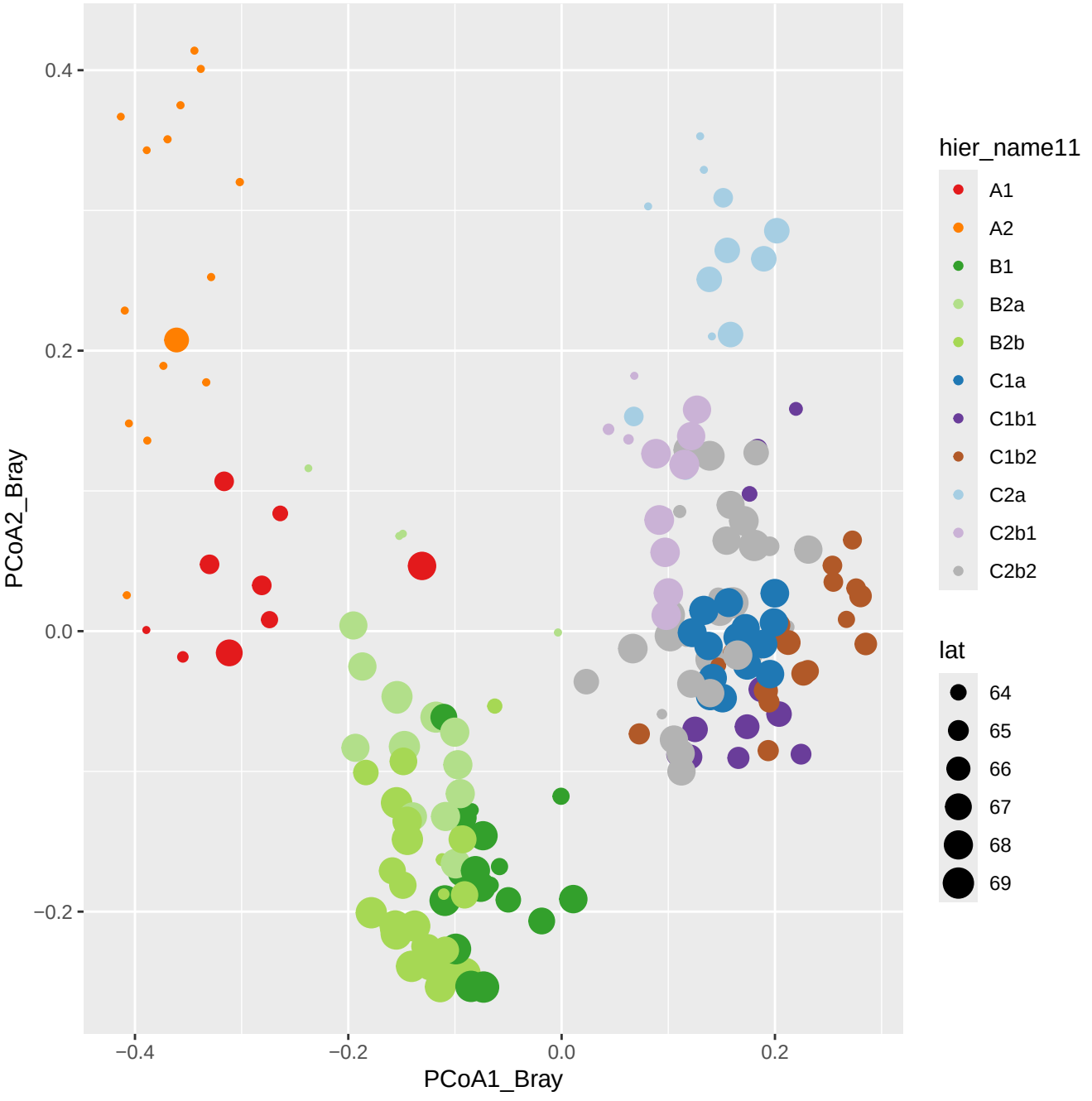

depth_m
A
B
C
50
40
30
20
10
0
depth_m
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

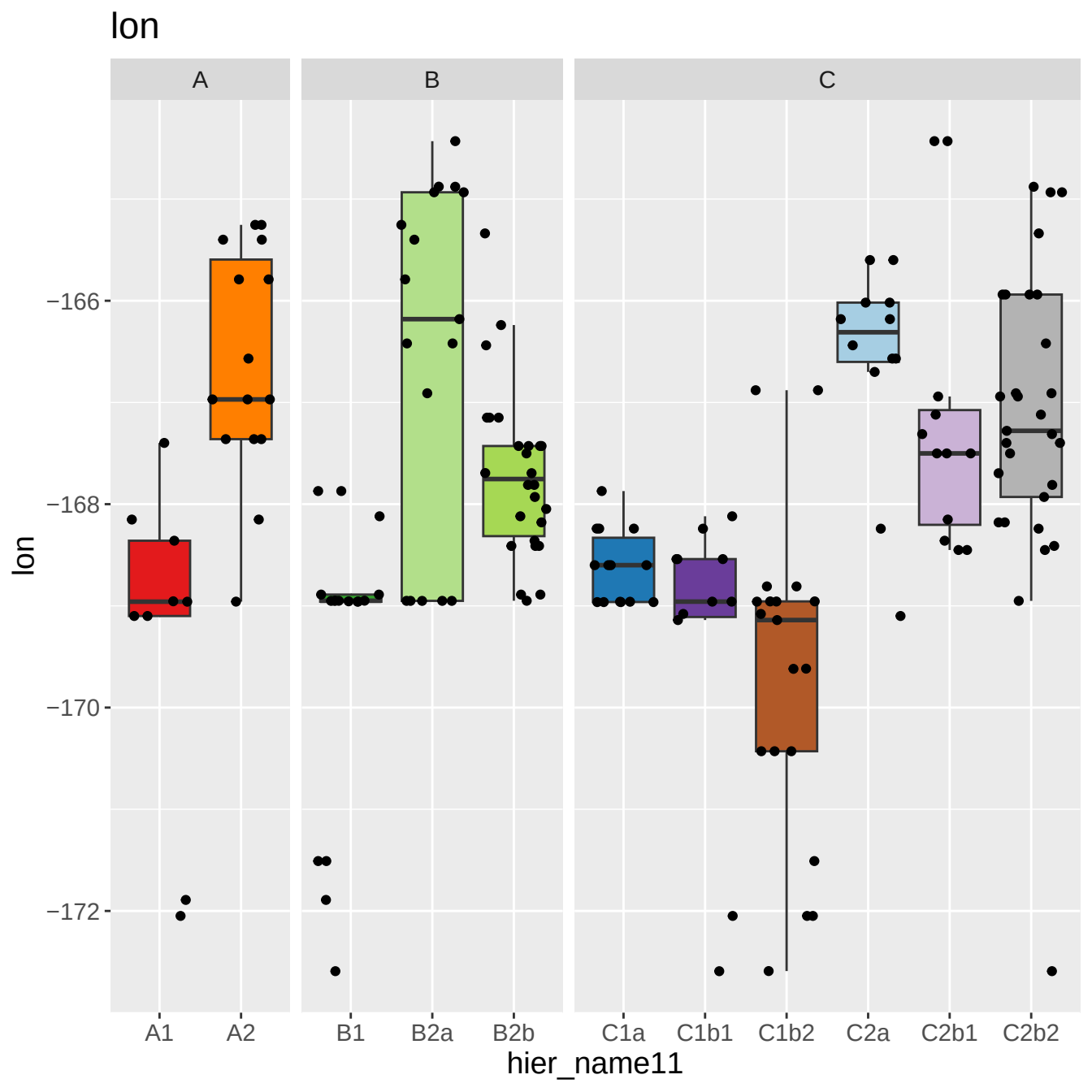

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
depth_m
10
20
30
40
50

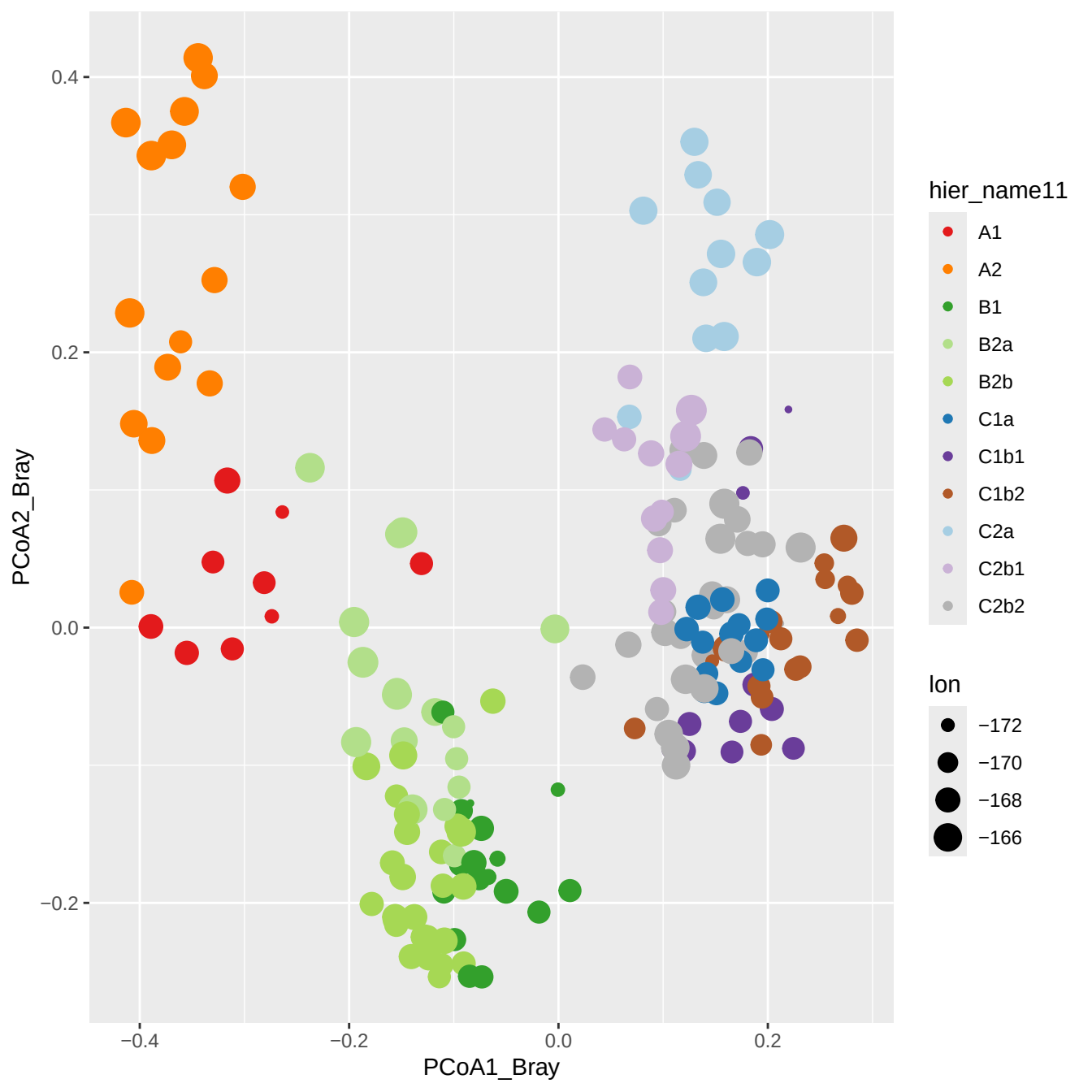

bottom_depth
A
B
C
60
50
40
30
20
bottom_depth
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

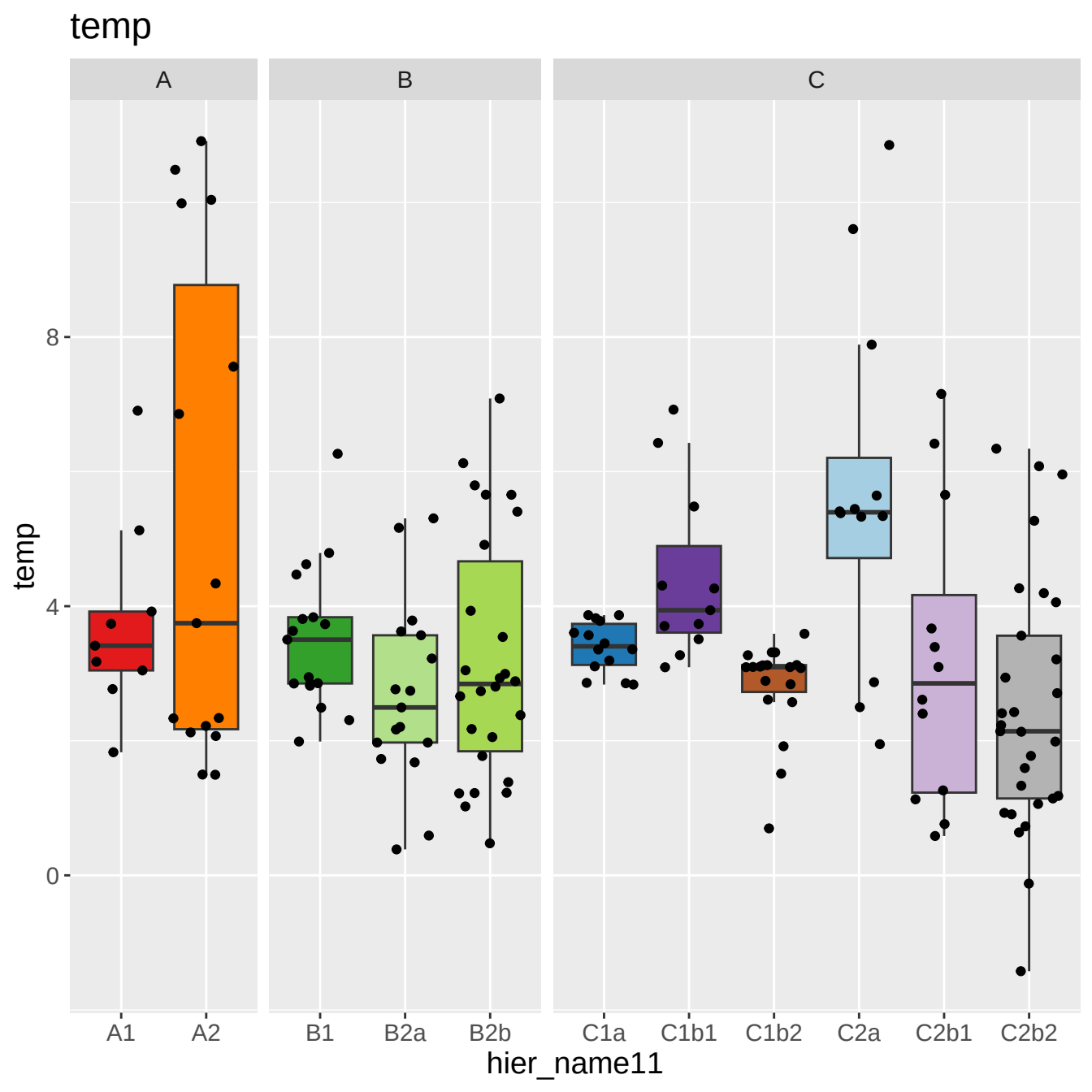

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
bottom_depth
20
30
40
50

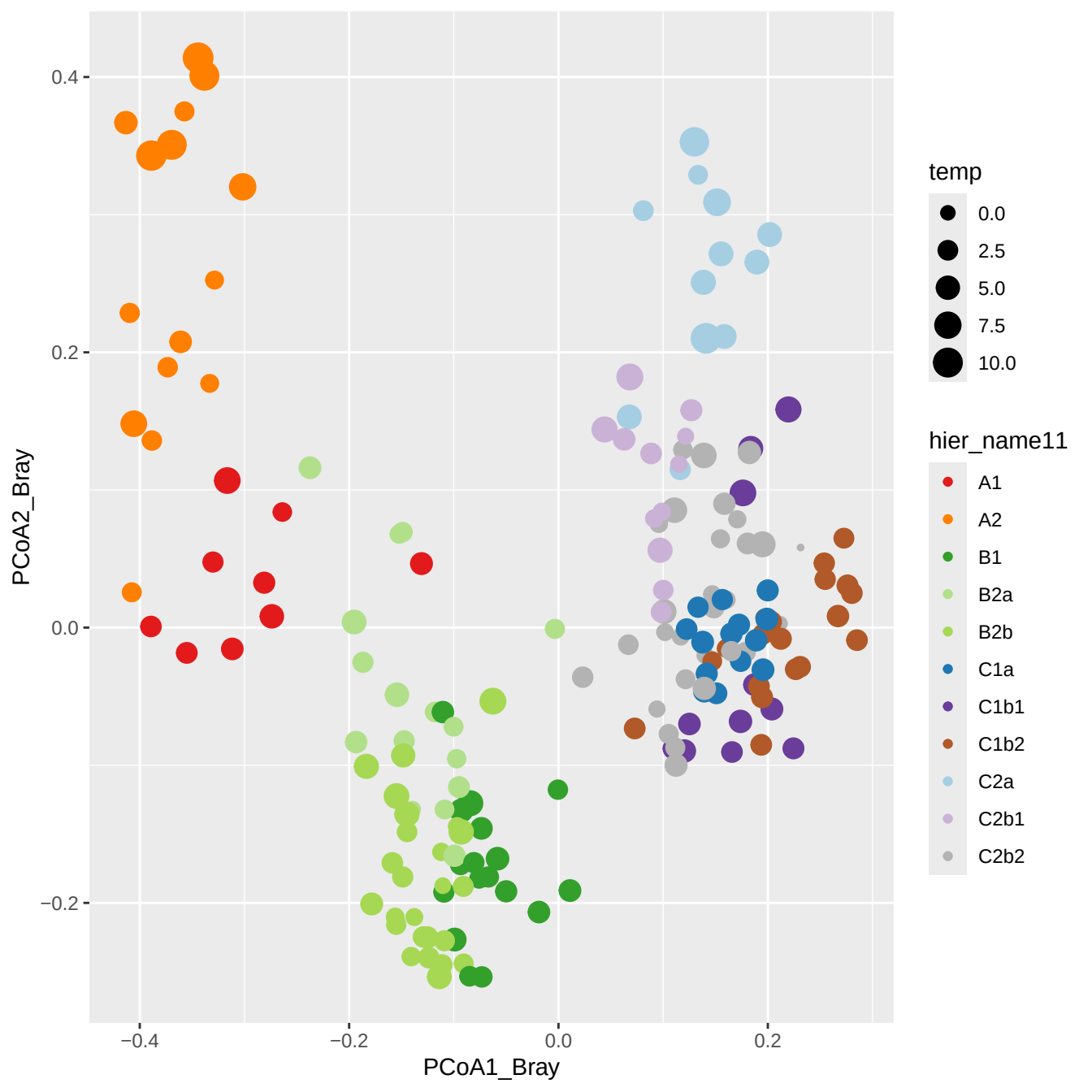

lat
A
B
C
69
68
67
66
65
64
63
lat
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
lat
64
65
66
67
68
69

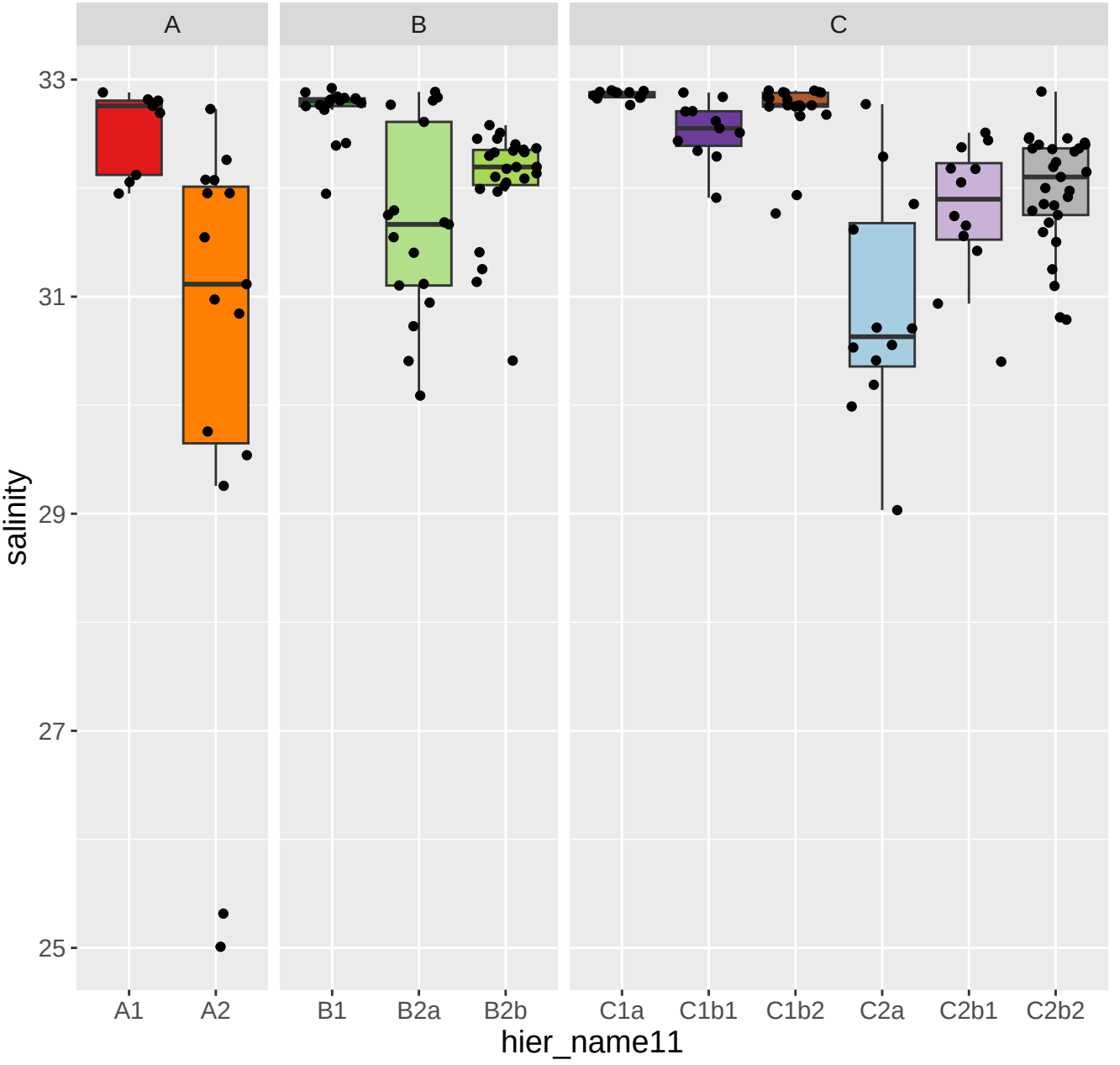

lon
A
B
C
−166
−168
−170
−172
lon
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

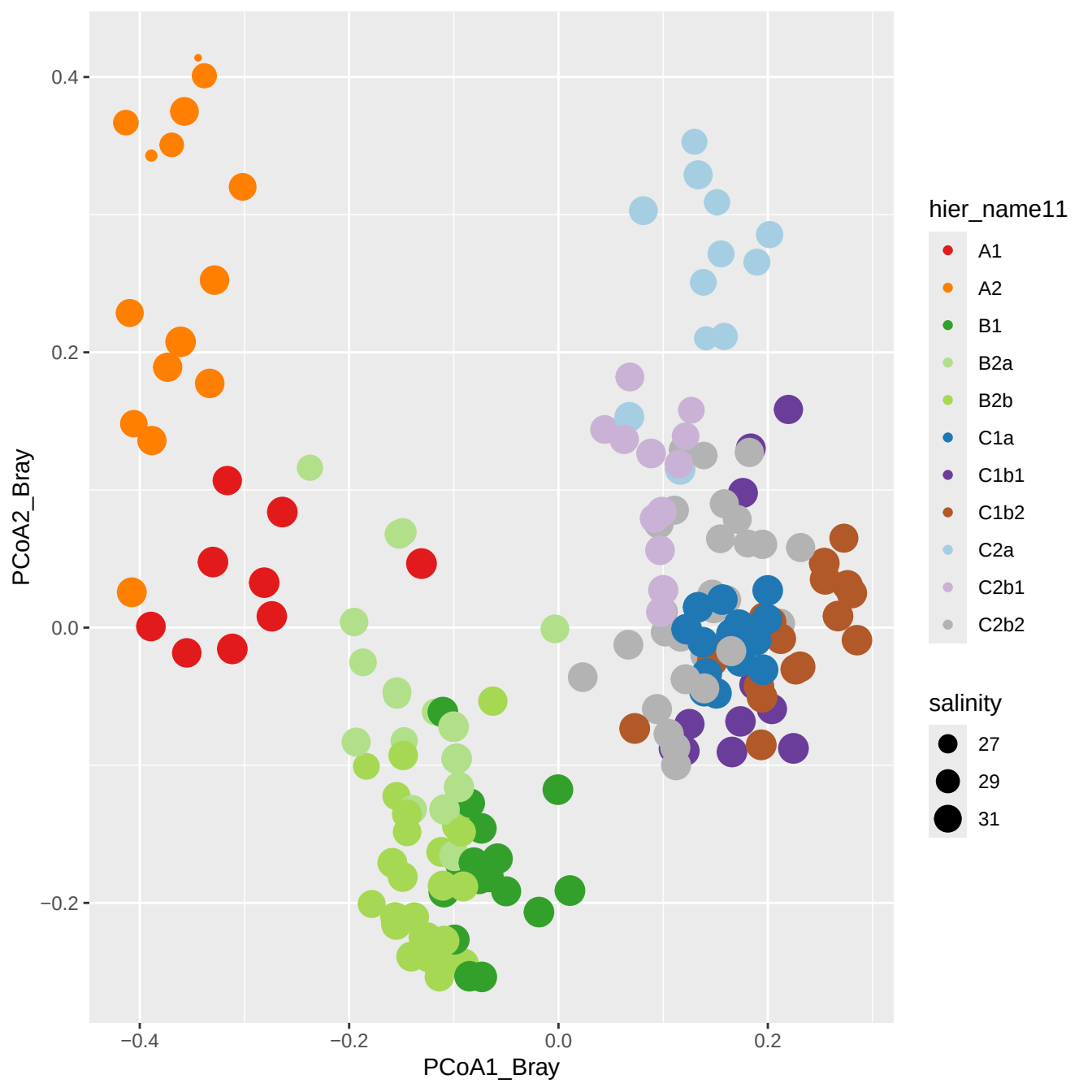

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
lon
−172
−170
−168
−166

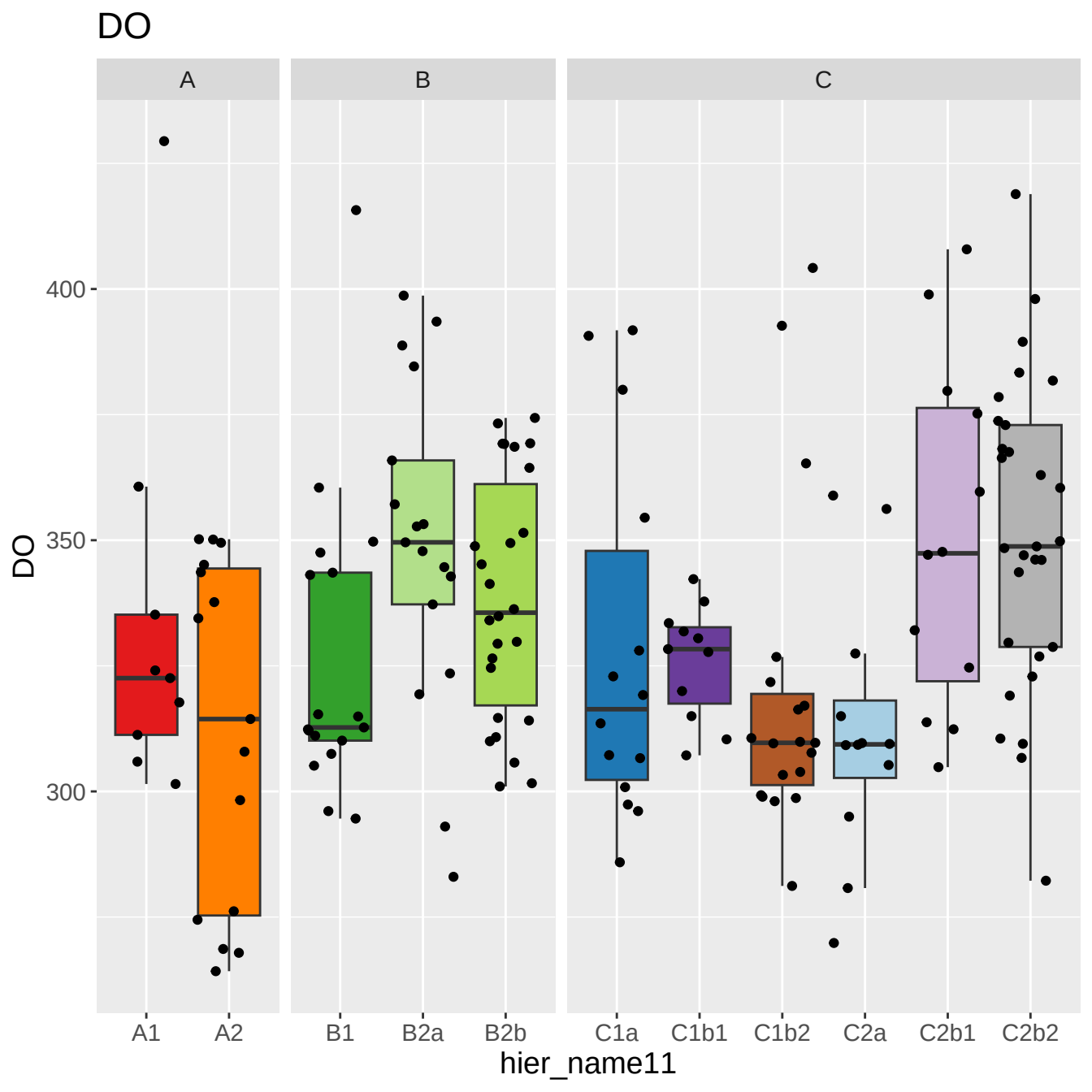

temp
A
B
C
8
4
0
temp
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

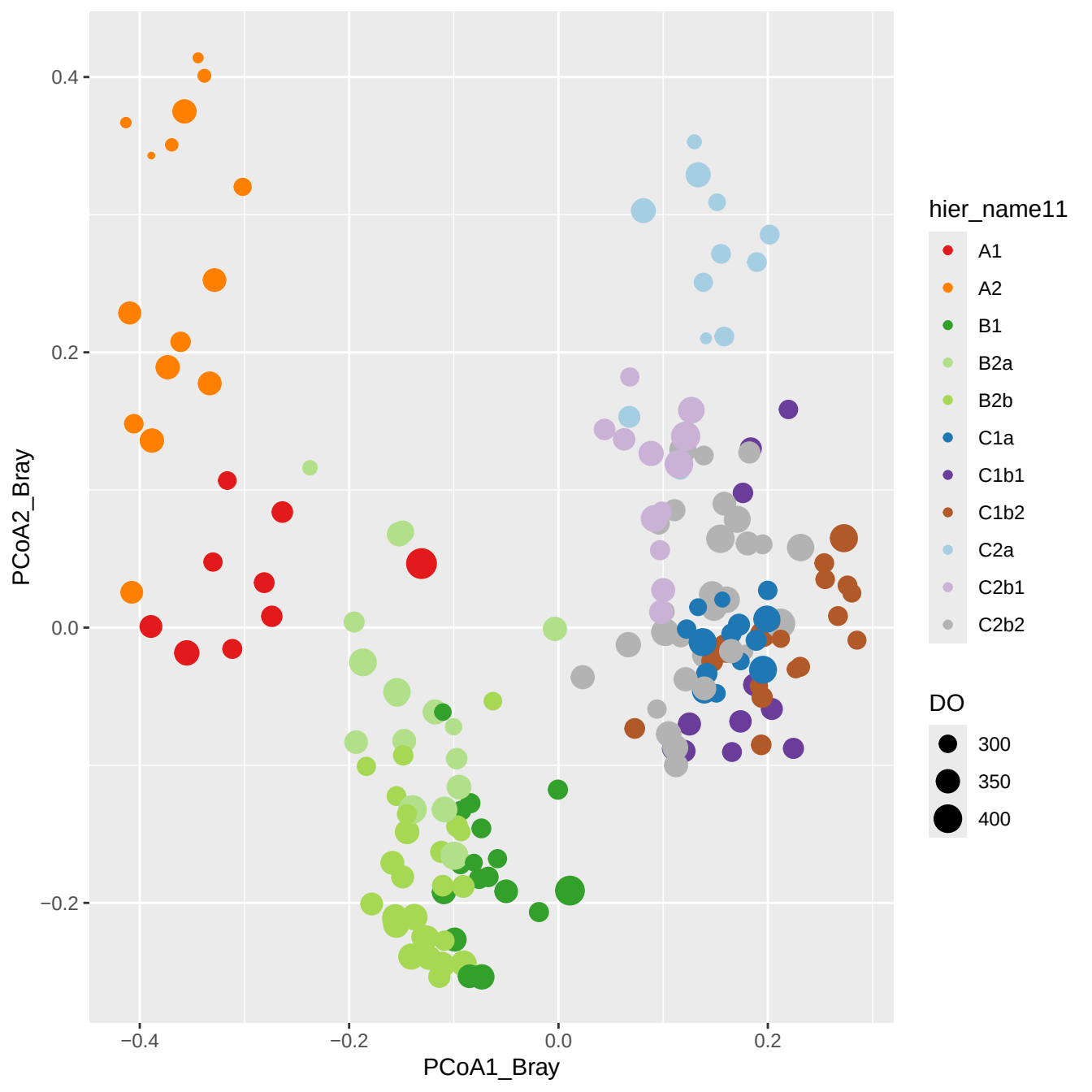

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
temp
0.0
2.5
5.0
7.5
10.0
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

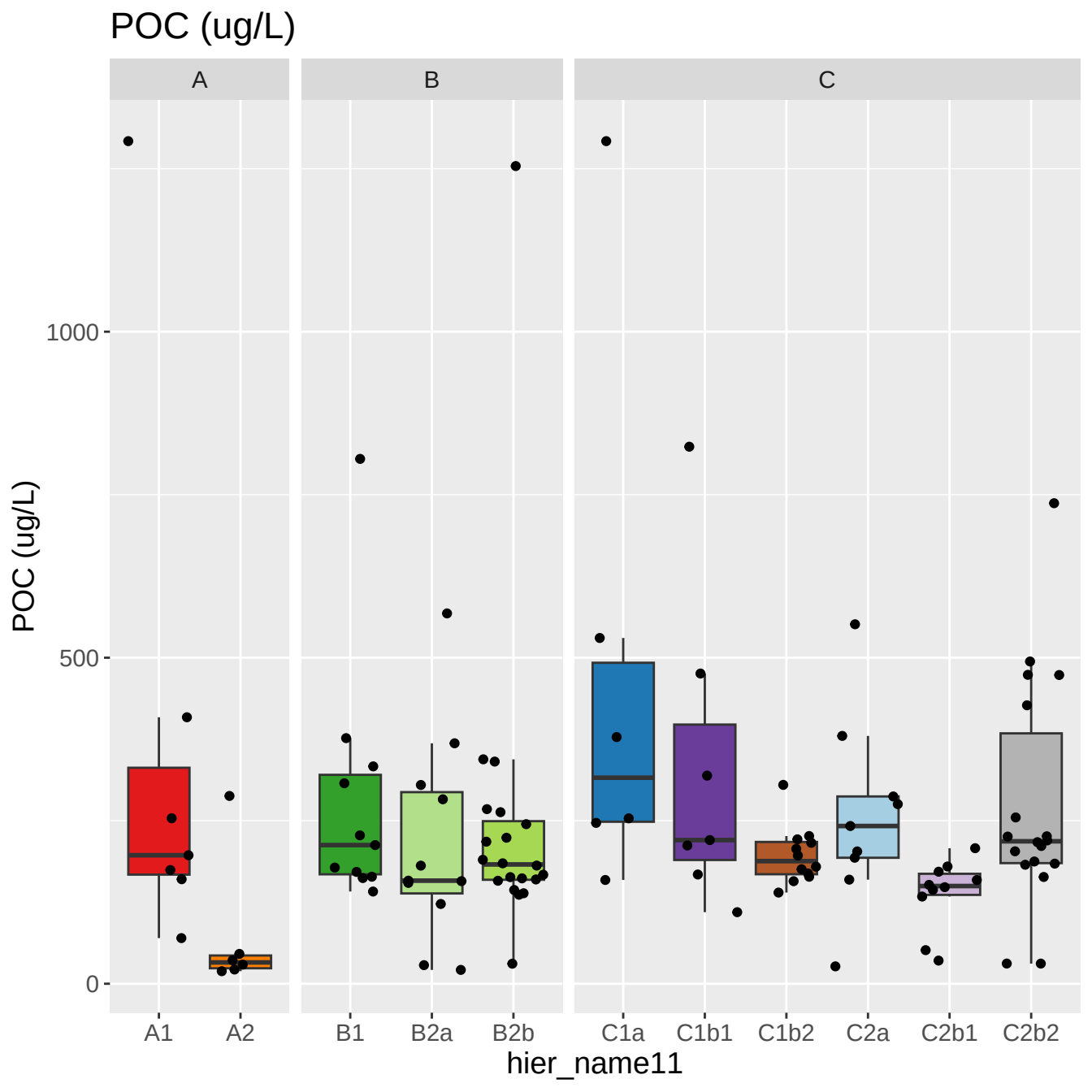
salinity

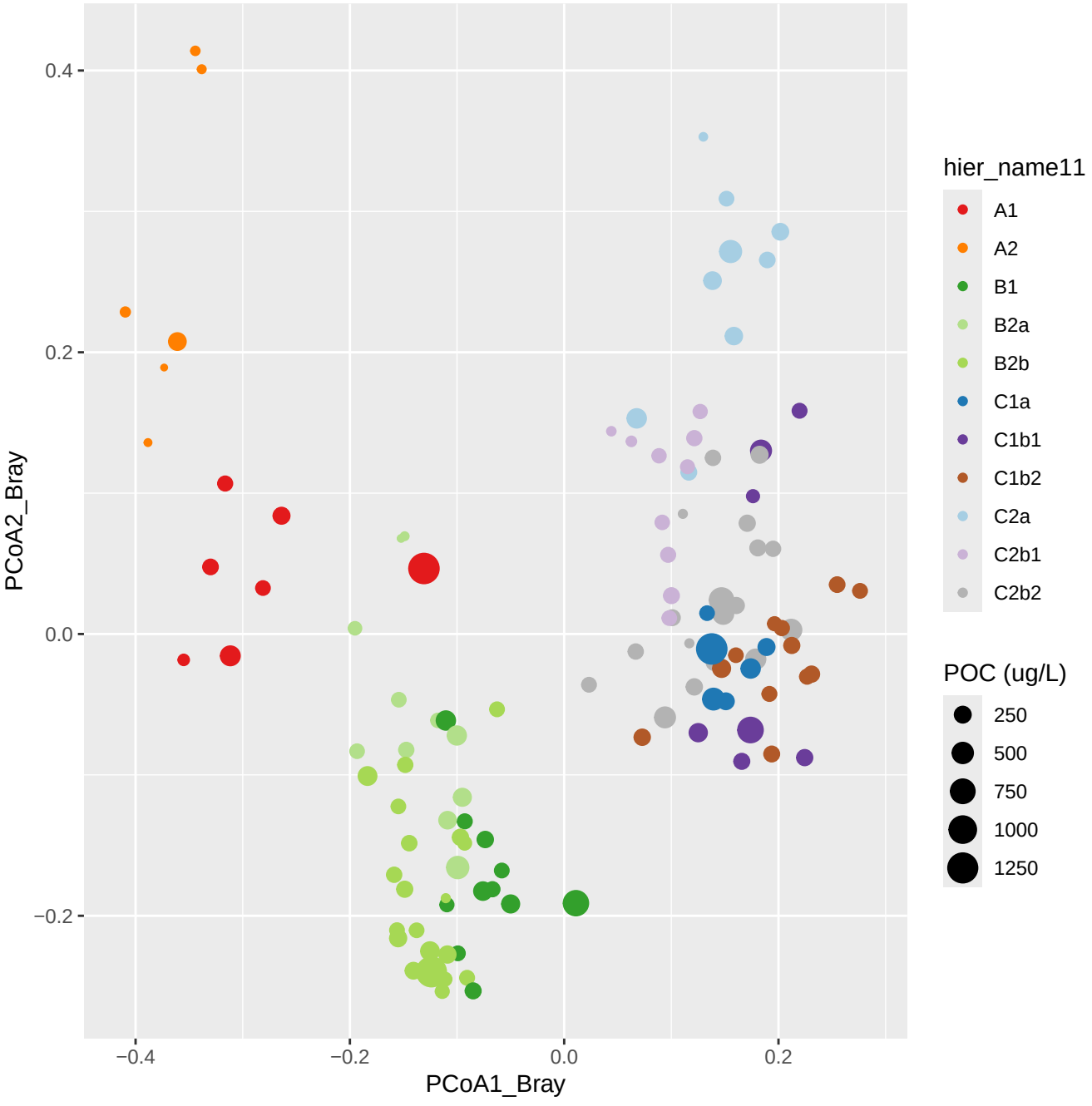

A
B
C
33
31
29
27
25
salinity
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

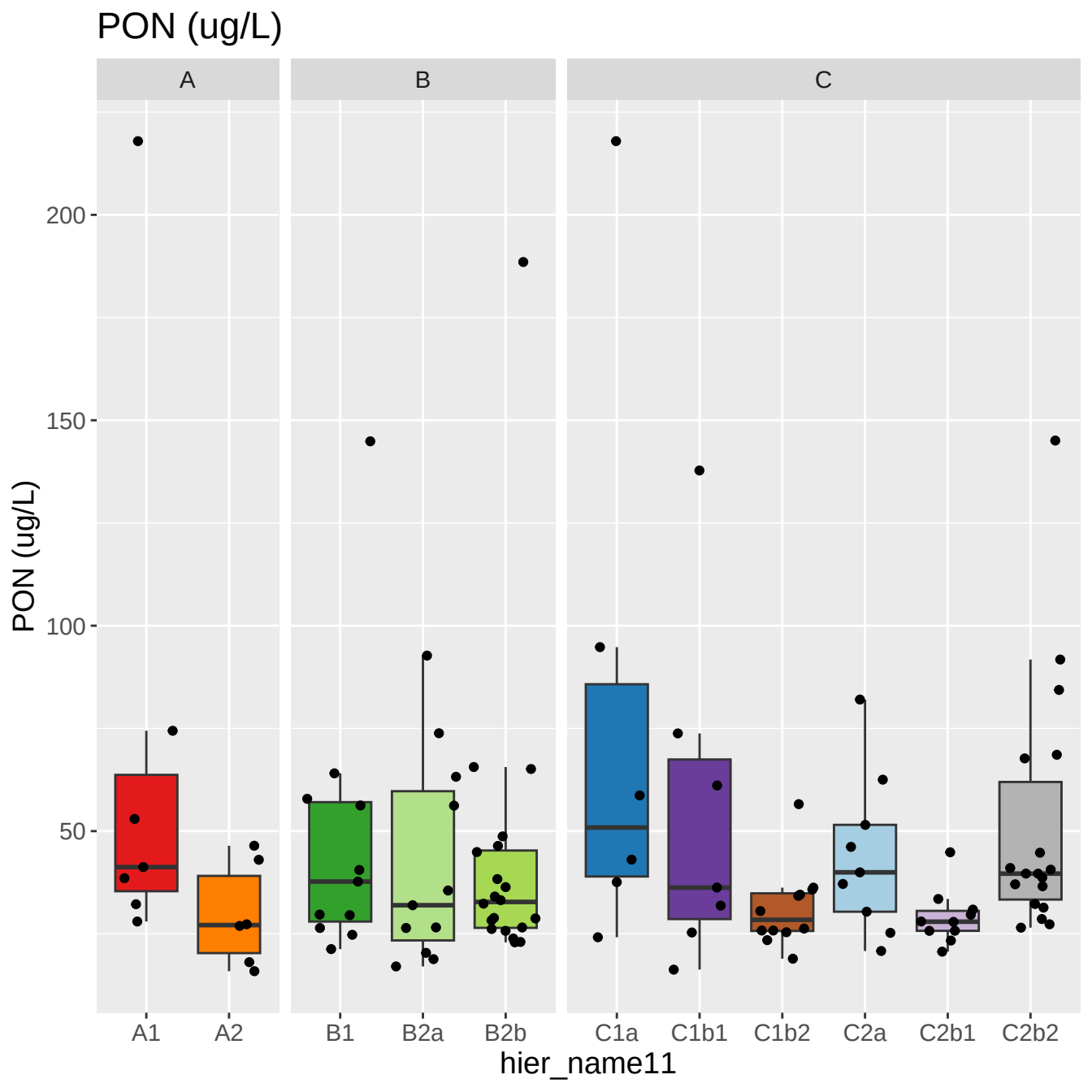

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
salinity
27
29
31

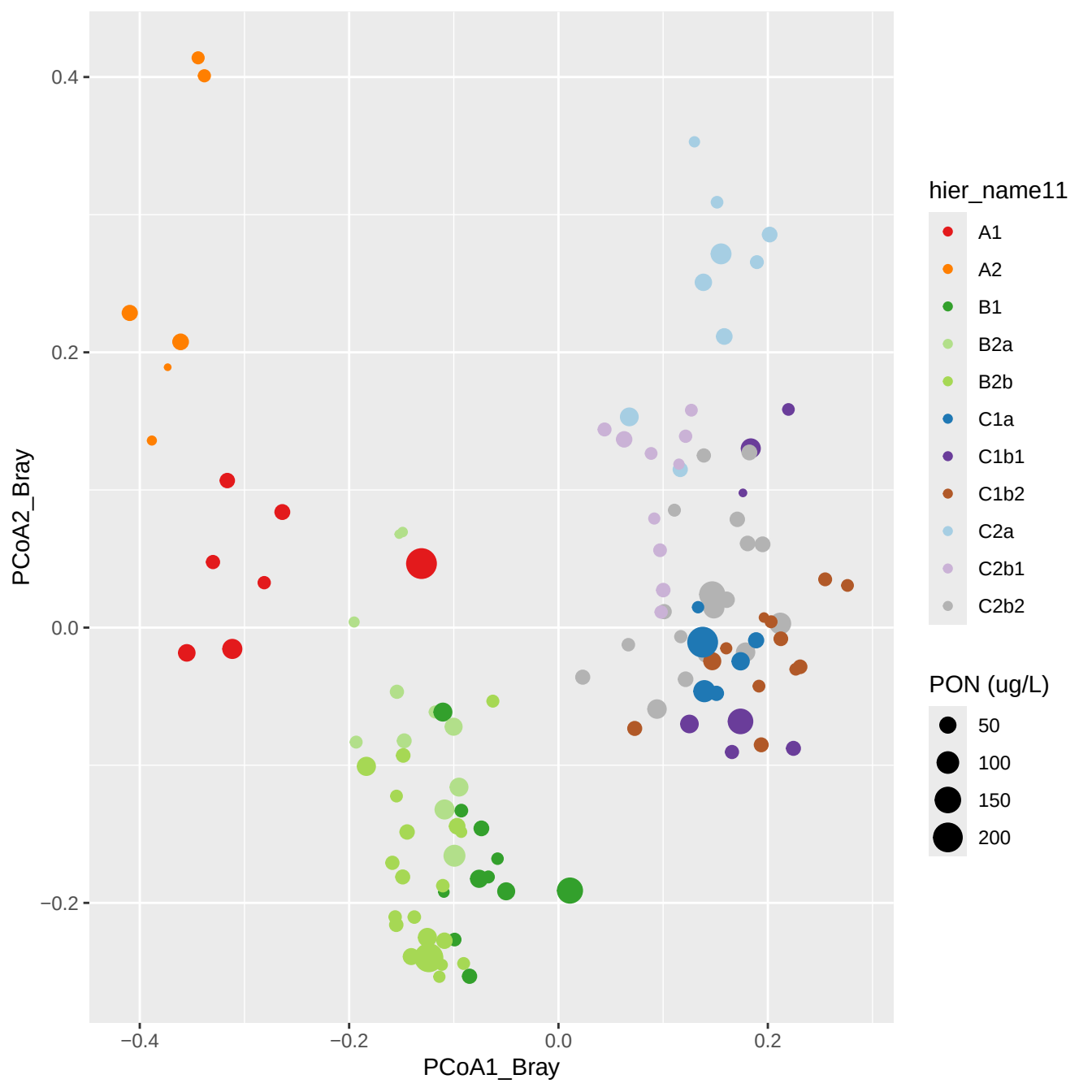

DO
A
B
C
400
350
300
DO
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

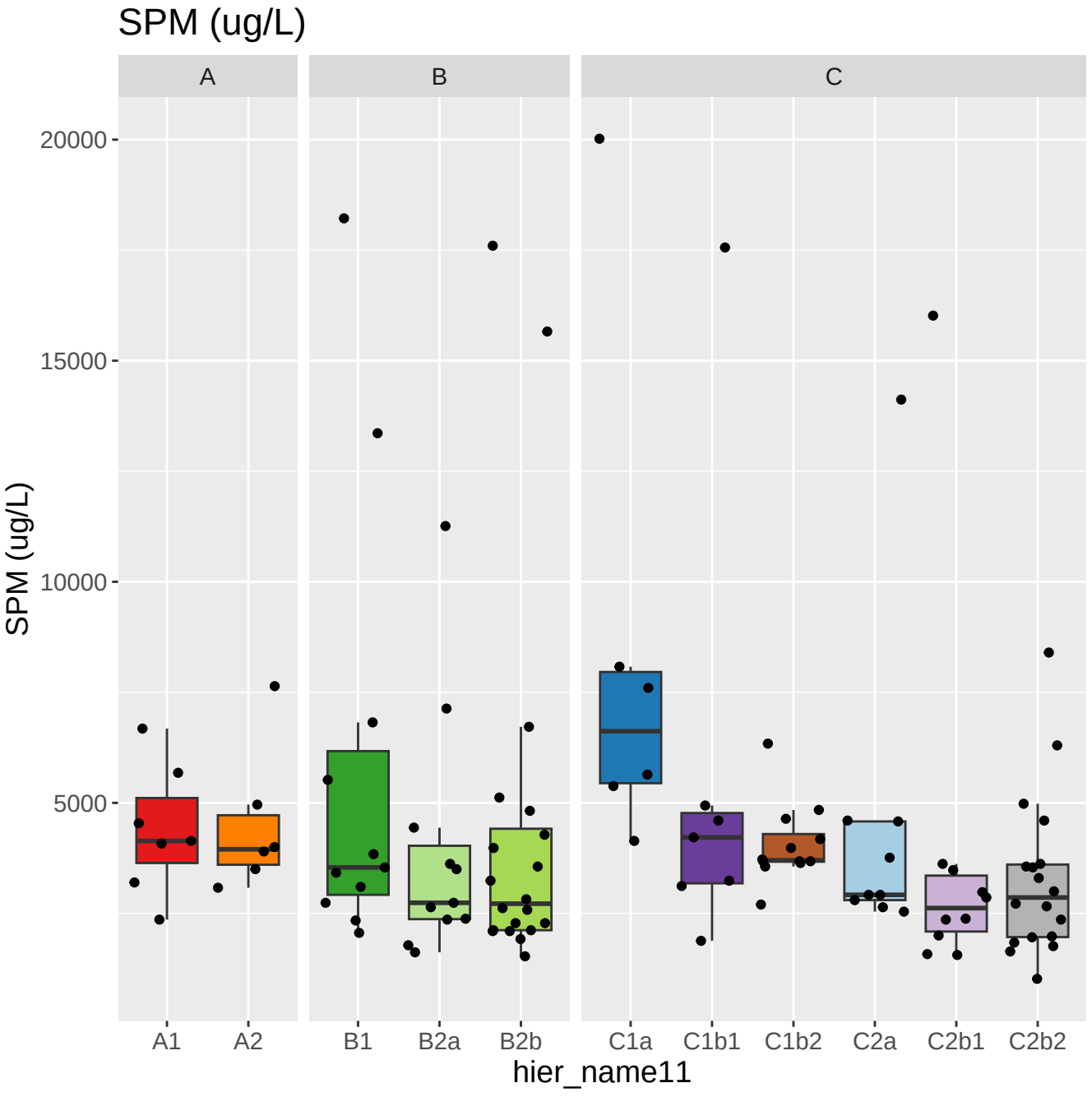

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
DO
300
350
400

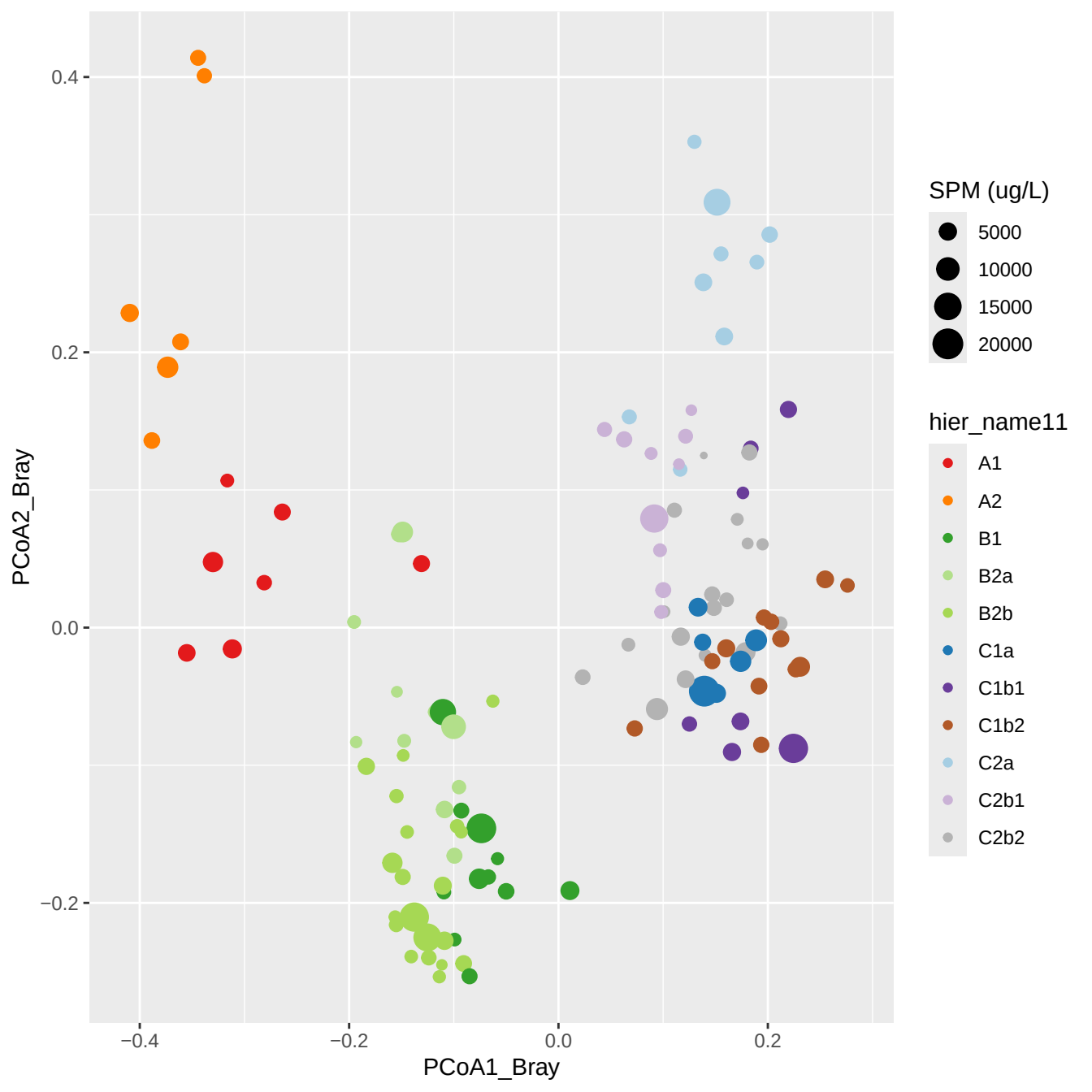

POC (ug/L)
A
B
C
1000
500
0
POC (ug/L)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

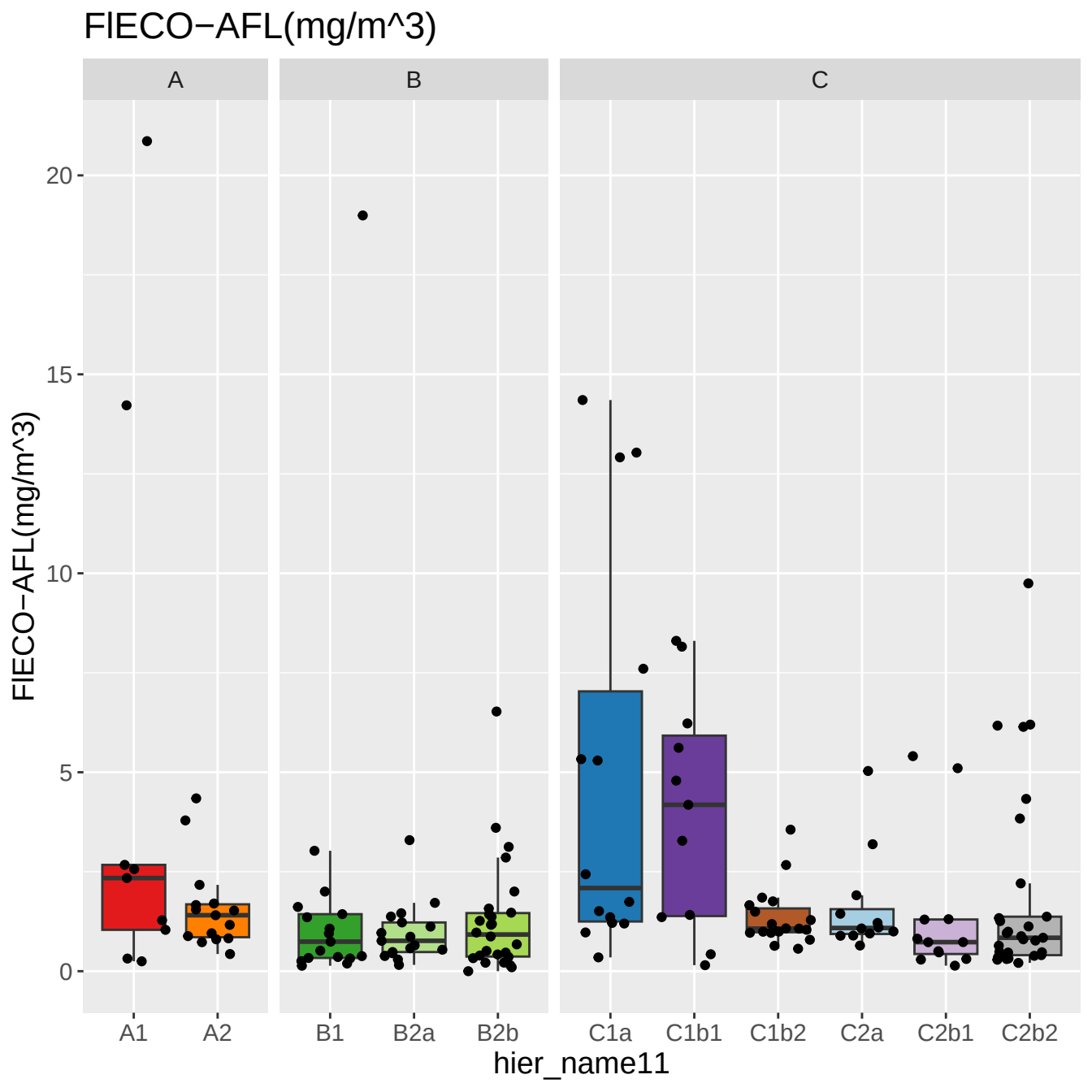

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
POC (ug/L)
250
500
750
1000
1250

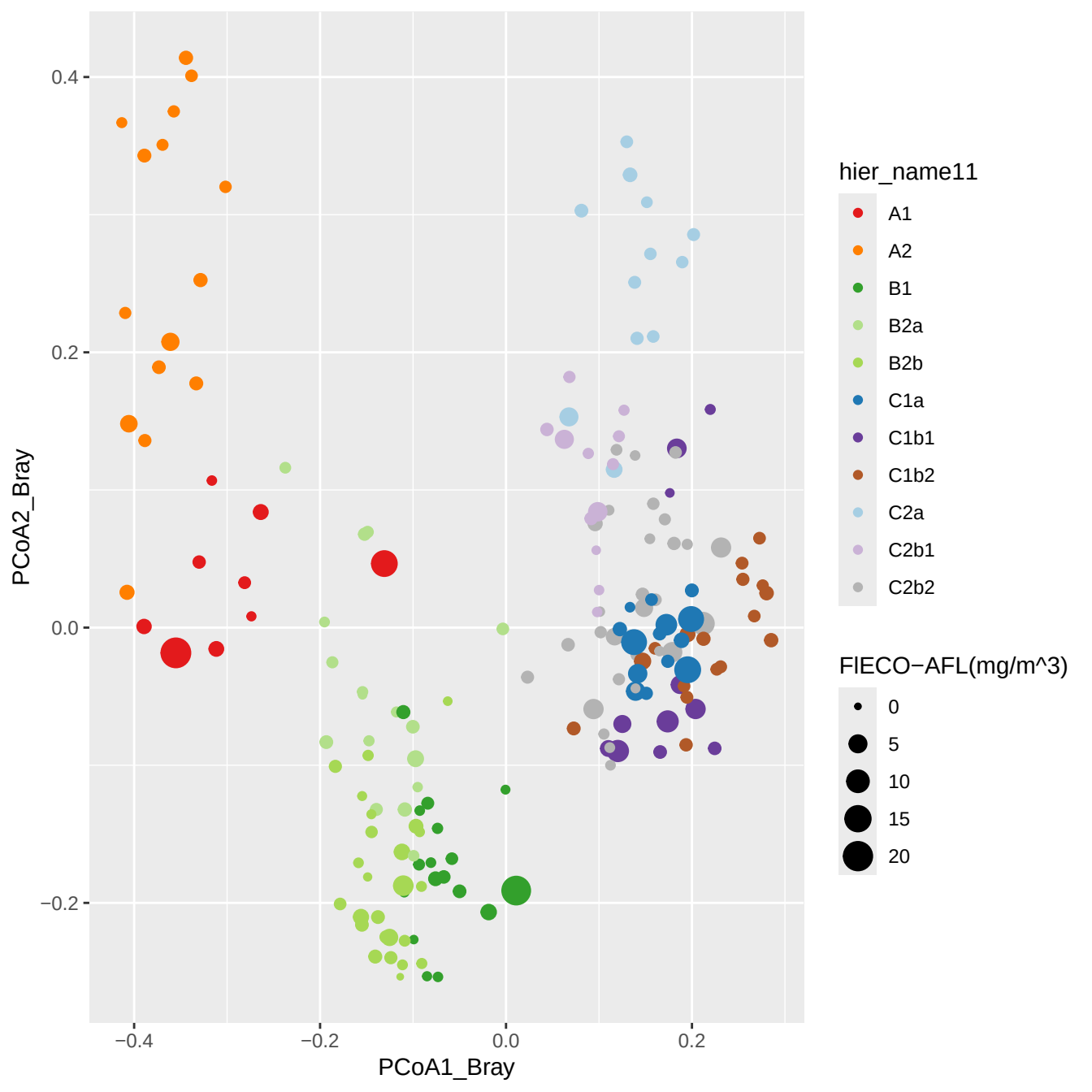

PON (ug/L)
A
B
C
200
150
100
50
PON (ug/L)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

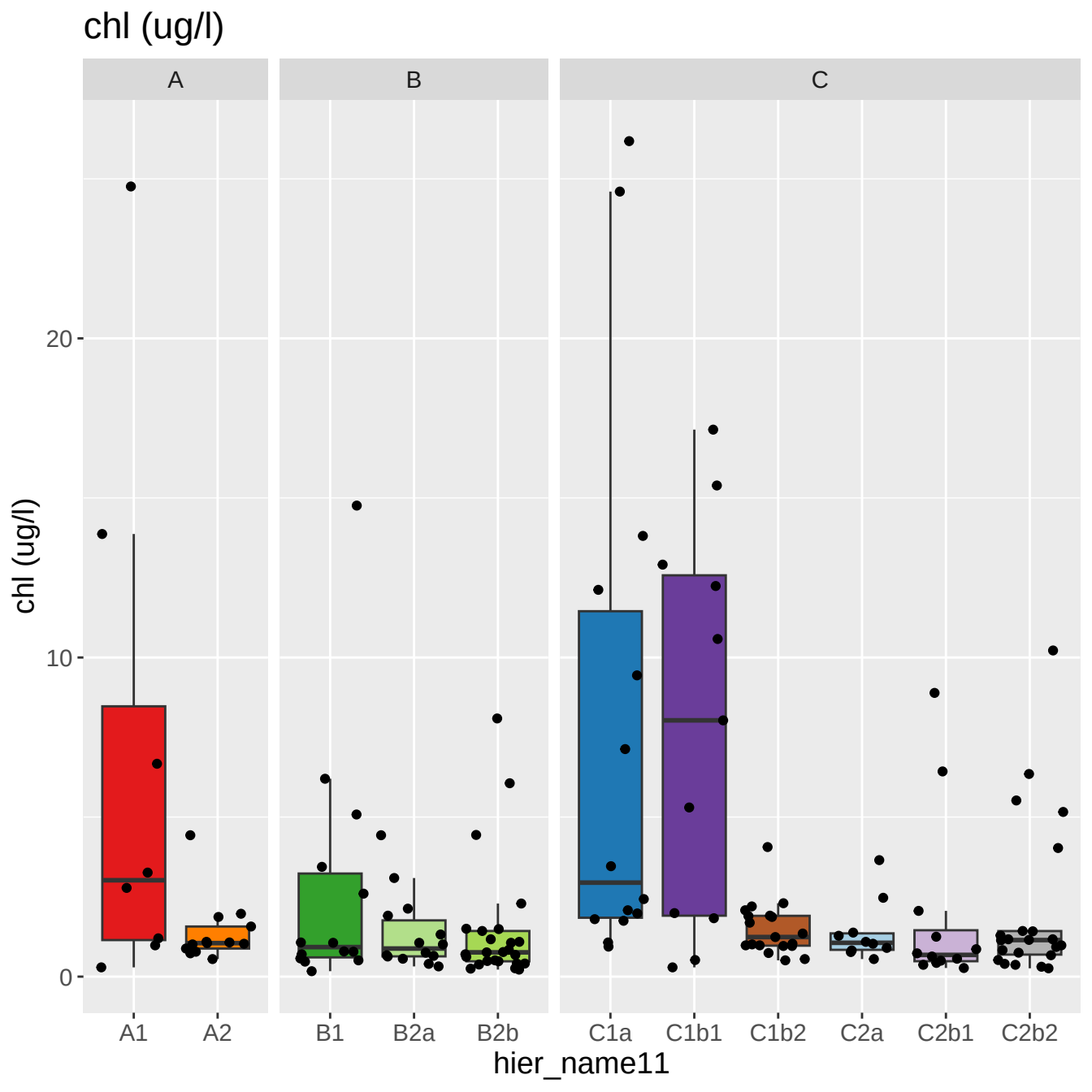

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
PON (ug/L)
50
100
150
200

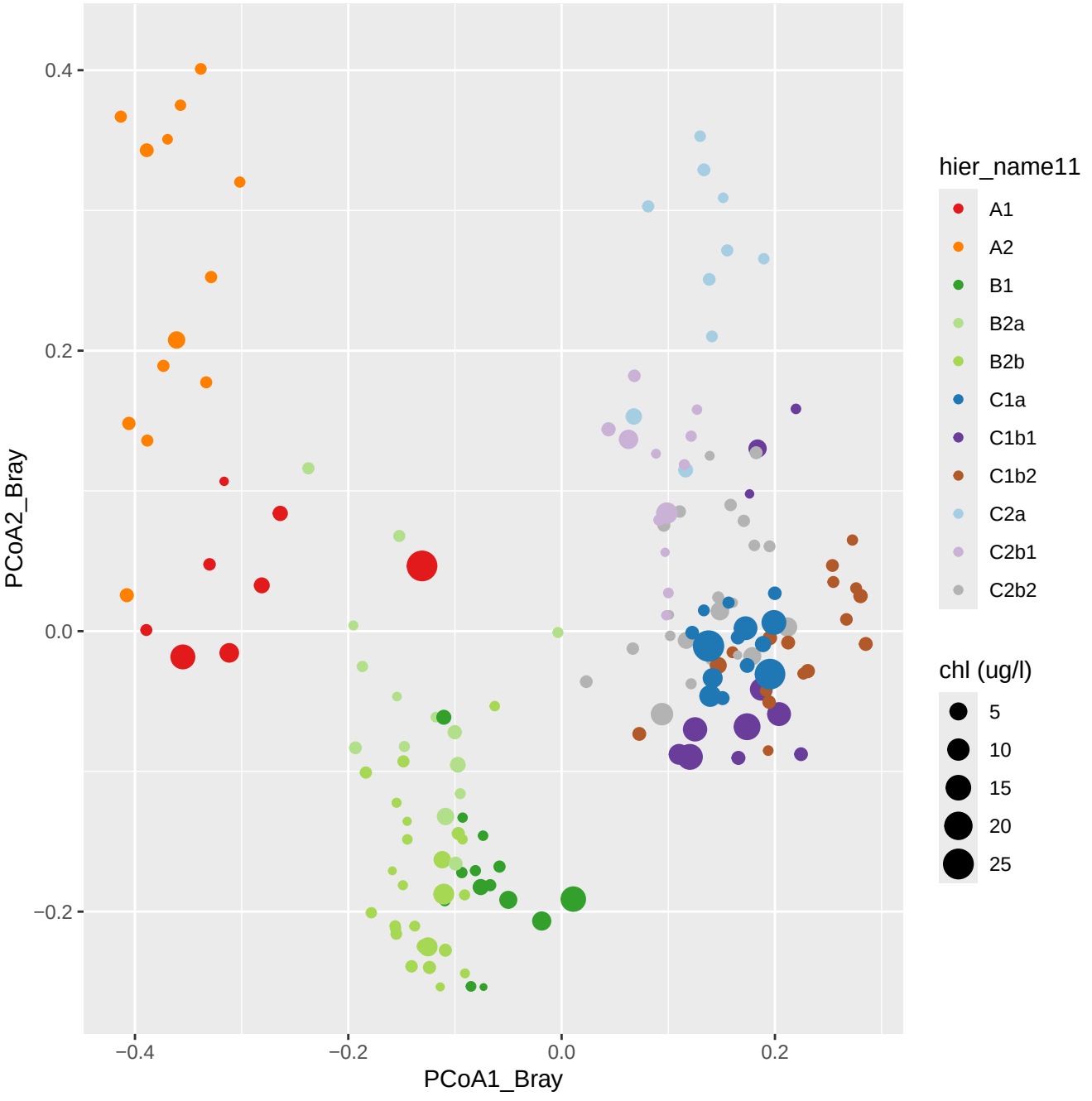

SPM (ug/L)
A
B
C
20000
15000
10000
5000
SPM (ug/L)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

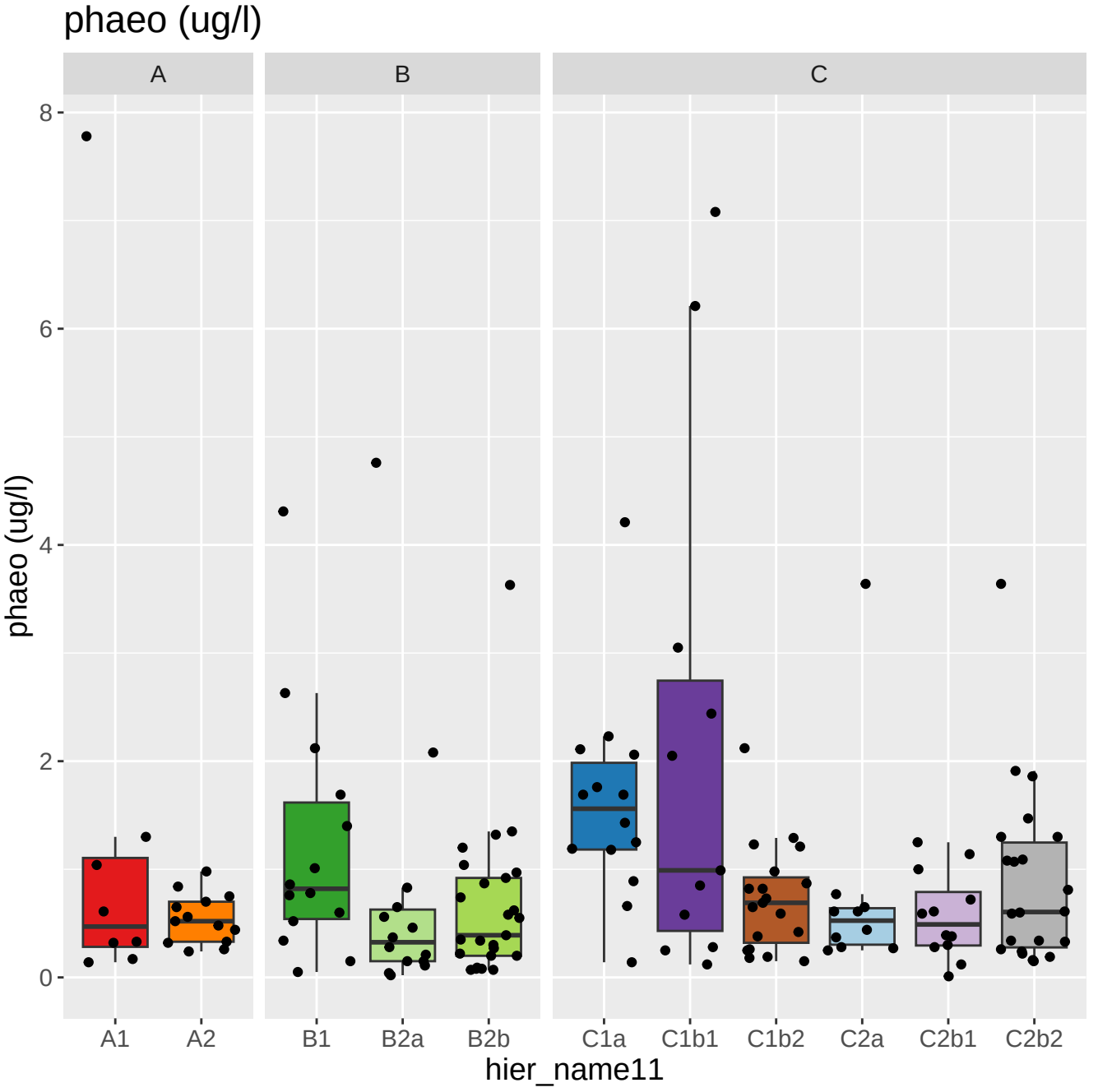

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
SPM (ug/L)
5000
10000
15000
20000
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

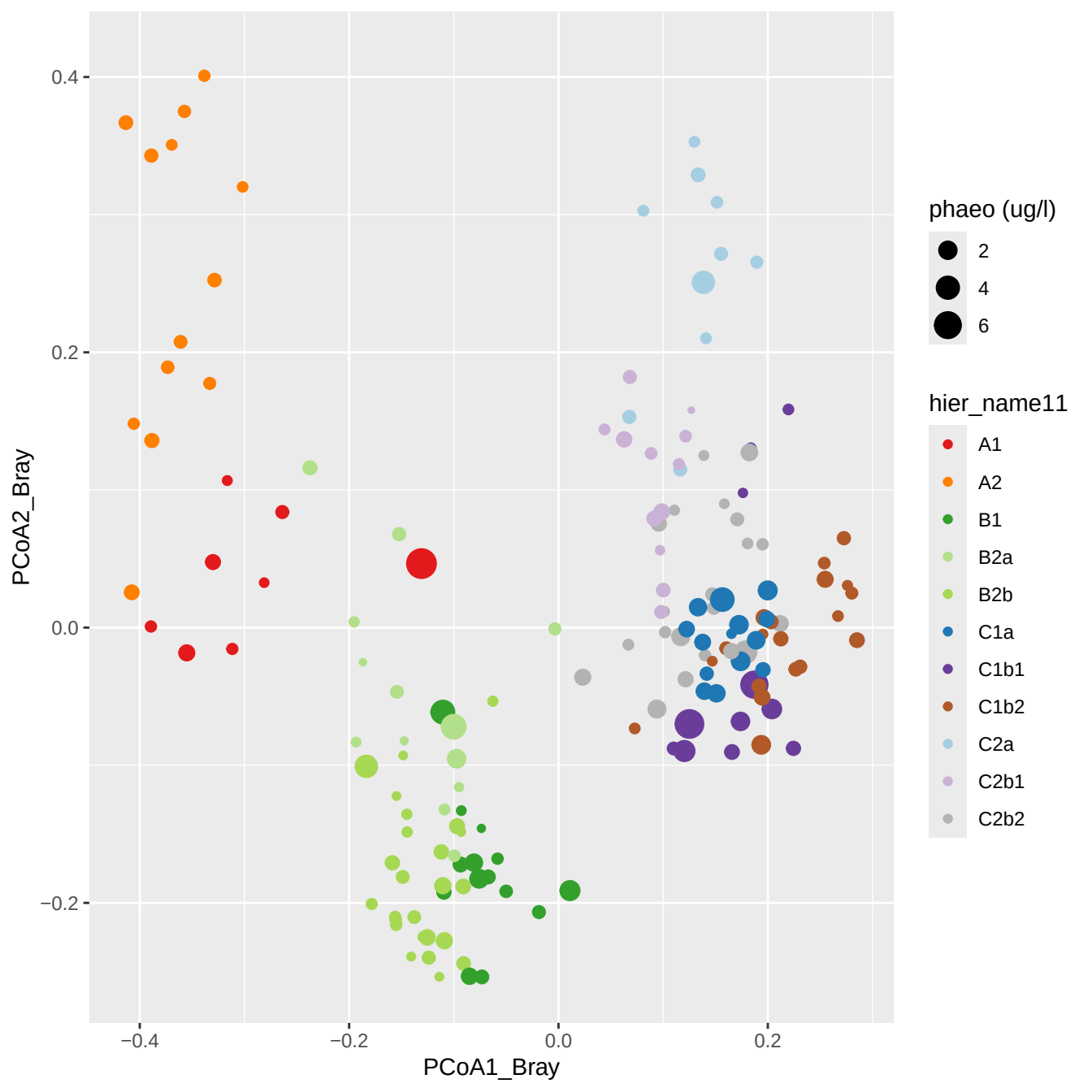

FIECO−AFL(mg/m^3)
A
B
C
20
15
10
5
0
FIECO−AFL(mg/m^3)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

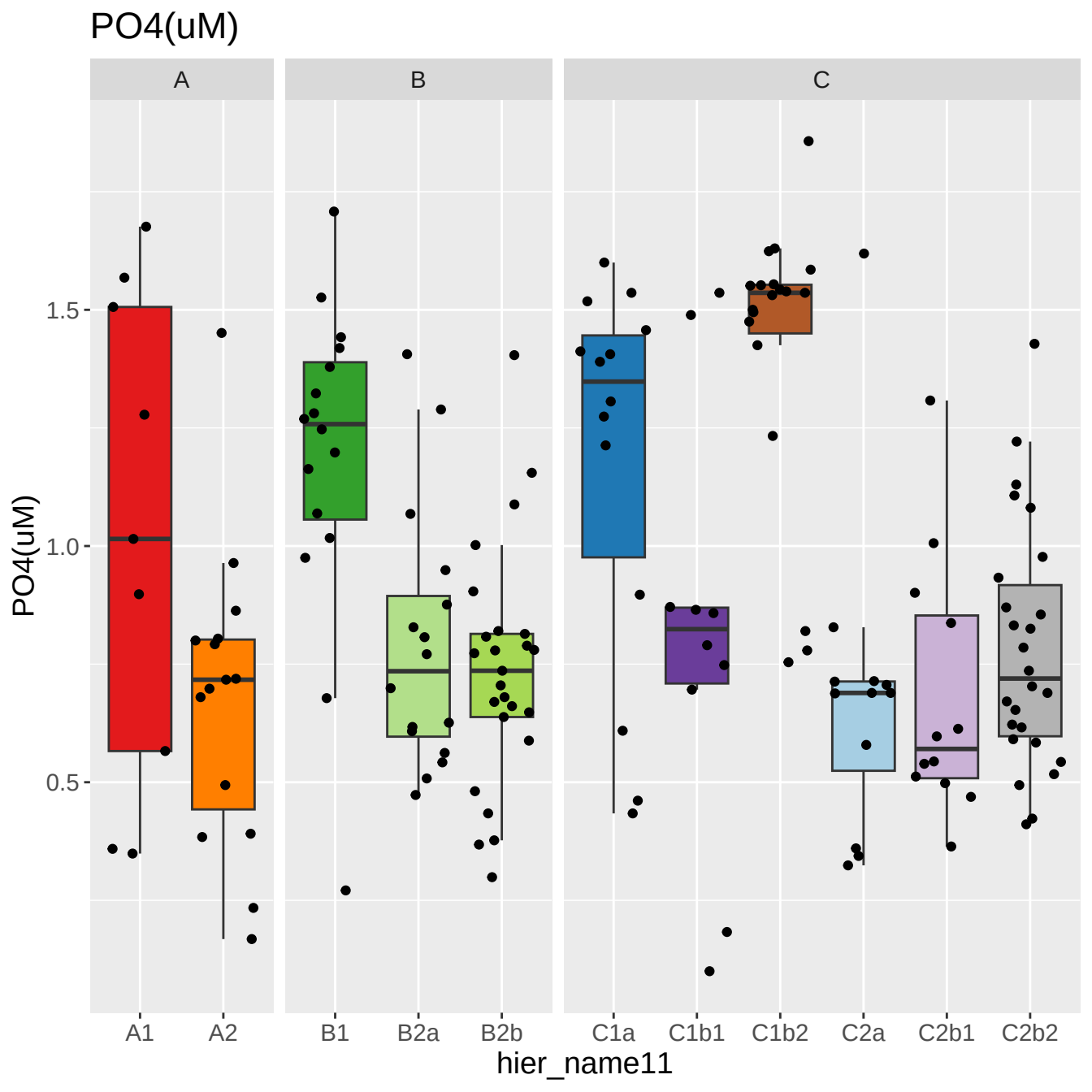

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
FIECO−AFL(mg/m^3)
0
5
10
15
20

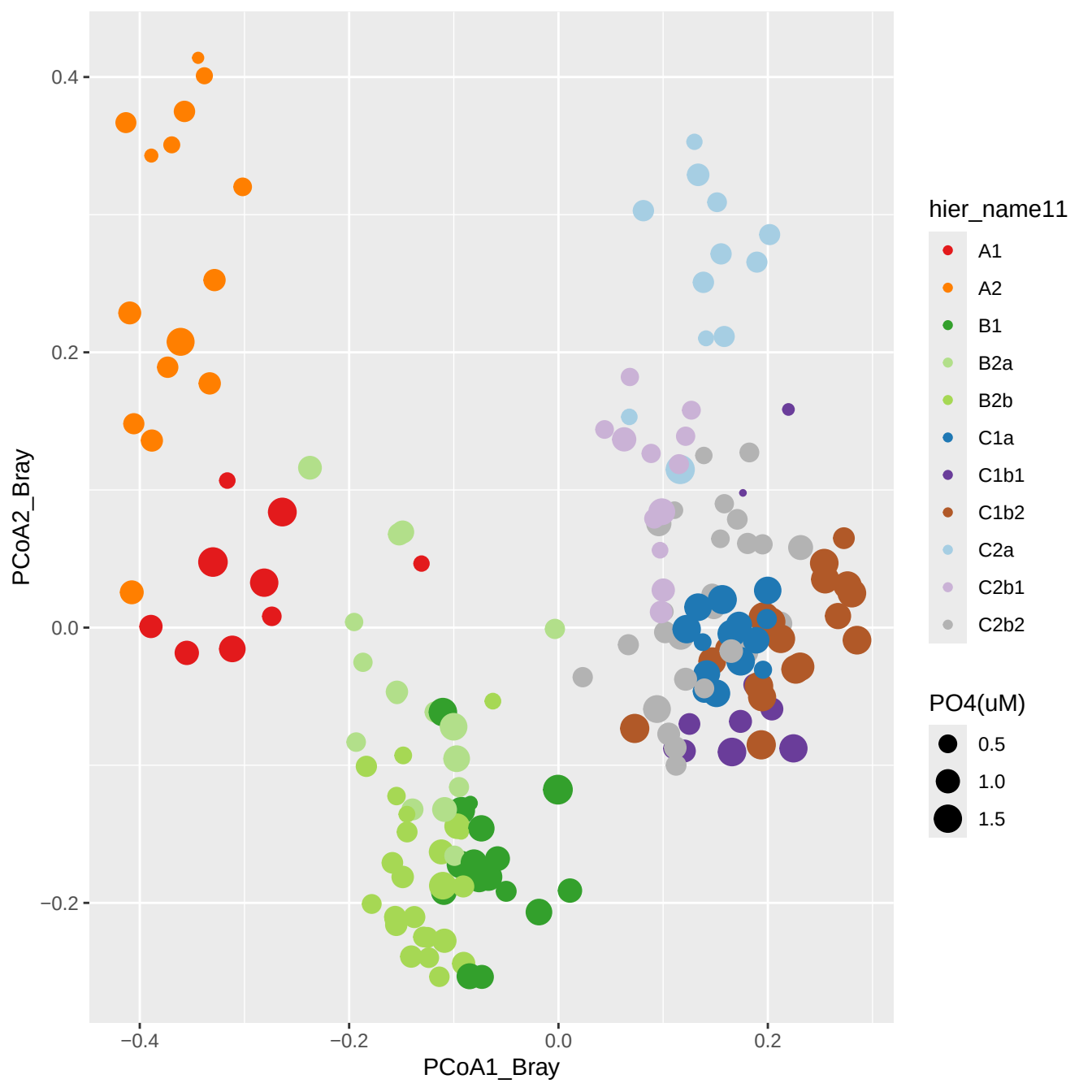

chl (ug/l)
A
B
C
20
10
0
chl (ug/l)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

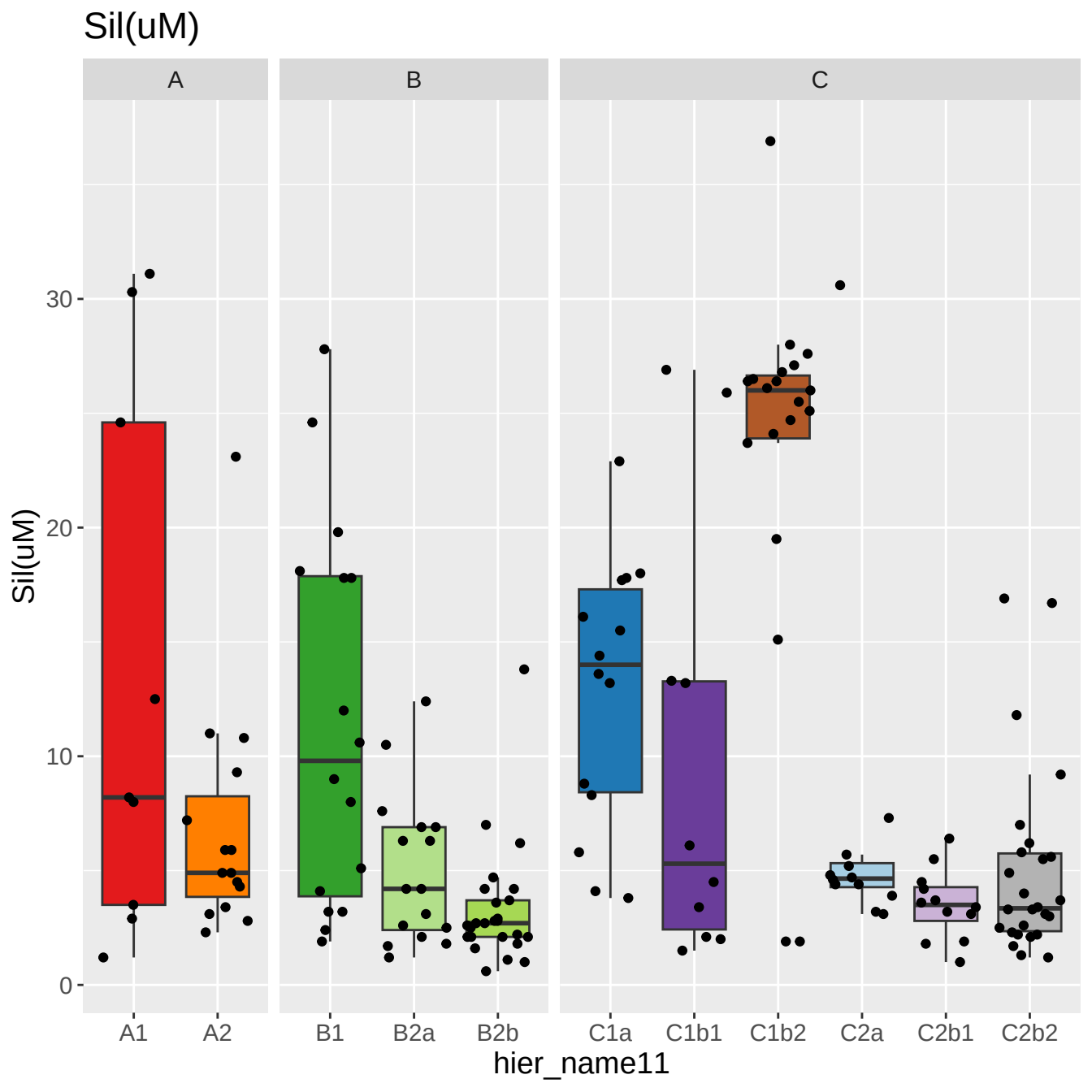

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
chl (ug/l)
5
10
15
20
25

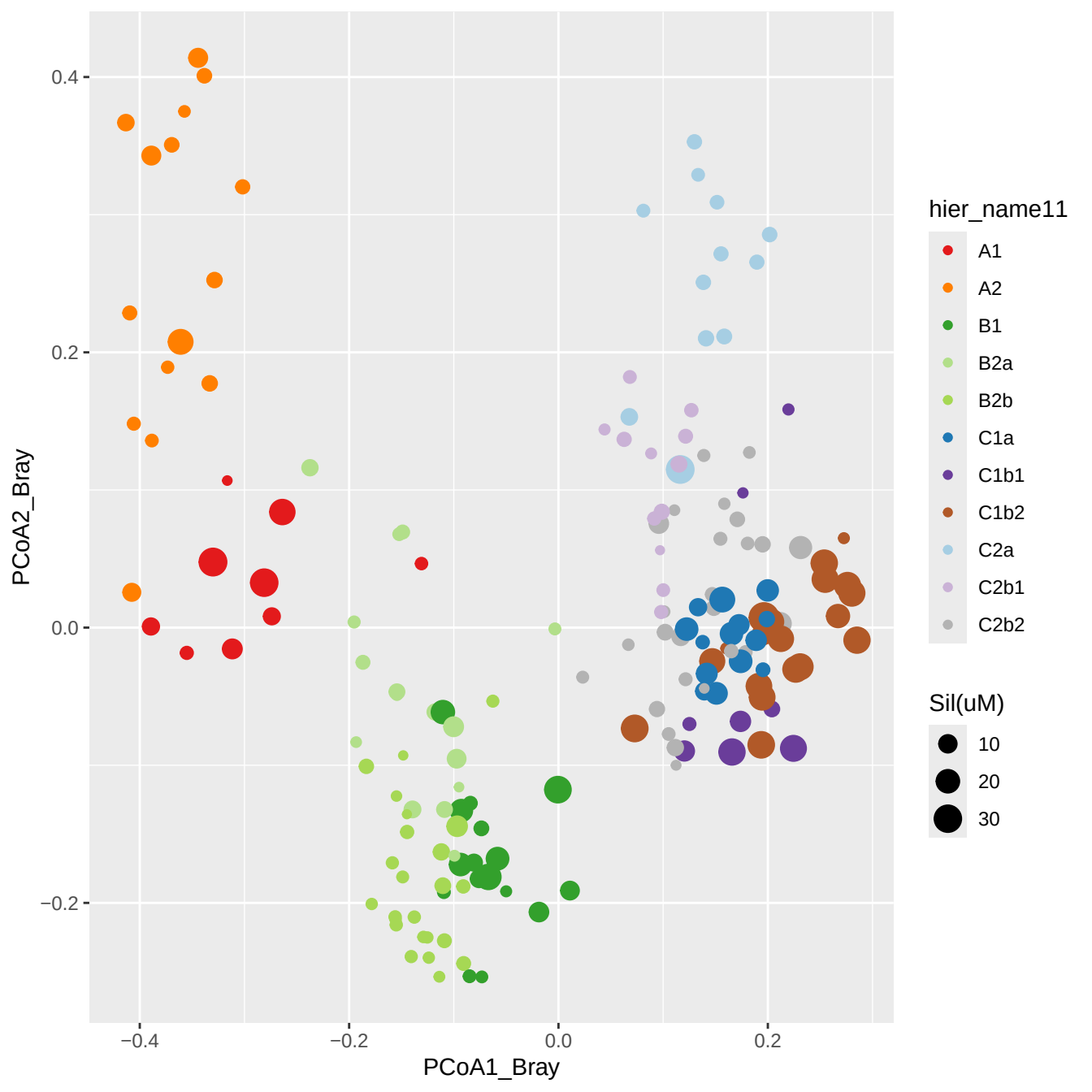

phaeo (ug/l)
A
B
C
8
6
4
2
0
phaeo (ug/l)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

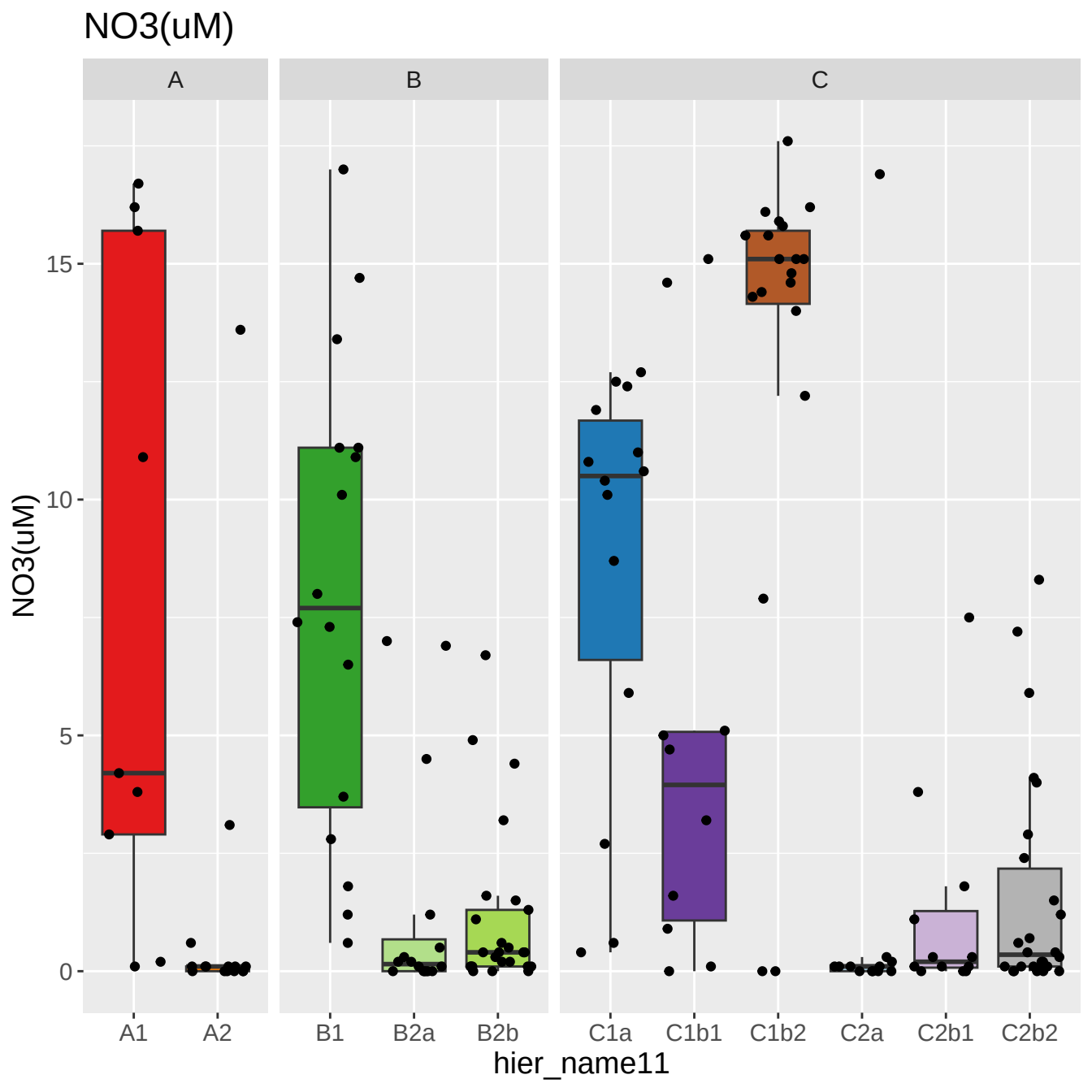

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
phaeo (ug/l)
2
4
6
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2

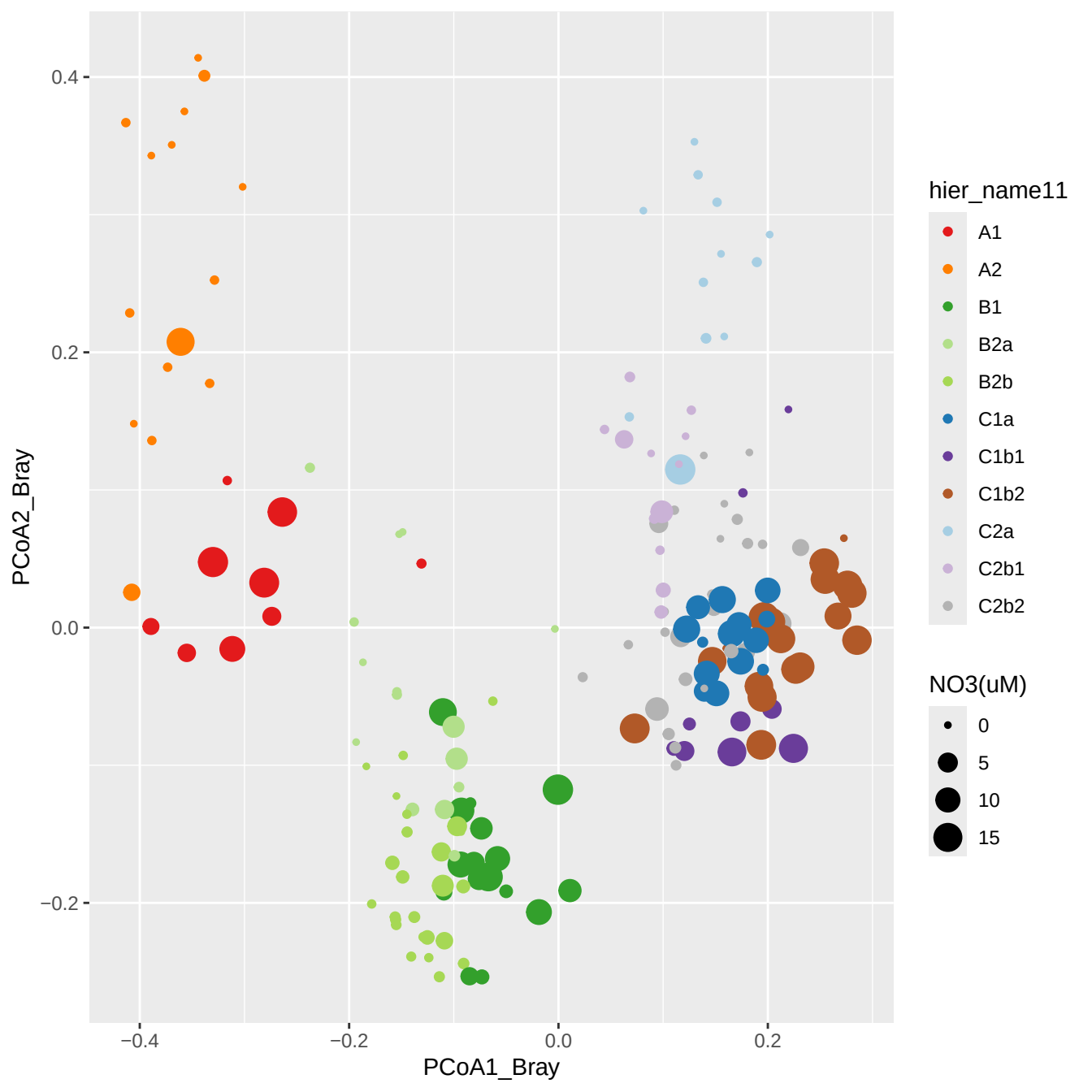

PO4(uM)
A
B
C
1.5
1.0
0.5
PO4(uM)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

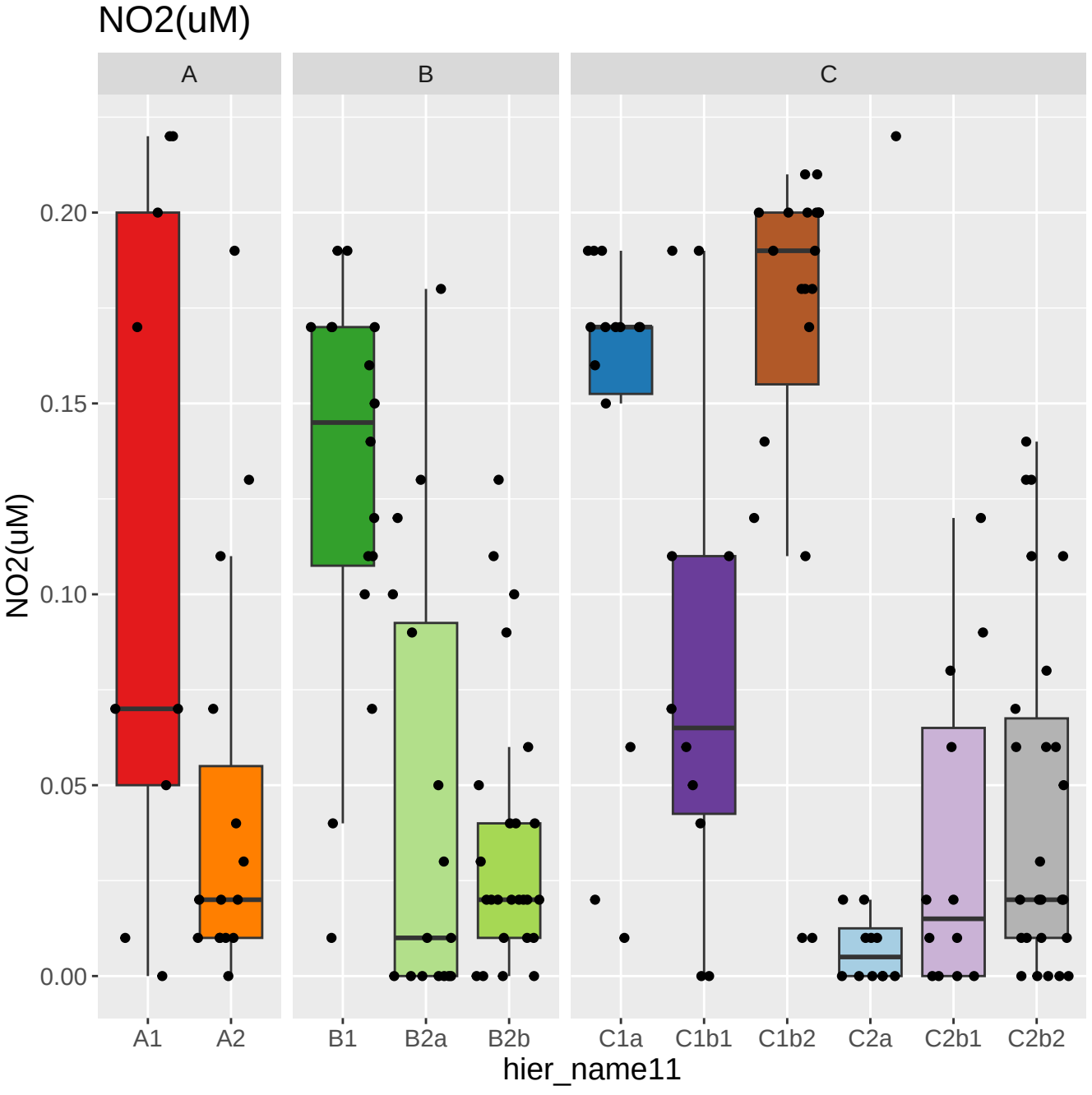

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
PO4(uM)
0.5
1.0
1.5

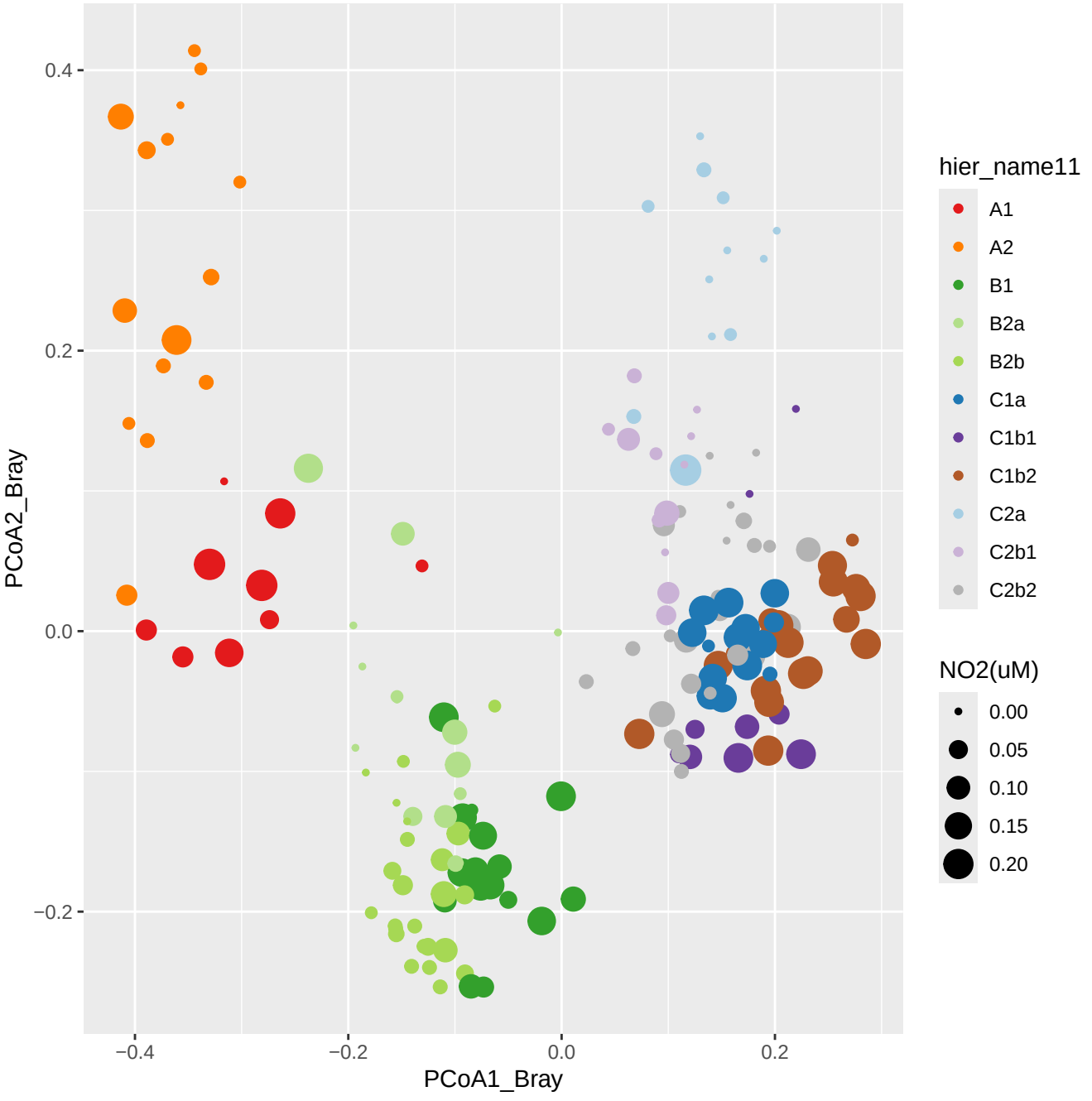

Sil(uM)
A
B
C
30
20
10
0
Sil(uM)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

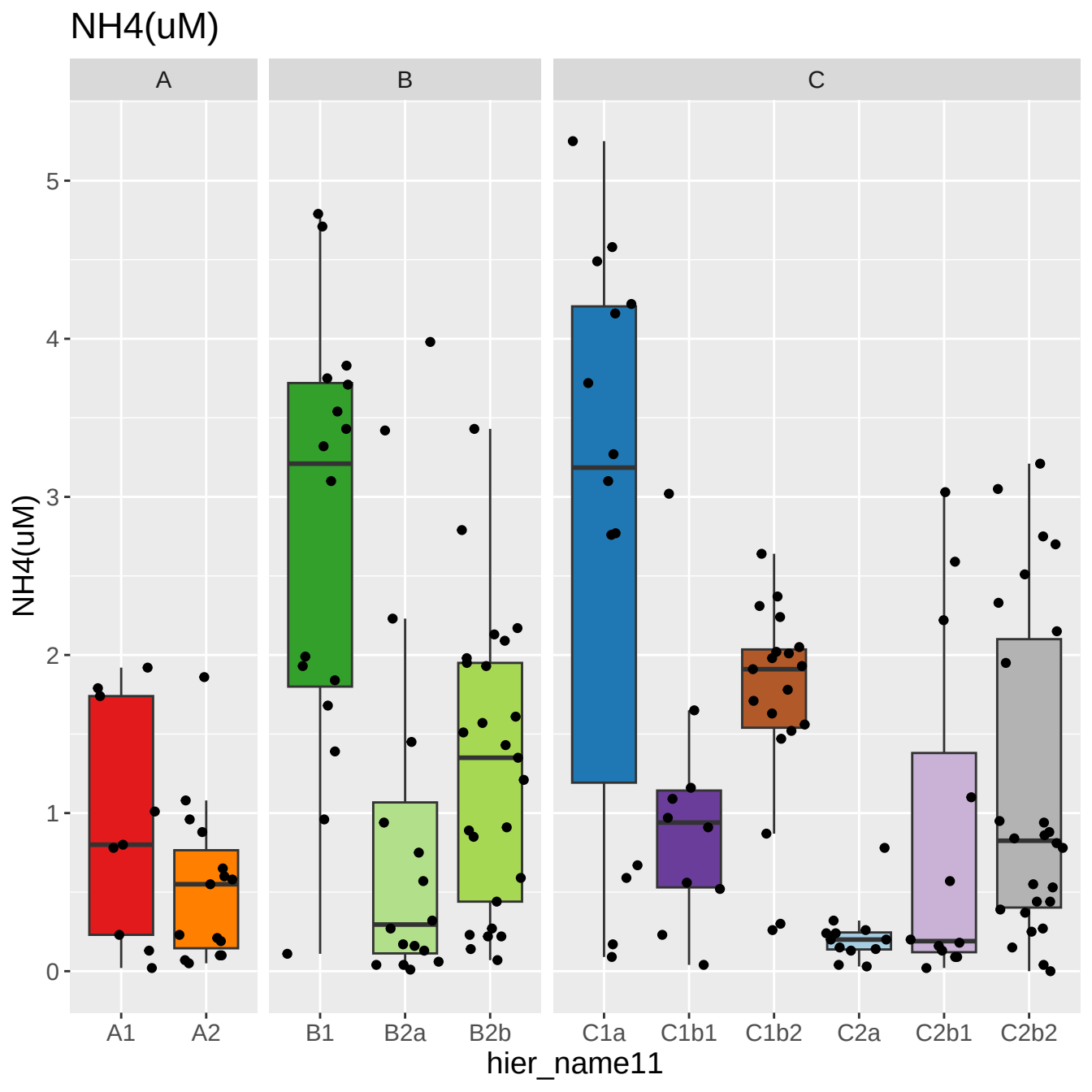

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
Sil(uM)
10
20
30

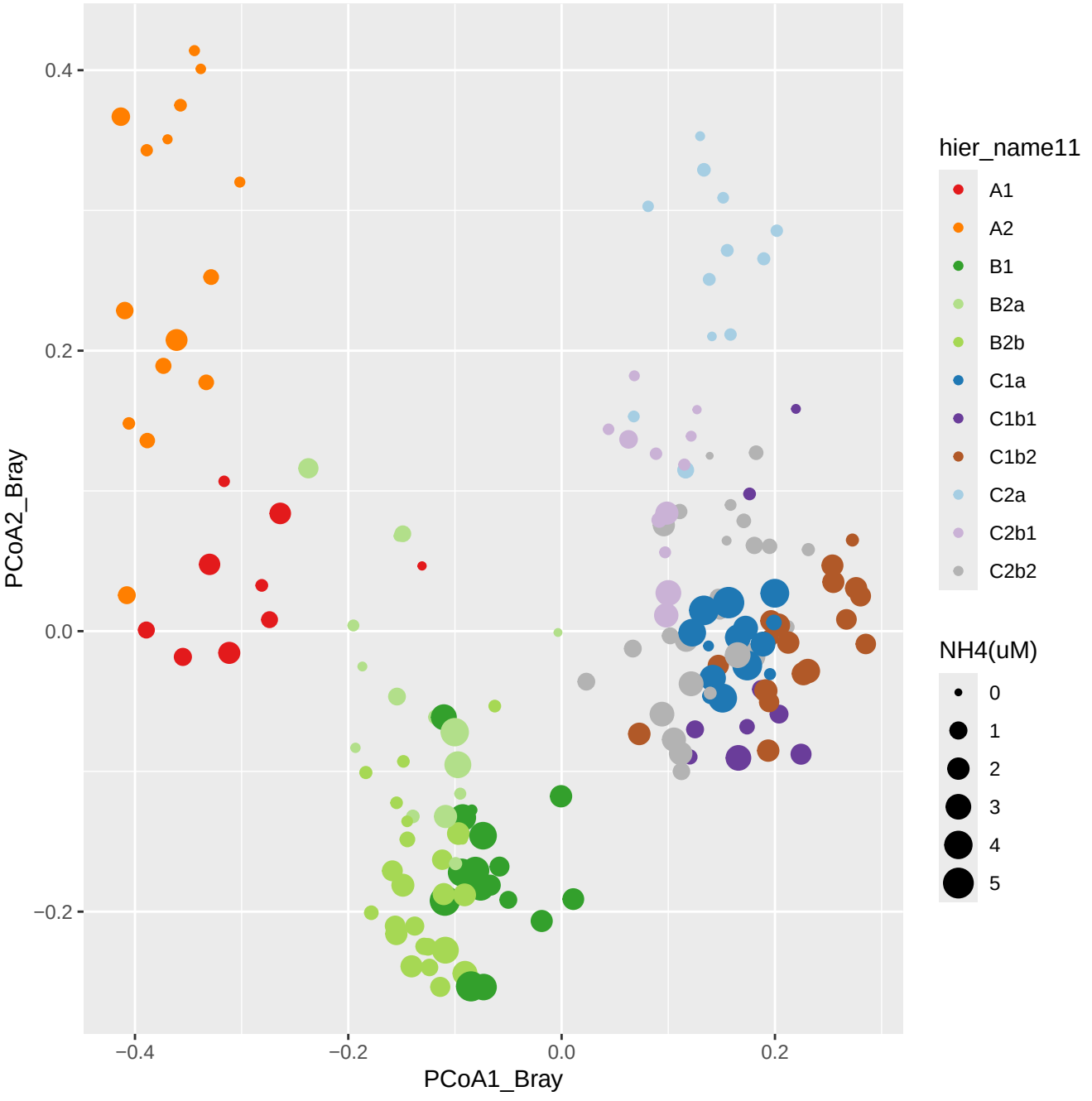

NO3(uM)
A
B
C
15
10
5
0
NO3(uM)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

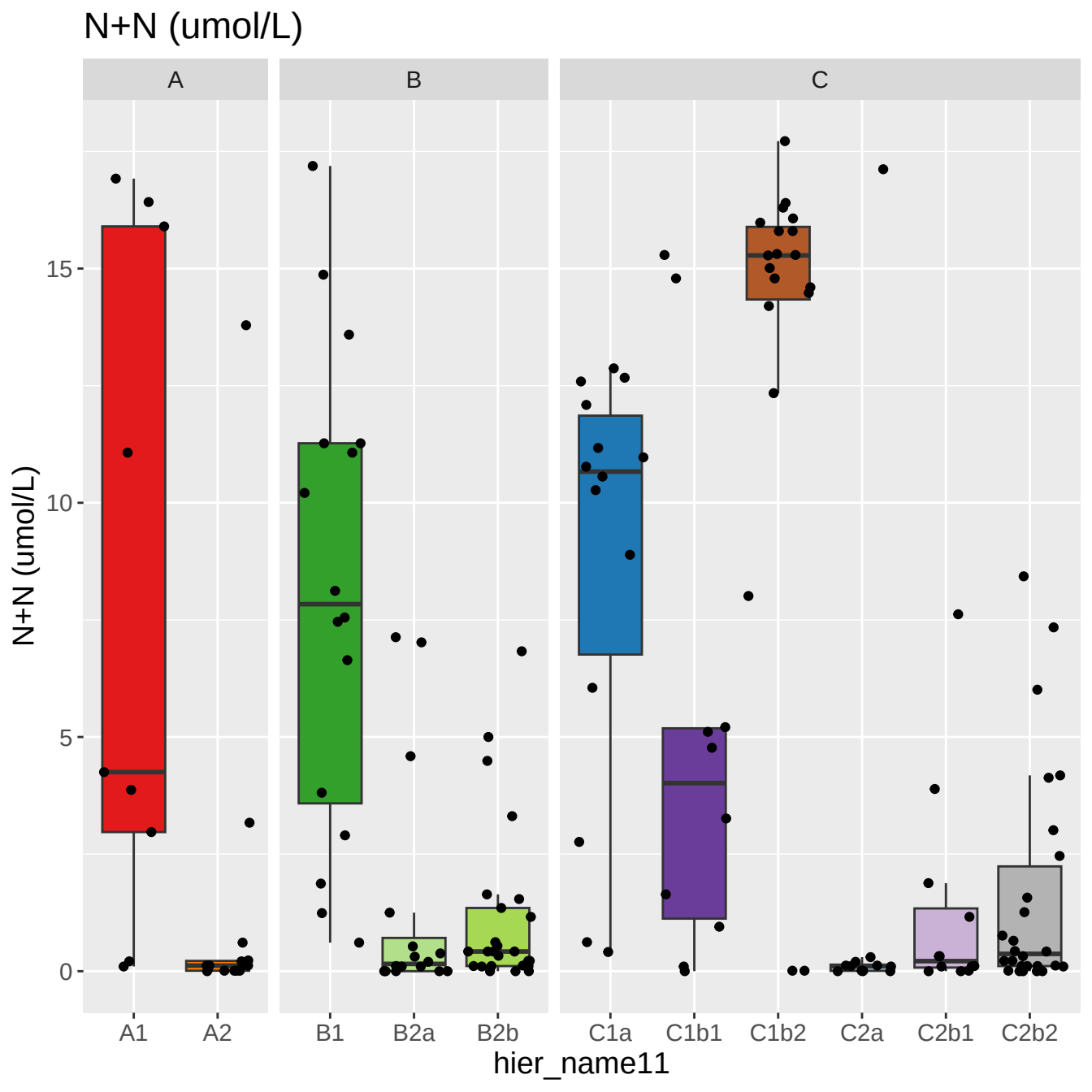

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
NO3(uM)
0
5
10
15

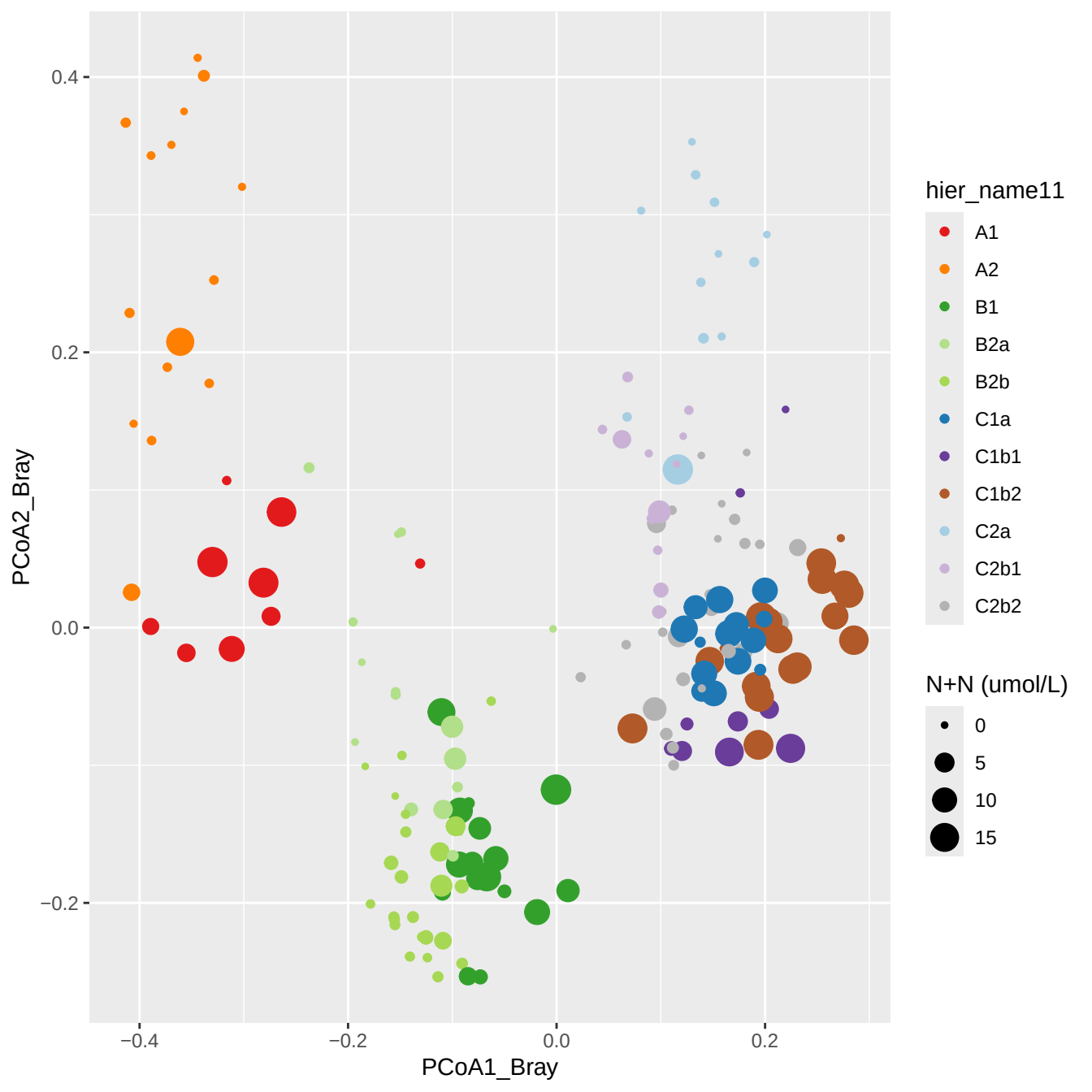

NO2(uM)
A
B
C
0.20
0.15
0.10
0.05
0.00
NO2(uM)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

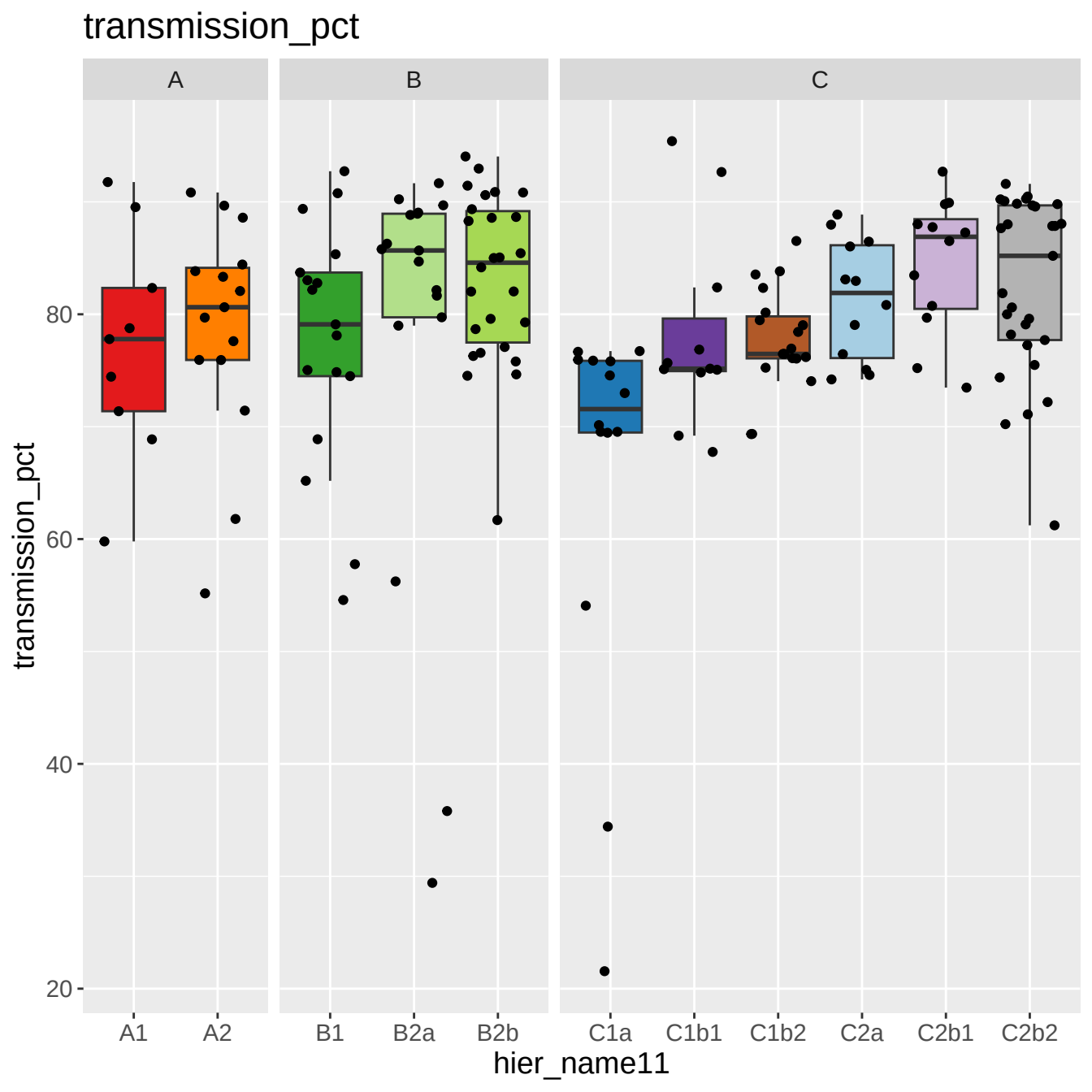

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
NO2(uM)
0.00
0.05
0.10
0.15
0.20

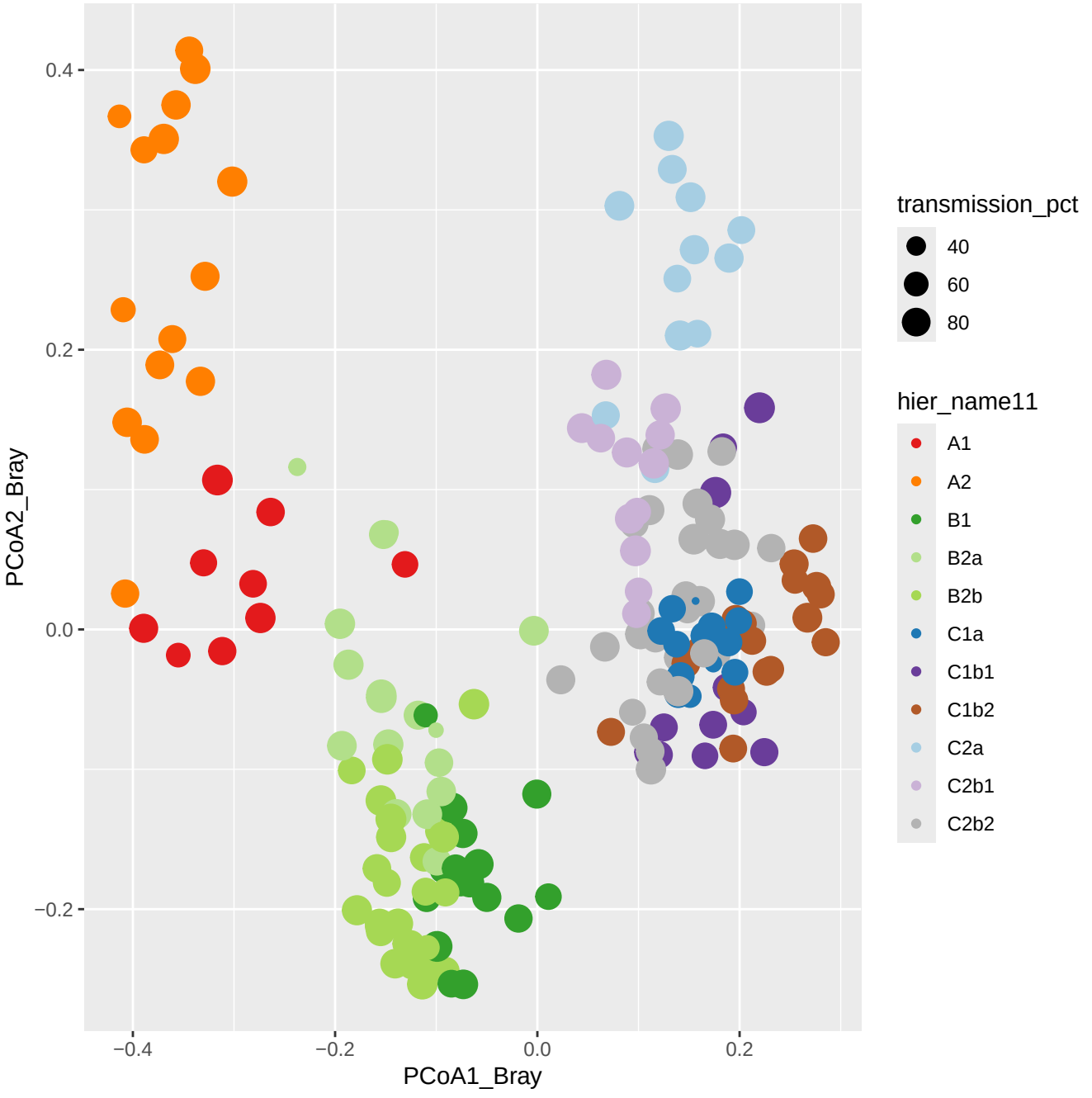

NH4(uM)
A
B
C
5
4
3
2
1
0
NH4(uM)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

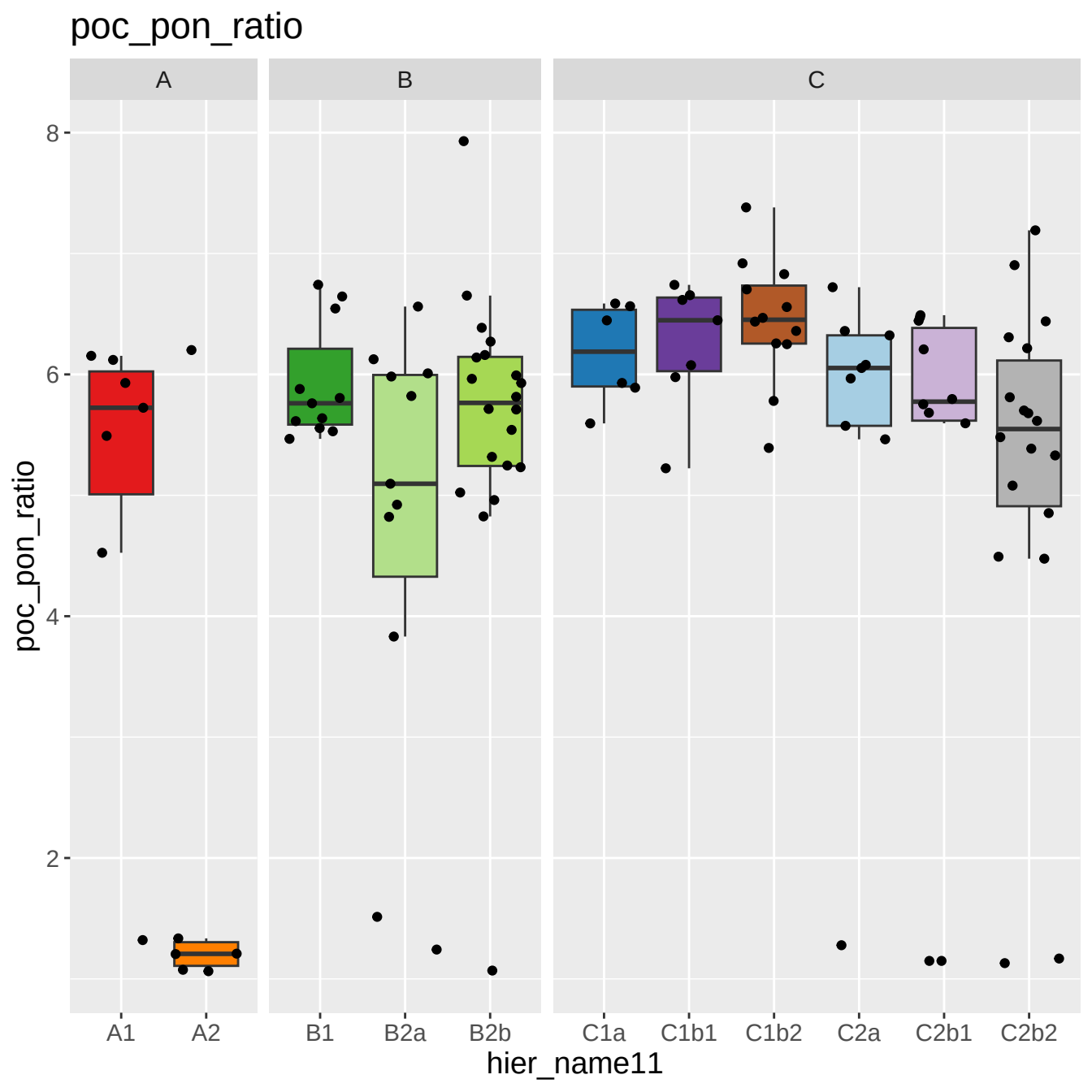

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
NH4(uM)
0
1
2
3
4
5

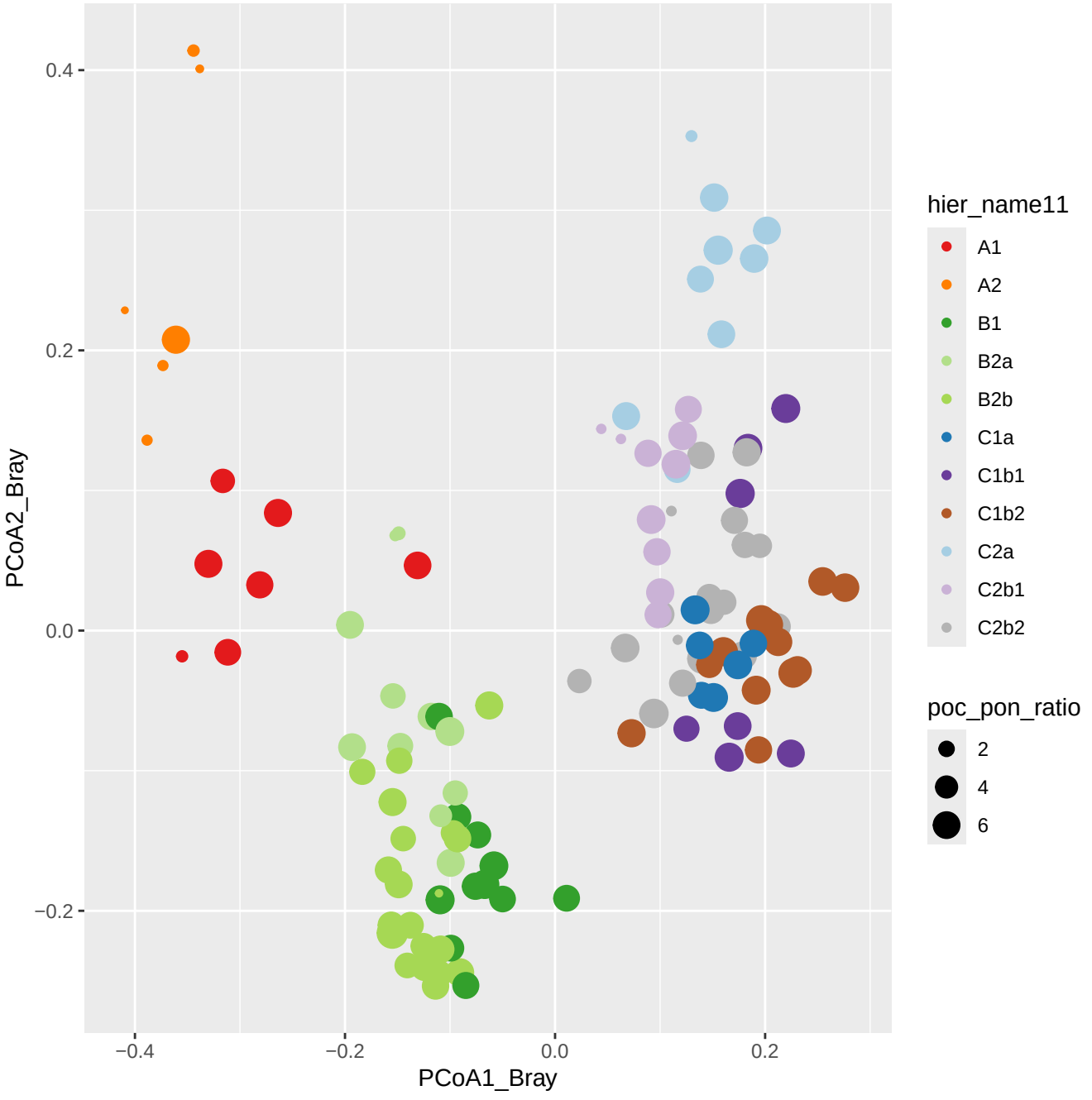

N+N (umol/L)
A
B
C
15
10
5
0
N+N (umol/L)
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11

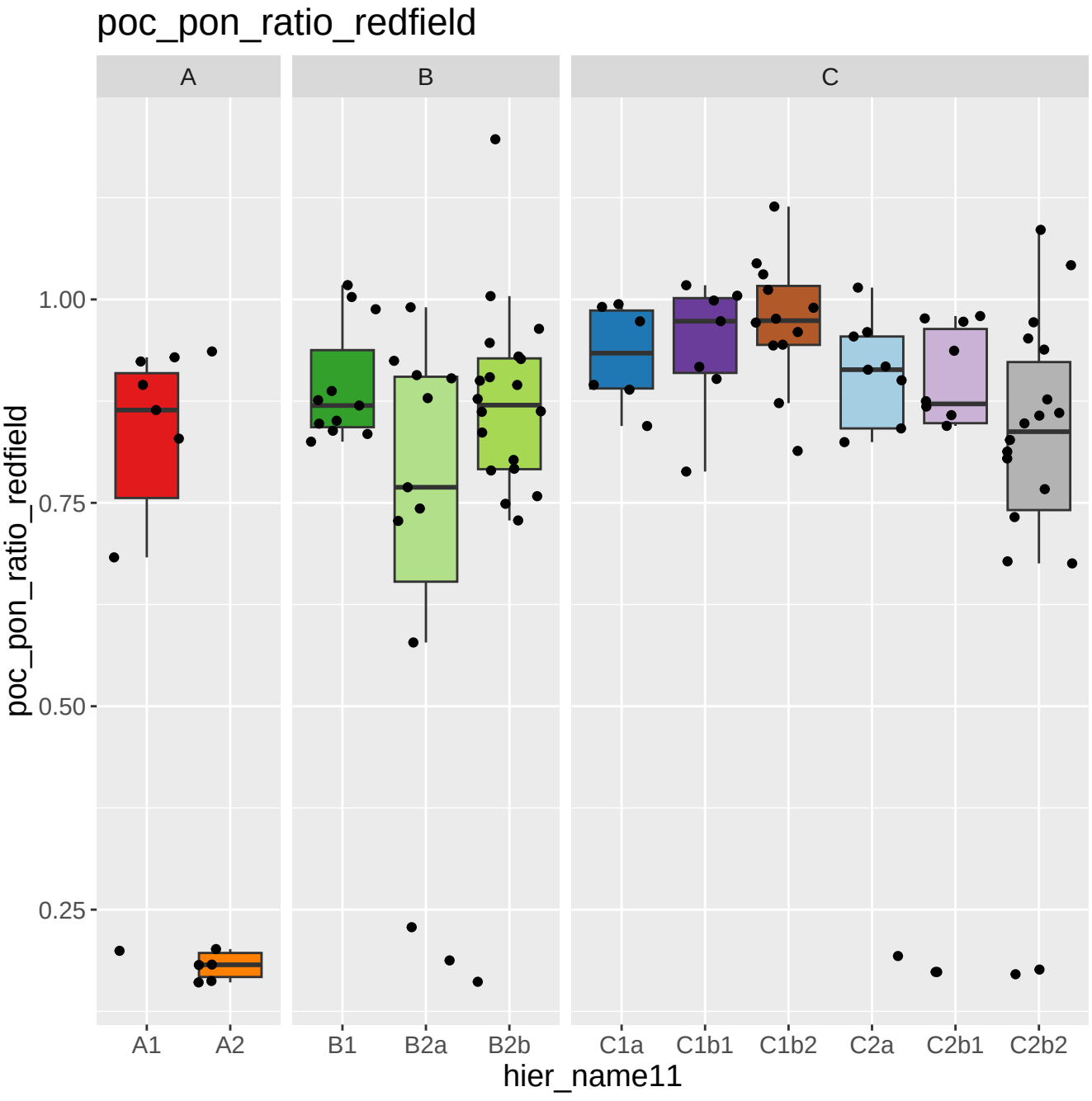

0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
N+N (umol/L)
0
5
10
15


transmission_pct
A
B
C
80
60
40
20
transmission_pct
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
transmission_pct
40
60
80
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2


poc_pon_ratio
A
B
C
8
6
4
2
poc_pon_ratio
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11


0.4
0.2
0.0
−0.2
PCoA2_Bray
−0.4
−0.2
0.0
0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
poc_pon_ratio
2
4
6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B
C
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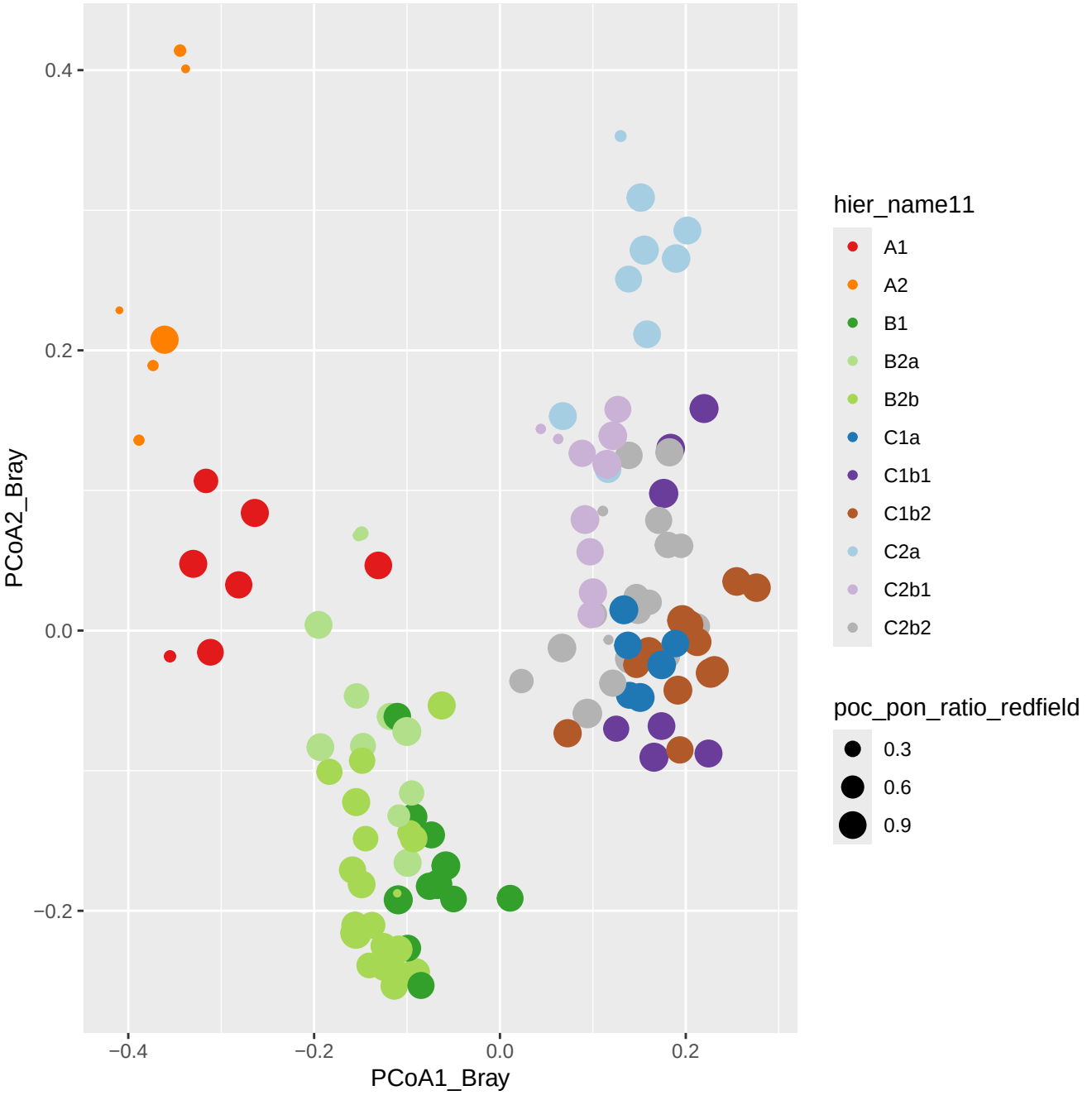
0.75
0.50
0.25
poc_pon_ratio_redfield
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
hier_name11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0.0
−0.2
PCoA2_Bray
−0.4
−0.2
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0.2
PCoA1_Bray
hier_name11
A1
A2
B1
B2a
B2b
C1a
C1b1
C1b2
C2a
C2b1
C2b2
poc_pon_ratio_redfield
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0.6
0.9